CAMPINAS–MIAMI MATHEMATICAL MEETING 2026

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Contents

Conference information.

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Organizing Committee.

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1. Collapsing Flat SU(2)-Bundles to Spherical 3-Manifolds

Speaker.
Eder M. Correa (Campinas, Brazil)

Title.
Collapsing Flat SU(2)-Bundles to Spherical 3-Manifolds.

Abstract.
In this talk, we present a geometric mechanism for the emergence of spherical 3-manifolds from the superspace of Riemannian metrics associated with flat SU(2)-bundles over closed orientable hyperbolic surfaces. We show that any spherical 3-manifold can be realized as a boundary point in the Gromov-Hausdorff closure of this superspace through a collapsing process. Furthermore, we discuss how the convergence toward the spherical limit is controlled by the order of the fundamental group of the target manifold and the systolic geometry of the hyperbolic base. Finally, we illustrate these results by showing how the Bolza surface provides the optimal error bound for the convergence toward the Poincaré homology sphere.

Main theme.
Spherical 3-manifolds appear as Gromov–Hausdorff boundary points of superspaces of suitable SU(2)-bundles over closed hyperbolic surfaces.

Spherical manifolds.

Definition 1.1. A Riemannian manifold M^n is *spherical* if it is complete and locally isometric to the round sphere S^n. Equivalently, it has constant sectional curvature $+1$ and universal cover S^n.

If M is connected, then

$$M \cong S^n/\Gamma$$

where $\Gamma < \mathrm{Isom}(S^n)$ is finite and acts freely.

In dimension 3 we use

$$S^3 \cong \mathrm{SU}(2),$$

via the unit quaternions. Thus one gets spherical quotients

$$\mathrm{SU}(2)/\Gamma$$

when $\Gamma < \mathrm{SU}(2)$ is finite and acts by left multiplication.

The finite subgroups of SU(2) are the binary cyclic, binary dihedral, binary tetrahedral, binary octahedral and binary icosahedral groups. This is the ADE list through the McKay correspondence.

See the constant curvature discussion in Section 5, especially Equation 5.3.1. For the ADE side, see Section 3.

Warning.
The general spherical space-form statement is S^3/Γ, with $\Gamma < \mathrm{SO}(4)$ finite and free. The ADE statement is the clean single-action case $\Gamma < \mathrm{SU}(2)$.

Main theorem.

Theorem 1.2. *Let*

$$S = \mathrm{SU}(2)/\Gamma$$

be a spherical 3*-manifold, with* $\Gamma < \mathrm{SU}(2)$ *finite. Then there exists a closed orientable hyperbolic surface* Σ *and a flat principal* $\mathrm{SU}(2)$*-bundle*

$$\mathrm{SU}(2) \to P \to \Sigma$$

such that S *occurs as a boundary point of the Gromov–Hausdorff closure of*

$$\mathcal{S}(P) = \mathrm{Met}(P)/\,\mathrm{Diff}(P).$$

Equivalently, there are smooth metrics g_i *on* P *such that*

$$(P, d_{g_i}) \to (S, d_{g_0})$$

in Gromov–Hausdorff distance.

Here g_0 is the spherical quotient metric. I am recording the theorem in the form understood from the talk. The exact statement should be checked against the speaker's paper.

The point is that a fixed smooth manifold P can carry metrics which collapse, in the Gromov–Hausdorff sense, to a lower dimensional spherical space form.

Holonomy and monodromy.

Let $\pi : P \to M$ be a principal G-bundle with connection θ. Fix $m \in M$ and $u \in P_m$. If γ is a loop based at m, let $\widetilde{\gamma}$ be its horizontal lift starting at u. Then $\widetilde{\gamma}(1)$ lies in P_m, so there is a unique $g_\gamma \in G$ with

$$\widetilde{\gamma}(1) = u \cdot g_\gamma.$$

The holonomy group is

$$\mathrm{Hol}_u(\theta) = \{g_\gamma\}.$$

The restricted holonomy group $\mathrm{Hol}^0_u(\theta)$ is obtained from null-homotopic loops. For a connected base it is the identity component of the holonomy group.

The monodromy map is

$$m_\theta : \pi_1(M, m) \to \mathrm{Hol}_u(\theta)/\,\mathrm{Hol}^0_u(\theta), \qquad [\gamma] \mapsto [g_\gamma].$$

So the quotient by restricted holonomy records the discrete part of holonomy seen by the fundamental group.

Ambrose–Singer theorem.

Let Ω_θ be the curvature form of θ. The Ambrose–Singer theorem says that

$$\mathfrak{hol}_u(\theta) = \mathrm{Lie}\,\mathrm{Hol}^0_u(\theta)$$

is generated by curvature. More exactly, take a point $v \in P$ reachable from u by a horizontal curve, and take horizontal vectors $X, Y \in T_vP$. Then the curvature value

$$\Omega_\theta(v)(X, Y) \in \mathfrak{g}$$

transported back to u contributes to $\mathfrak{hol}_u(\theta)$. The span of all such values is the holonomy Lie algebra.

For a vector bundle connection ∇, this is the formula

$$\mathfrak{hol}_x(\nabla) = \mathrm{span}\{P_\gamma^{-1} R_y(X, Y) P_\gamma\}.$$

Here $\gamma : x \to y$, P_γ is parallel transport, and $X, Y \in T_yM$.

Thus curvature gives the infinitesimal part of holonomy, while monodromy records the residual discrete part.

Riemann–Hilbert correspondence.

For the smooth discussion here, the relevant case is the flat case:

$$\Omega_\theta = 0.$$

Then parallel transport depends only on the homotopy class of the path. Choosing $m \in M$ and $u \in P_m$ gives the holonomy representation

$$\rho_\theta : \pi_1(M, m) \to G, \qquad [\gamma] \mapsto g_\gamma.$$

The Riemann–Hilbert correspondence identifies flat principal G-bundles with representations of the fundamental group:

$$\{\text{flat principal } G\text{-bundles on } M\} \,/ \cong \longleftrightarrow \mathrm{Hom}(\pi_1(M, m), G)/G.$$

The group G acts on representations by conjugation.

Thus two flat bundles with connection (P, θ) and (Q, ω) are equivalent if there exists a principal G-bundle isomorphism

$$\varphi : P \to Q$$

such that

$$\theta = \varphi^* \omega.$$

On the representation side this says that the holonomy representations ρ and ρ' are conjugate:

$$\rho' = g\rho g^{-1}.$$

If a point in the fiber over m is fixed, the conjugacy ambiguity disappears.

Conversely, a representation

$$\rho : \pi_1(M, m) \to G$$

defines a flat principal bundle by

$$\widetilde{M} \times_\rho G.$$

This is the flat case of the Riemann–Hilbert dictionary.

Collapse sequence.

The paper uses a Kaluza–Klein type family of metrics

$$g_{P,n} = \frac{1}{n} \pi^*(h_\Sigma) + (\theta, \theta)_{\mathrm{SU}(2)}$$

on the total space P. As $n \to \infty$, the horizontal directions shrink and one expects the Gromov–Hausdorff limit to be the spherical quotient determined by the holonomy.

Gromov–Hausdorff distance.

Let (X, d_X) and (Y, d_Y) be compact metric spaces. A correspondence is a subset

$$R \subset X \times Y$$

whose two coordinate projections are surjective.

A correspondence is like a generalized graph. The graph of a map $f : X \to Y$ is a correspondence only when f is surjective. In general, R behaves like a multivalued graph.

The distortion of R is

$$\mathrm{dis}(R) = \sup\{|d_X(x, x') - d_Y(y, y')| : (x, y), (x', y') \in R\}.$$

The Gromov–Hausdorff distance is

$$d_{GH}(X, Y) = \frac{1}{2} \inf_R \mathrm{dis}(R),$$

where R runs through all correspondences between X and Y.

A small-distortion correspondence is an almost distance-preserving matching.

Isometry classes.

The Gromov–Hausdorff distance is invariant under isometry. If

$$X \cong X', \qquad Y \cong Y',$$

then

$$d_{GH}(X, Y) = d_{GH}(X', Y').$$

Hence d_{GH} descends to isometry classes:

$$d_{GH}([X], [Y]) = d_{GH}(X, Y).$$

Moreover, for compact metric spaces,

$$d_{GH}([X], [Y]) = 0 \quad \iff \quad X \cong Y.$$

So compact metric spaces modulo isometry form a genuine metric space.

Superspace.

Let P be a smooth manifold and denote by $\mathrm{Met}(P)$ the space of smooth Riemannian metrics on P. The diffeomorphism group acts by pullback. The superspace of P is

$$\mathcal{S}(P) = \mathrm{Met}(P)/\,\mathrm{Diff}(P).$$

Thus two metrics define the same point when they differ by a diffeomorphism.

Assume now that P is compact. Each metric g defines the compact metric space (P, d_g), so there is a map

$$\mathcal{S}(P) \to \mathcal{GH}, \qquad [g] \mapsto [(P, d_g)].$$

It is well-defined because if $g = \phi^* g'$, then

$$\phi : (P, d_g) \to (P, d_{g'})$$

is an isometry.

Conversely, by Myers–Steenrod, an isometry between smooth Riemannian manifolds is smooth. Thus superspace embeds in the Gromov–Hausdorff moduli space of compact metric spaces.

Riemannian submersion lemma.

A technical lemma used in the proof gives a Riemannian submersion

$$f : (P, g) \to (\mathrm{SU}(2)/\Gamma, g_0).$$

Here g_0 is the quotient of the round metric on $\mathrm{SU}(2) \cong S^3$.

Recall that $f : (P, g) \to (N, h)$ is a Riemannian submersion if f is a submersion and

$$df_p : (\mathrm{Ker}\, df_p)^\perp \to T_{f(p)}N$$

is an isometry for every $p \in P$. Thus

$$T_pP = \mathrm{Ker}\, df_p \oplus (\mathrm{Ker}\, df_p)^\perp$$

is the vertical-horizontal splitting.

For horizontal vectors X, Y one has

$$g(X, Y) = g_0(df(X), df(Y)).$$

This lemma is the bridge between the principal-bundle construction and the Gromov–Hausdorff collapse: the ADE quotient is not only a holonomy target, but also the metric quotient sitting below (P, g).

Examples.

The Poincare homology sphere is

$$\Sigma(2,3,5) \cong S^3/I^*$$

where $I^* < \mathrm{SU}(2)$ is the binary icosahedral group of order 120. It double covers the usual icosahedral rotation group $A_5 < \mathrm{SO}(3)$. Thus the quotient is by the binary icosahedral group, not by the ordinary icosahedral group.

It is a homology sphere:

$$H_*(\Sigma(2,3,5), \mathbb{Z}) \cong H_*(S^3, \mathbb{Z}).$$

but it is not simply connected. Its fundamental group is I^*.

The Bolza surface is the smooth completion of

$$y^2 = x^5 - x.$$

It is a two-fold cover of $\mathbb{P}^1$ branched over

$$0,\ \infty,\ 1,\ -1,\ i,\ -i.$$

By Riemann–Hurwitz, a double cover of $\mathbb{P}^1$ branched over six points has genus

$$g = \frac{6-2}{2} = 2.$$

The Bolza surface is the unique *global* maximum of the systole function on the moduli space $\mathcal{M}_2$, after giving every Riemann surface its uniformizing hyperbolic metric. In other words, among closed hyperbolic surfaces of genus 2, it maximizes the length of the shortest closed geodesic.

Points to check.

1. Is the theorem stated for all spherical 3-manifolds, or only for the single-action quotients $\mathrm{SU}(2)/\Gamma$?
2. In the construction, which representation $\rho : \pi_1(\Sigma) \to \Gamma$ is chosen?
3. What is the exact collapse sequence g_i on P?
4. Where is the Riemannian submersion lemma used in the proof?

2. Updates on flows of geometric structures

Speaker.

Henrique Sa Earp (Campinas, Brazil)

Title.

Updates on flows of geometric structures.

Abstract.

Aiming at a public with interests among Riemannian and complex geometry, Lie groups and parabolic PDEs, I will recall some results towards a general analytical theory for flows of H-structures, obtained with Fadel, Loubeau and Moreno. Then, as a concrete example, I will present some recent progress on $SU(2)$-flows on 4-manifolds, initiating a classification programme of "parabolic" flows, based on a representation-theoretic method originally due to Bryant in the context of G_2 geometry, obtained with Fowdar.

Baby summary.

The talk asks which geometric structures can be moved by a well-posed PDE. First one sets up the general H-structure framework. Then one studies $SU(2)$ examples and uses representation theory to list the allowed flow terms.

H-structures.

Definition 2.1. Let M^n be a smooth manifold and let $H \subset \mathrm{GL}(n,\mathbb{R})$. An *$H$-structure* on M is a reduction of the frame bundle $\mathrm{Fr}(M)$ from $\mathrm{GL}(n,\mathbb{R})$ to H. Equivalently, it is a section of $\mathrm{Fr}(M)/H$. If ξ_0 is a reference tensor, then $H = \mathrm{Stab}(\xi_0)$.

$$\begin{tikzcd}
P := \mathrm{Fr}(M) \arrow[dd] \arrow[dr, "H"] & \\
& N := P/H \arrow[dl, "\pi"'] \\
M \arrow[ur, dashed, bend right, "\sigma"'] &
\end{tikzcd}$$

and there is an associated tensor bundle picture

$$\mathcal{F} \subset \bigoplus \mathcal{T}^{p,q} \to M, \qquad \pi^*\mathcal{F} \to N.$$

Here TP splits into vertical and horizontal parts. Write

$$\begin{tikzcd}
TP \arrow[dd] \arrow[dr] & \\
& TN := \nu \oplus \mathcal{H} \arrow[dl, "\pi"'] \\
M \arrow[ur, dashed, bend right, "\sigma"'] &
\end{tikzcd}$$

to record the tangent-bundle picture.

In prose: an H-structure is a way to equip M with extra geometric data that is modeled on a fixed tensor or algebraic structure with symmetry group H. The section $\sigma : M \to P/H$ picks, at each point, an adapted frame modulo H. Equivalently, one can encode the same information by a tensor field ξ on M whose stabilizer is exactly H; the bundle P/H is the space of all such local choices.

Example 2.2. For an H-structure σ with torsion tensor T, one may consider the energy

$$E(\sigma) = \int_M |T|^2.$$

The harmonic flow is the negative gradient flow of this energy. Its purpose is to deform σ toward a more special or torsion-free structure.

Then it is natural to consider the Ricci flow of H-structures.

Definition 2.3. Let $H \subset \mathrm{SO}(n)$. A *harmonic H-structure* is a critical point of

$$E(\xi) = \frac{1}{2}\int_M |T|^2.$$

Here ξ is the corresponding H-structure and T is its torsion tensor.

Theorem 2.4 (ε-regularity)**.** *Let (M,g) be closed and let $\xi(t)$ be a harmonic H-flow on $M\times[0,\tau)$. Fix $0<r_M<\operatorname{inj}(M,g)$ and $E_0\in(0,\infty)$. If the local energy of $\xi(t)$ on a geodesic ball $B_{r_M}(y)$ is small enough, then torsion stays controlled on a smaller ball for a short time.*

Theorem 2.5 (Energy gap)**.** *Let (M,g) be a closed Riemannian n-manifold admitting a compatible H-structure. Then there is a constant $\varepsilon(M,g,H)>0$ such that if a compatible harmonic H-structure ξ satisfies*

$$E(\xi)=\|\nabla\xi\|^2_{L^2(M)}<\varepsilon(M,g,H),$$

then ξ is torsion-free:

$$\nabla\xi=0.$$

The upshot is that the flow is being used to drive the structure into the small-energy regime where the ε-regularity theorem forces torsion to disappear. So the long-time picture is meant to end at a torsion-free limit.

One can also reverse the procedure. Let (M,g) be such that there is no torsion-free structure in the homotopy class. Then the question becomes whether the flow still has a useful limit, or whether it must break down.

I will recall results toward a general theory for flows of H-structures, obtained with Fadel, Loubeau and Moreno. Then I will present recent results on SU(2)-flows on 4-manifolds, starting a classification programme for parabolic flows. The method is representation-theoretic and goes back to Bryant in G2 geometry; the recent work is joint with Fowdar.

The key concept to construct these flows is the Fadel–Loubeau–Moreno– SE 24 theorem, and also the Karigiannis 08 theorem: the "diamond action". In this language the evolution equation is

$$\partial_t\eta=A\diamond\eta$$

with

$$A\equiv A(t)=S(t)+C(t),\qquad S(t)\in S^2,\qquad C(t)\in\Lambda^2.$$

The freedom in choosing A is the freedom to choose a differential invariant of the H-structure. Bryant's 2003 point is that, for $H\subset\mathrm{O}(n)$, the space $V_k(\mathfrak{h})$ of kth order invariants is characterized by

$$V_k(\mathfrak{h})\oplus(\Lambda^1\otimes S^{k+1})=(S^2\oplus\mathfrak{h}^\perp)\otimes S^k,$$

where $S^k=S^k(\Lambda^1)$ and $\mathfrak{h}^\perp=\Lambda^2/\mathfrak{h}$. The first space $V_1(\mathfrak{h})=\Lambda^1\otimes\mathfrak{h}^\perp$ is the intrinsic torsion. The next piece $V_2(\mathfrak{h})$ is spanned by ∇T and Rm, with overlap coming from the Bianchi-type identity

$$\nabla T\diamond\eta=\nabla^2\eta+\text{l.o.t.}$$

so the lower-order terms are not independent.

In the simplest case $H=\mathrm{O}(n)$, one has

$$V_1(\mathfrak{so}(n))=0.$$

Also,

$$V_2(\mathfrak{so}(n))=\mathbf{R}\oplus S^2_0(\mathbf{R}^n)\oplus W,$$

and

$$\operatorname{End}(\mathbf{R}^n)_{\mathfrak{so}(n)}=\mathbf{R}\oplus S^2_0(\mathbf{R}^n).$$

SU(2) flows.

Any $SU(2)$-flow can be written as

$$\partial_t \omega = A \diamond \omega.$$

The point of the theory is that the possible choices of A form a finite dimensional space of differential invariants. So one can classify the flows. For $SU(2)$ this leads to a list of quadratic torsion functionals. There are 15 of them, built from terms of the form

$$\int_M g(a_i, a_j)\,\mathrm{vol} \qquad \text{and} \qquad \int_M g(a_i, J_j a_k)\,\mathrm{vol}.$$

Among them are the Dirichlet energy

$$\int_M |T|^2\,\mathrm{vol}$$

and the complex-geometric energy

$$\int_M |N_J|^2\,\mathrm{vol}.$$

So the classification problem is really a representation-theoretic bookkeeping problem: write down the allowed invariant tensors, then choose a linear combination that gives a parabolic flow.

In other words, the flow is not arbitrary. It is built from the finite list of quadratic expressions that the representation theory allows. The hard part is to choose the coefficients so that the evolution equation is still geometric and parabolic. That is why the talk keeps returning to invariant tensors, torsion, and curvature.

Favorite flow.

Consider the two-parameter rich flow

$$\partial_t \omega = \Big(-2\,\mathrm{Ric}(g) - \lambda_1 \mathcal{L}_{\sum_{i=1}^{2} J_i a_i} g + \lambda_2 \sum_{i=1}^{3} \mathrm{div}(a_i)\Big) \diamond \omega.$$

It has short-time existence and uniqueness when

$$-4\sqrt{5} < \lambda_2 + \lambda_1/2 < 0.$$

This is the favorite flow.

Questions and outlook.

Can one add terms involving W^+ to the flow and still keep short-time existence? Are there good choices of such curvature terms? The Lie-derivative term can be more generally taken as

$$\sum_{i,j} \lambda_{ij} \mathcal{L}_{J_i a_j} g.$$

What lower-order terms should one choose? Are there flows preserving torsion classes, for instance Laplacian-type flows? What about long-time existence, singularity formation, and solitons? More broadly, can the community classify the good flows for other groups H?

Audience questions.

Is one tacitly assuming the flow is quasi-linear? Is that crucial?

The answer is that one wants to control low-order derivatives of the torsion. Once those are under control, the rest follows. So the point is to control the derivatives of the norm of the torsion.

Would it make sense to study flows that are not quasi-linear? For example, one could take the Riemann curvature tensor, contract it in many ways, and ask what happens.

Why would one do that? The lower-order terms depend on which PDE one chooses. The hope is to use this as technology for solving geometric PDEs. That is the perspective here.

Baby summary.

The talk is about moving geometric structures by PDEs. The structure has torsion, and the flow tries to reduce it. Representation theory tells you which terms you are allowed to use. In the best cases, the flow is well-posed and drives the structure toward something more rigid. The $SU(2)$ case is especially rich, because there are many explicit quadratic terms to choose from.

3. Atom applications and MMP

Speaker.

Ludmil Katzarkov

Title.

Atom applications and MMP.

Abstract.

In this talk we introduce the theory of Atoms. Applications and connection with MMP will be discussed.

Birational invariants.

(1) LG-modules.
(2) Orlov spectra.
(3) Differential equations to the monads.
(4) Road map to the twistor examples.

Orlov spectrum.

Take a generator g of $D^b(X)$ and generate by taking sums, summands, triangles and shifts. Let $t(g)$ be the minimum number of triangles needed to get to any object in $D^b(X)$.

The Orlov spectrum is the list of generation times of $D^b(X)$. In the shorthand from the talk,

$$D^b(X) = \langle t_1(g), \dots, t_n(g) \rangle.$$

(1) $D^b(\mathbb{P}^1)$ has Orlov spectrum $(1, 2)$.
(2) $D^b(E)$ has Orlov spectrum $(1, 2, 2, 4)$.

Conjecture: this number is the dimension of the variety. This was confirmed in dimension 1.

There are gaps in the spectrum. A conjecture by Ballard, Favero and Katzarkov says that if the gap is greater than $\dim X - 2$, then X is not rational.

Differential equations.

Now we take a shortcut and use differential equations. Keep in mind the Landau–Ginzburg model, but something else will happen. Take a Fano manifold X and consider it as an algebraic variety, but also as a symplectic manifold.

Main objective.

If X is a variety over $\mathbb{C}$, then birational geometry asks when

$$\mathbb{C}(X) \cong \mathbb{C}(x_1, \dots, x_n), \qquad n = \dim X.$$

This is the rationality problem in its simplest form.

Mirror symmetry.

For smooth X, homological mirror symmetry says

$$D^b(X) \leftrightarrow \mathrm{FS}(Y, W),$$

where Y is an open symplectic manifold and W is a function. The category on the right is built from Lagrangian submanifolds, and morphisms are given by Floer homology.

Example 3.1. For $X = \mathbb{P}^2$ one has

$$D^b(\mathbb{P}^2) = \langle \mathcal{O}, \mathcal{O}(1), \mathcal{O}(2) \rangle.$$

The mirror is

$$Y = (\mathbb{C}^*)^2, \qquad W = x + y + \frac{1}{xy}.$$

The observation made by them is that the singularity theory on the symplectic side recovers, my guess, the birational data on the algebraic side.

Blow-up.

Now turn to birational geometry. What happens when you blow up a point of $\mathbb{P}^2$?

$$D^b(\mathrm{Bl}_p \mathbb{P}^2) = \langle \mathcal{O}, \mathcal{O}(1), \mathcal{O}(2), E \rangle,$$

and on the symplectic side we get I_8 (singularity).

Cubic threefold.

Next example: $X_3 \subset \mathbb{P}^4$. The classical approach is by studying

$$H^{2,1}(X_3)/H_3(X_3, \mathbb{Z}),$$

that is, the intermediate Jacobian. This gives a moduli space called the Theta divisor

$$\Theta.$$

Example 3.2. From the HMS point of view, we have a symplectic manifold and a potential

$$W = \frac{(x+y+z)^3}{xyz} + z.$$

Some singular K3 surfaces appear. There is also a complicated thing at infinity.

Here one writes

$$D^b(X_3) = \langle \mathcal{A}, E_1, E_2 \rangle.$$

Theorem 3.3 ((-, Prizholkovsky, 2005))**.** *Let X be an equidimensional Fano with* $\operatorname{Pic} X = \mathbb{Z}$*, and suppose X is not $\mathbb{P}^3$. Consider the monodromy action on near-by fibers of the symplectic side. The monodromy is a 3×3 matrix. If μ is strictly unipotent, then X is rational.*

Differential equations.

Now we shall take a shortcut: we use differential equations. Keep in mind the Landau–Ginzburg model, but something else will happen. Take a Fano manifold X and consider it as an algebraic variety, but also as a symplectic manifold.

We have de Rham cohomology, and also a noncommutative Hodge package

$$H_{nc} = \bigoplus_{\lambda_i} H_{\lambda_i}.$$

The formula for E_{λ_i} in the notes is too unclear to reconstruct exactly, so I do not guess it here.

Theorem 3.4 (Kontsevich, Pantev, Yu)**.** *The object*

$$\bigoplus_{\lambda_i} H_{\lambda_i}$$

is a birational invariant.

The quantum differential equation is

$$\left(\frac{\partial}{\partial u} - \frac{K}{u^2} + \frac{G}{u}\right)\psi(u) = 0.$$

where

$$\psi = u^{\delta} g(1, \dots).$$

Theorem 3.5. *Let $X \subset \mathbb{P}^N$ be a Fano manifold of dimension ≥ 3. Then*

$$\delta_{\max} = \dim X - 1 \cdots$$

If MMP works, and we can ... then you might be able to conclude that this 7-dimensional cubic is Picard as well.

G-equivariant Hodge atoms.

Part 1.

Review of coarse Hodge atoms.

Let $X/\mathbb{C}$ be a smooth projective variety of dimension at least 2. For instance, one can think of a compact Kähler manifold. Let $G < \mathrm{Aut}(X)$ be a finite subgroup. We call X a G-variety if G acts generically freely on X. The pair (X, G) is non-G-linearizable if there is no linear G-action on $\mathbb{P}^n$ and no G-equivariant birational map between X and $\mathbb{P}^n$.

The equivariant version of the coarse Hodge atoms keeps track of the G-action on the Hodge package. The natural coarse version is obtained by decomposing the atomic package into G-isotypical pieces.

Application.

These invariants give obstructions to G-linearizability. In particular, they produce many examples of non-G-linearizable G-actions on projective varieties.

4. Recent progresses in the theory of atoms

Speaker.

Leonardo Cavenaghi

Title.

Recent progresses in the theory of atoms.

Abstract.

In this talk we report on new results related to the theory of atoms obtained in collaboration with L. Katzarkov and M. Kontsevich. They will appear in the text titled "9.5 lectures in the theory of atoms", in the final stage of preparation.

Upshot.

The point is to package the Hodge-theoretic and quantum information into atoms. These atoms carry both the representation-theoretic and birational content.

Relation with Mumford–Tate groups.

This fits the notes in 6. There the Mumford–Tate group controls Hodge symmetries. The same Hodge package is also developed in the complex-geometry notes. Here the Hodge atoms are a more refined bookkeeping device: they split the Hodge package into pieces on which $\mathrm{Hod}(\daleth)$ and its equivariant variants act.

G-equivariant atoms and quantum cohomology.

Theorem.

For any multi-degree hypersurface

$$X := X(1,1,1,1) \subset (\mathbb{P}^1)^4,$$

for instance the blow-up of $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$, the invariant $\mathbb{Z}/2$-action by swapping the first and second, and the third and fourth factors, is not $\mathbb{Z}/2$-linearizable. The same holds for

$$X \times \mathbb{P}^1,$$

where $\mathbb{Z}/2$ acts trivially on the $\mathbb{P}^1$ factor.

Coarse atoms.

For this we introduce a coarse version of the theory of G-equivariant atoms.

Notation.

Take your projective complex variety X and its rational cohomology

$$V := H^*(X, \mathbb{Q}),$$

seen as a supervector space.

- Coordinates:

  $$\mathbf{t} = \{t_a\}$$

  formal dual super coordinates.
- Novikov ring: to track effective curve classes $\beta \in \overline{\mathrm{NE}}(X, \mathbb{Z})$. The small Novikov ring is

  $$\mathcal{N}_X = \left\{ \sum c_\beta q^\beta : \beta \in \overline{\mathrm{NS}}(X, \mathbb{Z}) \right\}$$

  And then

  $$\mathrm{Nov}_X = \mathcal{N}_X[[t_a, a\text{is even}]] \hat{\otimes} \Lambda(ta, a \text{ is odd})$$

We “enrich” the ring with those power series.

- Coefficient ring: formal power series.
- Take a homogeneous basis of V:

$$\{\Delta_0, \Delta_1, \dots, \Delta_\rho, \Delta_{\rho+1}, \dots, \Delta_r\}$$

Think that $t_a \leftrightsquigarrow \Delta_a$ are (super) coordinates.

Question: what is a generating function to the GW invarirants of X? A $\phi_X \in \mathrm{Nov}_X$ with properties! (WDVV equations)

The Gromov–Witten potential allows us to define a new product in this ring that recovers the classical Poincaré cup product.

- Classical part: intersection theory.
- Quantum correction: curve counting, enumerative geometry.

Big quantum cohomology.

- Space:

$$V_{\mathrm{big}} = V \otimes_{\mathbb{Q}} \mathcal{N}_X.$$

- Product: third-order differential operator.
- Deformation: full deformation in $\mathbf{t}$.

Small quantum cohomology.

- Space:

$$V \otimes \mathcal{N}_X.$$

- Product: [missing]
- Deformation: [missing]

Example 4.1. For $X = \mathbb{P}^2$, the cohomology has three generators:

$$\{t_0, t_1, t_2\} = \{\Delta_0, \Delta_1, \Delta_2\} = \{1, H, P\},$$

Now since $H_2(\mathbb{P}^2, \mathbb{Z}) = \mathbb{Z}[\ell]$, we think $q^\beta \leftrightsquigarrow q^d$. The Gromov–Witten potential is

$$\phi_X(\mathbf{t}, q) = \underbrace{\frac{1}{2}t_0^2 t_1 + \frac{1}{2}t_0 t_1^2}_{\text{classical part}} + \underbrace{\sum_{d=1}^{+\infty} N_d (t_2^{3d-1})/(3d-1)! q^3 e^{dt_1}}_{\text{quantum part}}$$

where N_d is the number of rational curves of degree d passing through $3d-1$ points in general position in $\mathbb{P}^1$. We can write the term $q^3 e^{dt_1}$ as

$$q^d e^{dt_1} = (qe^{t_1})^d.$$

As a first remark note that, a priori, $\phi_{\mathbb{P}^2} \in \mathrm{Nov}_X$, but because we have that element $q^d e^{dt_1}$, …, we can actually see that

$$\phi_{\mathbb{P}^2} \in \mathbb{Q}[[qe^{t_1}, t_2]].$$

A second remark says how to compute the quantum part:

$$N_d = \left\langle \underbrace{p, \dots, p}_{3d-1} \right\rangle_{0,3d-1,\beta=[Z?]}$$

Kontsevich noted (a while ago) that WDW $\iff$ recursion formula for computing N_d.

We have chosen

$$\Delta_1, \dots, \Delta_\rho$$

separately. Assume that this is a basis of $\mathrm{NS}(X) \subset H^2(X, \mathbb{Q})$. Goal: quantum multiplication, very geometrically, builds products on the tangent bundle of some space.

For $\mathbb{P}^1$, the GW potential is an entire function.

More generally, GW potential has a positive radius of convergence [check this...].

Questions.

1. Analytic germ.
2. Tangent space product.
3. Logarithmic tangent bundle.

Now consider

$$q^\beta = \prod_{i=1}^{\ell} q_i^{\int \Delta_i} \beta$$

and

$$\mathcal{N}_X[[(t_a)_{a \notin \{1, \dots, \ell\}}]] \otimes_{\mathbb{Q}} \Lambda(t_a; a \text{ is odd}) = \mathrm{Nov}_X^{\mathrm{red}}.$$

The idea is to take a space such that its ring of functions is precisely $\mathrm{Nov}_X^{\mathrm{red}}$, we call it $\mathrm{Spf}(\mathrm{Nov}_X^{\mathrm{red}}$, and do

$$\Delta_a \leftrightsquigarrow \frac{\partial}{\partial t_a} \; \& \; \Delta_a \leftrightsquigarrow q_s \frac{\partial}{\partial q_i}$$

$$a \notin \{1, \dots, \ell\}, q_i \frac{\partial}{\partial q_i} q^\beta) = \left(\int_\beta \Delta_i \right) q^\beta.$$

Passing from the formal picture to the rigid picture, one can show how to actually have a positive radius of convergence for $\phi_X \leftrightsquigarrow$ replace with $\mathrm{Spf}\mathrm{Nov}_X^{\mathrm{red}}$.

Now we are ready to define the quantum multiplication. It's given by derivatives of third order:

$$\Delta_a \star \Delta_b := \sum_{c,d} \frac{\partial^3}{\partial t_a \partial t_b \partial t_c} \eta^{dd^c} \Delta_c.$$

Big quantum product relations.

General logarithmic tangent bundle.

Idea: to have a germ of a manifold such that the tangent space is the cohomology vector space and the quantum product is an associative product in this vector space.

For $X/\mathbb{C}$ smooth projective, let

$$V := H^*(X, \mathbb{Q}).$$

Let Φ_X be the Gromov–Witten potential. Then the third derivatives $\partial^3 \Phi_X$ yield a unital, commutative, associative product on the tangent bundle of a germ

$$M \subset H^*(X, \mathbb{C}).$$

This is the Frobenius-manifold picture behind quantum cohomology.

The problem is that convergence is most of the time hypothetical. The strategy is to introduce an auxiliary non-Archimedean field $\daleth = \overline{\mathbb{Q}}((T^Q))$ (which is a Puiseaux seris ring) of characteristic 0 to ensure convergence. The idea is to replace the germ M by a manifold B_X where we do have convergence for the Gromov–Witten invariants. More precisely, for $b \in B_X$,

$$T_b B_X \simeq H^*(X, \daleth, \star_b).$$

That is: there exists a concrete supermanifold with analytic topology B_X, for which we claim: $T_b B_X$ has the structure of a commutative associative algebra (WDVV).

Definition 4.2. The manifold B_X is called an *A-model manifold.*

Now we put an Euler vector field on B_X. If $\star$ is a tensor field on B_X which is actually a Higgs field, then Eu scales the product. That is,

$$\mathcal{L}_{\mathrm{Eu}}(\star) = \star.$$

Equivalently, we have an automorphism

$$k_b := \mathrm{End}_{\mathrm{Eu}}(\star_b) \in \mathrm{End}\big(H^*(X, \daleth)\big).$$

That is the starting point of atom theory, from our perspective. There are other starting points.

Disjoint unions.

Thom–Sebastiani principle. If a variety is a disjoint union

$$X \sqcup Y,$$

its Gromov–Witten potential uncouples as a sum of the factors:

$$\Phi_{X \sqcup Y} = \Phi_X \oplus \Phi_Y.$$

This is important because we need to study blow-ups, the main operations in birational geometry.

Theorem 4.3 (Iritani, 2003)**.** *Let $\widetilde{X}$ be the blow-up of X along a smooth subvariety $Z \subset X$ of codimension $r \geq 2$. There exists a canonical analytic identification between open domains*

$$\psi : B_{\widetilde{X}} \supset U \xrightarrow{\sim} V \subset B_X \times B_Z \times \cdots \times B_Z.$$

Moreover, one can choose the open domains so that the Euler vector field and the product are preserved:

$$\psi^*(*) = *, \qquad \psi^*(\mathrm{Eu}) = \mathrm{Eu}\,.$$

Hodge action and fixed locus.

$\mathrm{Hod}(\daleth)$ acts on B_X preserving the quantum product $*$ and the Euler field Eu. On coordinates:

- U_1 (Nëron–Severi): trivial action, corresponding to algebraic classes.
- U_2, U_{odd}: scaling by Hodge weights.

Definition 4.4. The *Hodge fixed locus*

$$B_X^{\mathrm{Hod}} \subset B_X$$

is the subvariety where $\mathrm{Hod}(\daleth)$ acts trivially. Geometrically, it is the locus where all non-(p,p) coordinates vanish.

$$\mathcal{HS} \simeq \operatorname{Rep}\operatorname{Hod}$$

for $\mathbb{Q}$-polarizable pure Hodge structures.

We can decompose the tangent space to B_X using generalized eigenspaces:

$$T_b B_X = \bigoplus_{\lambda \in \operatorname{Spec}(k_b)} V_\lambda .$$

The catch is that the group is pro-reductive. This means we can write, in fact,

$$T_b B_X = \bigoplus_{\lambda \in \operatorname{Spec}(k_b)} \bigoplus_{\gamma} V_\lambda \cap V_\gamma .$$

The summands

$$\bigoplus_{\gamma} V_\lambda \cap V_\gamma$$

are called *coarse Hodge atoms*.

Each atom α corresponds to a $\overline{\mathbb{Q}}$-linear representation

$$E^\alpha$$

of $\operatorname{Hod}_{\overline{\mathbb{Q}}}$, independent of the point b.

Numerical invariants.

The Hochschild grading $(p-q)$ on $E^\alpha \otimes_{\overline{\mathbb{Q}}} \mathbb{C}$ defines:

1. Rank of Hodge classes:
$$\rho_\alpha = \dim\left(V_\lambda \cap H_{p,q}\right).$$
2. Hodge polynomial. It is characterized by the coefficient:
$$P_\lambda(t) = \dim\left(V_\lambda \cap H^{p-q=n}(X)\right).$$

Theorem 4.5 (KKPY Non-rationality criterion)**.** *Let X be an n-dimensional smooth complex projective variety. If there is a coarse Hodge atom in the Hodge atomic composition of X that does not appear in the Hodge atomic composition of any variety of dimension $\leq n-2$, then X is not rational.*

Presence of G-actions.

The action on the variety defines an action on the cohomology. Now one does not look at $\operatorname{Hod}(\daleth)$ alone, but at

$$\operatorname{Hod}(\daleth) \times G,$$

which acts on B_X. Restricting to

$$B_X^{G \times \operatorname{Hod}},$$

a coarse G-atom α is the isomorphism class of a $G \times \operatorname{Hod}(\daleth)$- representation on $\mathcal{B}_{\lambda,b}$.

G-invariant Hodge classes.

$$\rho_\alpha^G := \dim_{\mathbb{C}} \left(E^\alpha \otimes_{\overline{\mathbb{Q}}} \mathbb{C}\right)^{G \times \operatorname{Hod}}.$$

Equivariant constraint.

Since the algebraic unit 1 is always $G \times \operatorname{Hod}$ invariant, we have

$$\rho_\alpha^G \geq 1.$$

Obstructions to G-linearizable actions.

Atomic constraints in low dimensions (≤ 1).

Atoms originating from points (G/H) or curves ($G \times_H C$) satisfy:

- Rank bound:
$$\rho_\alpha^G \in \{1, 2\}.$$
- Rigidity: if ρ_α^G is constant, there is only one invariant Hodge class.

Example 4.6. The hypersurface
$$X(1,1,1,1) \subset (\mathbb{P}^1)^4.$$
Rationality: X is rational, the blow-up of $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ along an elliptic curve E. The G-action is $G = \mathbb{Z}/2$, swapping factors $1 \leftrightarrow 2$ and $3 \leftrightarrow 4$.

The atomic decomposition of X yields three primary coarse G-atoms:

λ	ρ	ρ^G	$P_\alpha(t)$
16	1	1	1
4	4	2	4
0	5	3	$t + 3 + t^{-1}$

Note that
$$\rho^G_{\text{total}} = 1 + 2 + 3 = 6,$$
tracking the G-invariant Hodge classes.

The atom of the elliptic curve must contribute. It contributes with:

- points $(1,1)$ or $(2,1)$.
- curve E: $(2,2)$.

Target atom $\lambda = 0$:
$$(\rho, \rho^G) = (5, 3).$$

If the action was linearizable, its atoms must be constructible from atoms of $\mathbb{P}^3$ and G-invariant subvarieties of dimension ≤ 1.

The contradiction: ...

Theorem 4.7 (Rationality criterion)**.** *If X is a smooth complex projective variety of dimension $n \geq 2$, and the maximal generic spectral dimension on the Hodge locus B_X^{Hod} satisfies*
$$\mu(X) > n - 2,$$
then X is not rational.

Quantum differential equation.

Over the non-Archimedean field, recall that we have a manifold and then we put the quantum differential equation.

There is a bundle $(\mathcal{H}, \nabla)$ with a flat connection ∇ and we have the equation
$$\nabla_{u\frac{\partial}{\partial u}} = \frac{\partial}{\partial u} + \frac{1}{u}\mathrm{Eu} \star + \mathrm{Gr}$$
where Gr is the grading operation. This is the *quantum differential equation.*

We solve it via by an inf
$$\psi(u) = u^{-5} e^{\lambda/y}(\psi + \dots)$$

Definition 4.8.
$$\dim_{\text{Spec}}(\lambda) = -2\min\left\{ s \in \mathbb{Q} : \substack{\exists\psi(u) \text{ solving} \\ \text{QDE associated to } \lambda \text{ w exp. } s} \right\}$$

Theorem 4.9 (inf). *At a very general point*

$$\dim_{\mathrm{Spec}}(0)$$

obstructs rationality. If X is a smooth cubic threefold then $\lambda \in \{0, \pm 1\}$. Solution to QDE on the direction of 0 $s_1 = -1/6$, $s_2 = -5/6$. $\dim_0 = 5/3 > 3 - 2 = 1$ *then X is irrational.*

After a big analysis, we get for the cubic, indeed,

$$\frac{5}{3}.$$

5. Finite simple subgroups of the real Cremona group of rank three

Speaker.

Antoine Pinardin

Title.

Finite simple subgroups of the real Cremona group of rank three.

Abstract.

Very little is known about the classification of finite subgroups of Cremona in dimension three. It is natural to start with the case of simple groups, and this step was achieved by Prokhorov in 2009 over the field of complex numbers. In the work I will present, we show that the only non-cyclic finite simple subgroups of the real Cremona group of rank three are A5 and A6. This is a joint project with I. Cheltsov and Y. Prokhorov.

Upshot. To prove the theorem of classification of non-cyclic finite simple subgroups of $\mathrm{Cr}_3(\mathbb{R})$, i.e. Theorem **??**, which says they are A_5 and A_6, we introduce group actions on the varieties. Note that there is no mention of such actions in the notes of Opega II.

AI-generated proof strategy outline.

The bridge theorem identifies a finite subgroup of the Cremona group with a rational G-variety, modulo equivariant birational change of model. From there, equivariant MMP reduces the problem to a G-Mori fiber space, and in the cases discussed here to a G-Fano model. The big diagram records this passage from the original action to the fiber/base decomposition. Once that reduction is in place, the remaining possibilities are checked against the known Fano-type examples and the Sarkisov-link factorization picture.

Notes.

Take a field of characteristic 0. Define

$$\mathrm{Cr}_n(k) = \mathrm{Bir}(\mathbb{P}^n) = \mathrm{Aut}_k(k(x_1, \dots, x_n)).$$

This is a huge group. A question of Serre is whether any finite subgroup injects in $\mathrm{Cr}_3(\mathbb{C})$.

For $n = 1$, $k = \mathbb{C}$, we have

$$\mathrm{Cr}_1(\mathbb{C}) = \mathrm{PGL}_2(\mathbb{C}),$$

whose finite subgroups were classified by Klein:

$$Z_n,\ D_n,\ A_4,\ S_4,\ A_5.$$

For $n = 2$, $k = \mathbb{C}$:

- Bertini, 1877.
- Bangalo–Beauville, 2000: subgroups isomorphic to $\mathbb{Z}_2$ up to conjugation.
- Belzance, 2006: finite abelian subgroups.
- Dolgachev–Iskovskikh, 2009: “all” (there are conjugation subtleties) finite subgroups of $\mathrm{Cr}_2(\mathbb{C})$.

	A_5	A_6	$\mathrm{PSL}_2(\mathbb{F}_7)$
Conj. classes	3	1	2
Models	$\mathbb{P}^2$, dP_5, $\mathbb{P}^1 \times \mathbb{P}^1$	$\mathbb{P}^2$	$\mathbb{P}^2$, $dP_2 \overset{2:1}{\to} \{x^3y+y^3z+z^3x=0\} \subset \mathbb{P}^2$ i.e. a double cover of Klein’s quartic

For $n = 3$, $k = \mathbb{C}$, this is widely open.

- Progress for abelian groups, 2025.
- (Prokhorov, 2009.) Noncyclic finite simple subgroups:
    - A_5: infinitely many.
    - A_6: finitely many? (not known!)
    - $\mathrm{PSL}_2(7)$: finitely many? (not known!)
    - A_7: one, $\mathbb{P}^3$.
    - $\mathrm{SL}_2(8)$: one, Prokhorov’s threefold.
    - $\mathrm{PSpin}_4(3)$: two, $\mathbb{P}^3$ and Birkhodt’s quartic.

Further notes:

- For Prokhorov’s threefold:
$$\mathrm{LGr}(4, 9) \cap \mathbb{P}^8 \subset \mathbb{P}^{15}.$$
- For Birkhodt’s quartic:
$$\{x_0^4 - x_0 \sum_{i=1}^{3} x_i^3 + \cdots\} \subset \mathbb{P}^4.$$
- For A_5: book: “Cremona groups and the icosahedron”, Cheltsov and Yomdin, 2016.

Purpose of today.

Take $k = \mathbb{R}$.

For $n = 1$, there is no non-cyclic finite group contained in $\mathrm{Cr}_1(\mathbb{R})$.

For $n = 2$, A_5 acts on $\mathbb{P}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, and dP_5. This is due to Yaskovsky.

For $n = 3$:

Theorem 5.1. *[Cheltsov, Prokhorov, Pinardin, 2025] The non-cyclic finite simple subgroups of* $\mathrm{Cr}_3(\mathbb{R})$ *are* A_5 *and* A_6*.*

Method of proof.

Definition 5.2. An *G-variety* is a pair (X, ρ) with X projective and $\rho : G \to \mathrm{Aut}(X)$.

Definition 5.3. An *G-birational map* is $\varphi : (X, \rho) \dashrightarrow (X', \rho')$ such that $\varphi\rho(g)\varphi^{-1} = \rho'(g)$ for all $g \in G$. If such a φ exists, then X and X' are *G-birationally equivalent.*

Theorem 5.4. *Finite subgroups of* $\mathrm{Cr}_n(k)$*, up to conjugacy, correspond to rational G-varieties for finite G, up to equivariant birational maps.*

Definition 5.5. We may assume X is a *G-Mori fiber space*. That is, X is a terminal G-variety, all invariant divisors are $\mathbb{Q}$-Cartier, and:

- There exists $\pi : X \to B$ surjective and G-equivariant with $\dim B < \dim X$.
- K_X is π-ample.
- $\operatorname{rk}\operatorname{Pic}^G(X)$
$$= 1 + \operatorname{rk}\operatorname{Pic}^{G_B}(B).$$
- The fibers are connected.

In particular, if $B = \mathrm{pt}$, then X is a minimal G-Fano variety.

Remark 5.6. Since

$$\begin{tikzcd}
1 \arrow[r] & K \arrow[r] \arrow[d] & G \arrow[r] & G_B \arrow[d] & 1 \\
 & \text{fibers of } \pi & & B &
\end{tikzcd}$$

we may assume that X is Fano.

For $\mathcal{L}_0$ and simple groups, we may assume X is Fano.

Remark 5.7. $\mathrm{Cr}_3(\mathbb{R}) \subset \mathrm{Cr}_3(\mathbb{C})$.

A_5 acts on:

- $\mathbb{P}^3$.
- The quadric
$$x_0^2 + x_1^2 + x_2^2 + x_3^2 = x_4^2 \subset \mathbb{P}^4.$$
- $\mathbb{P}^1 \times Q$, where
$$Q = \{x_0^2 + x_1^2 + x_2^2 = x_3^2\}.$$
- The Segre cubic.

A_6 acts on:

- The Segre cubic.
-
$$\begin{cases} \sum x_i^3 = 0, \\ \sum x_i = 0 \end{cases} \subset \mathbb{P}^5.$$

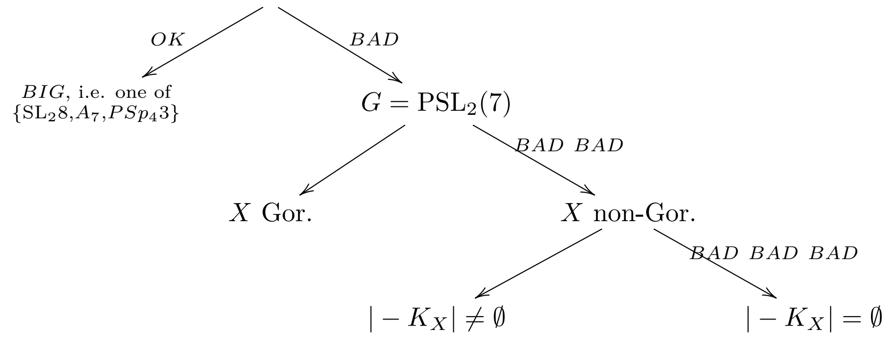
The group S acts on X by permuting the coordinates.

$$\begin{tikzcd}
 & \widetilde{X} \arrow[dl, "5 \text{ pts}"'] \arrow[dr, "\text{contract } 10 \text{ lines}"] & \\
\mathbb{P}^3 & & X
\end{tikzcd}$$

The first part of this project was obtaining the following result:

Proposition 5.8. *If X is a real rational threefold and*
$$G \in \{\mathrm{PSL}_2(7), \mathrm{SL}_2(8), A_7, \mathrm{PSpin}_4(3)\},$$
then there is no faithful action of G on X.

In the following diagram “BAD” means how difficult it was to obtain the result.



Remark 5.9 (Terpenan-Simmerman). Let

$$X_{\mathbb{C}} = \{x_0y_0 + x_1y_1 + x_1y_2 = 0\} \subset \mathbb{P}^2_{\mathbb{C}} \times \mathbb{P}^2_{\mathbb{C}}.$$

$X_{\mathbb{C}}$ has a real form $X_{\mathbb{R}}$ such that $\mathrm{PSL}_2(7) \subset \mathrm{Aut}(X_{\mathbb{R}})$.

Moreover, $X_{\mathbb{C}}$ is rational.

This means that $X_{\mathbb{C}}$ has a real form $X_{\mathbb{R}}$ with an action of a group — $\mathrm{PSL}_2(7)$. The variety $X_{\mathbb{C}}$ is rational, but $X_{\mathbb{R}}$ is not.

In this extra time we shall explain more details about the uppermost leftmost item in the big diagram above. This was deduced from the work of Prokhov (and something else).

Let I be BIG, i.e. $\in \{\mathrm{SL}_2(8), A_7, PSp_4(3)\}$. Now, A_7 acts on $X_{\mathbb{C}} = \mathbb{P}^3$, $\mathrm{SL}_2(8)$ acts on Prokhov's threefold, $LG(4,9) \cap \mathbb{P}^8 \subset \mathbb{P}^{15}$, and finally $PSp_4(3)$ acts on $\mathbb{P}^3$ and Birkhodt's threefold

$$\left\{x_0^4 - x_0\sum_1^3 x_i^3 + 3\prod_0^3 x_i\right\}.$$

Over $\mathbb{R}$, we want to list Fano threefolds X such that $G \subset \mathrm{Aut}(X)$.

6. On metric connections with parallel and closed skew torsion

Speaker.

Paul Schwahn

Title.

On metric connections with parallel and closed skew torsion.

Abstract.

This talk is divided in three parts. First, I review structural properties and examples of geometries with parallel skew torsion. Second, I take a closer look at those geometries where the torsion is in addition closed; this equips every tangent space with the structure of a metric Lie algebra. These manifolds always locally split as a product into well understood factors, allowing a complete local classification. Third, I investigate various G-structures defining a geometry of this type, with a focus on almost Hermitian structures, extending recent results of Barbaro-Pediconi-Tardini. This is joint work with Andrei Moroianu.

Table.

(M,g)	∇^τ	$\mathfrak{h}$	$\underline{\mathrm{Hol}(\nabla^\tau)}$	$\underline{\mathrm{Stab}}$ (τ)	$d\tau = 0$?
G/H isotropy-irred., non-symmetric	canonical	$\mathfrak{h}$	$\mathfrak{h}$	$\mathfrak{j}$ or $\mathfrak{g}_2$ (Berger)	No
$(H\times H)/H$, $\mathfrak{h}$, $H^{\mathbb{C}}/H$ (bi-invariant metric)	Cartan-Schouten	$\nabla^t,\ t=0$	$\mathfrak{h}$	0 on $(H\times H)/H$, $t=\pm 1$ or $\mathfrak{h}$	Yes

Special cases.

Geometry	∇^τ	$\mathfrak{h}$	$\underline{\mathrm{Stab}}(\tau)$	$d\tau = 0$?	
6-dimensional nearly Kähler	characteristic	$\underline{\mathfrak{su}(3)}$	$\underline{\mathfrak{so}(3)}$ or $\underline{\mathfrak{su}(3)}$ $(S^3\times S^3)$ $\underline{\mathfrak{u}(1)\oplus\mathfrak{u}(1)}$ $(\mathrm{SU}(3)/T^2)$ $\underline{\mathfrak{su}(2)\oplus\mathfrak{u}(1)}$ (twistor space)	$\underline{\mathfrak{su}(3)}$	No
7-dimensional nearly parallel G_2	connection	$\underline{\mathfrak{g}_2}$	$\mathfrak{so}(3)_{\mathrm{inv}}$ (Berger space) $\mathfrak{so}(4)$ (3-or(5)-Sos) or $\underline{\mathfrak{g}_2}$	$\underline{\mathfrak{g}_2}$	No
M^3	$\nabla^g + t\ \mathrm{vol}_g$	$\underline{\mathfrak{so}(3)}$	0 (S^3, irred.) $\underline{\mathfrak{u}}(1)$ (-Sasakian) or $\underline{\mathfrak{so}(3)}$	$\underline{\mathfrak{so}(3)}$	Yes

Curiosity: the $\mathfrak{g}_2$ (Berger) appearing in the first table is the very interesting space $\mathrm{SO}(5)/\mathrm{SO}(3)_{\mathrm{irr}}$.

Definition 6.1. A *geometry with parallel skew torsion* (PST) is a Riemannian manifold (M,g) with a metric connection ∇^τ whose torsion T^τ is skew, that is, $T^\tau \in \Omega^3(M)$, and

$$\nabla^\tau T^\tau = 0.$$

Notation.

We write

$$\nabla^\tau = \nabla^g + \tau, \qquad \tau \in \Omega^3(M), \qquad T^\tau = 2\tau.$$

For a form α,

$$\nabla^\tau_X \alpha = \nabla^g_X \alpha + (\tau_X)\cdot\alpha.$$

Example 6.2. (1) Manifolds with Killing spinors: nearly Kähler NK^6, nearly parallel G_2, (α)-Sasaki, and 3-(α,δ)-Sasaki.

(2) Naturally reductive homogeneous spaces. Let $M = G/H$ and assume that

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}.$$

This gives a canonical connection ∇ with

$$T(X,Y) = -[X,Y]_{\mathfrak{m}}, \qquad R(X,Y)Z = -[[X,Y]_{\mathfrak{h}}, Z].$$

Any G-invariant tensor is parallel. In particular,

$$\nabla T = 0, \qquad \nabla R = 0.$$

If g is a G-invariant metric with

$$g([X,Y]_{\mathfrak{m}}, Z) + g([X,Z]_{\mathfrak{m}}, Y) = 0,$$

then

$$(M, g, \tau = \tfrac{1}{2}T)$$

is a PST.

(3) Cartan-Schouten connections. Fix a compact Lie group G with a bi-invariant metric g. Set

$$\sigma(X,Y,Z) = g([X,Y],Z).$$

Then $\sigma \in \Omega^3(G)$. It is both left- and right-invariant. Now define

$$\nabla^t = \nabla^g + \tfrac{t}{2}\,\sigma.$$

For $t = \pm 1$, the connection ∇^t is flat. Moreover, ∇^{-1} parallelizes the left-invariant vector fields, and ∇^{+1} parallelizes the right-invariant vector fields.

Properties of PST.

Let (M, g, τ) be a PST. Then:

- $\mathrm{Hol}(\nabla^\tau) \subseteq \mathrm{Stab}_{O(n)}(\tau)$.
- R^τ is pair-symmetric, so

$$R^\tau \in \mathrm{Sym}^2 \Lambda^2 T^* M.$$

  Hence, if

$$\mathfrak{hol}(\nabla^\tau) \subseteq \mathfrak{h},$$

  then ∇^τ is an $\mathfrak{h}$-instanton. Then

$$\Lambda^2 \cong \mathfrak{h} \oplus \mathfrak{h}^\perp, \qquad R^\tau(\mathfrak{h}^\perp) = 0.$$

  That is, ∇^τ is an instanton.
- Cleiton–Moriano–Semmelman, 2021. Fix

$$\mathfrak{hol}(\nabla^\tau) \subseteq \mathfrak{h} \subseteq \underline{\mathrm{Stab}}(\tau) \subseteq \underline{\mathfrak{so}}(n).$$

  Then (M, g, τ) admits a foliation with totally geodesic leaves. The local leaf space is a PST

$$(N, g_N, \sigma) \cong \prod_i (N_i, g_i, \sigma_i),$$

  where $\underline{\mathrm{Stab}}(\sigma_i)$ acts irreducibly on TN_i. Moreover,

$$\tau = \tau_{\mathrm{leaf}} + \pi^* \sigma - A,$$

  where A is the O'Neill tensor.

Closed torsion.

What if we impose that the torsion is closed, i.e. $d\tau = 0$?

Lemma 6.3. *Let (M, g, τ) be a PST. Then the following are equivalent.*

- $d\tau = 0$.
- $\nabla^g \tau = 0$.
- R^τ *satisfies the first Bianchi identity.*
- *For all $X \in TM$, $(\tau_X) \diamond \tau = 0$.*
- *For all $p \in M$, (T_pM, g_p, τ_p) is a metric Lie algebra.*

Definition 6.4. PST geometries with closed torsion are called *PSCT geometries.*

Barbaro–Pediconi–Tardini, 2024, study Hermitian PSCT geometry.

Theorem 6.5. *Any PSCT geometry locally splits. More precisely,*

$$(M, g, \tau) \cong \prod_i (M_i, g_i, \tau_i) \times (\mathbb{R}^k, g_{\mathrm{Euc}}, 0) \times \prod_j (N_j, h_j, \eta_j),$$

where each (M_i, g_i, τ_i) is irreducible under $\underline{\mathrm{Stab}}(\tau_i)$ and $\tau_i \neq 0$, and each (N_j, h_j, η_j) is irreducible and non-flat.

Parallelized geometries.

Which G-structures are parallelized by a PSCT connection? A natural list is the Berger list:

$$U(m),\ SU(m),\ \mathrm{Spin}(m),\ \mathrm{Spin}(m) \times \mathrm{Spin}(1),\ G_2,\ \mathrm{Spin}(7).$$

Hermitian torsion classes.

For a $U(m)$-structure, the intrinsic torsion lies in

$$TM \otimes \Lambda^{(2,0)+(0,2)}.$$

The Gray–Hervella classes are:

$$W_1 = \Lambda^{3,0} \oplus \Lambda^{0,3}, \qquad W_3 = \Lambda^{2,1}_0 \oplus \Lambda^{1,2}_0, \qquad W_4 = TM \wedge \omega.$$

The class W_2 is the remaining Gray–Hervella summand. In this notation:

- $W_3 \oplus W_4$ iff integrable.
- $W_1 \oplus W_3 \oplus W_4$ iff skew torsion.
- $W_1 \oplus W_2 \oplus W_3$ iff semi-Kähler.
- W_1 iff nearly Kähler.

General PSCT manifolds.

Now we explain how a general PSCT manifold looks.

Theorem 6.6. *Let (M, g, τ, J) be a Hermitian PSCT manifold. Then locally*

$$(M, g, \tau) \cong \underbrace{(H, g_H, -\tfrac{1}{2}\sigma)}_{\textit{Lie group with } \nabla^{-1}} \times \prod_i \underbrace{(S_i, g_i, \alpha_i \operatorname{vol}_{g_i})}_{\textit{3-dimensional}} \times \prod_j \underbrace{(N_j, g_j, 0)}_{\textit{Kähler}}.$$

Each S_i is an α_i-Sasaki manifold, with Reeb vector field

$$\xi_i \in \mathfrak{X}(S_i).$$

This splitting is not preserved by J. Instead, J preserves

$$\xi_i^\perp \subset TS_i, \qquad \mathfrak{h} \oplus \operatorname{span}\{\xi_1, \ldots, \xi_r\}, \qquad TN.$$

Theorem 6.7 (Barbaro–Pediconi–Tardini, 2024)**.** *If (M, g, τ, J) is Hermitian PSCT, then there exists a Cartan subalgebra*

$$\underline{\mathfrak{t}} \subseteq \underline{\mathfrak{h}}$$

such that J preserves

$$\underline{\mathfrak{t}} \oplus \operatorname{span}\{\xi_1, \ldots, \xi_r\},$$

and it projects to an integrable complex structure on H/T.

Now assume $\dim M > 4$.

Theorem 6.8. *If (M, g, τ, J) is a PSCT structure in $W_1 \oplus W_4$ or in W_3, then it is Kähler.*

Theorem 6.9. *If (M, g, τ, J) is a PSCT structure in $W_1 \oplus W_3$, then*

$$(M, g, \tau) \cong (H, g_H, -\tfrac{1}{2}\sigma) \times \prod_i (N_i, h_i, 0),$$

and

$$\underline{\mathfrak{h}} \neq \mathfrak{su}(2) \oplus \mathbb{R}.$$

These manifolds are classified. The J are not classified. One can have different almost complex structures, but at least we know what the manifolds are.

7. FLAG VARIETIES AS GKM SPACES

Speaker.
Giovane Galindo

Title.
Flag varieties as GKM spaces

Abstract.

In this talk, we present flag varieties as GKM spaces and use their Lie-theoretic structure to describe the associated combinatorial data explicitly. This perspective enables computations in equivariant cohomology, including the equivariant Gromov–Witten invariants. This will facilitate simpler atom computations and enable the solution of problems in G-equivariant birational geometry. This is ongoing joint work with Ludmil Katzarkov.

For GKM varieties, we can compute GW invariants in the equivariant cohomology ring via a torus action and the graph associated to the GKM variety.

Big quantum cohomology.

Definition 7.1. A smooth projective variety X is given. A *genus zero stable map* is a map

$$f : (C; x_1, \ldots, x_n) \to X$$

where:

1. C is a projective connected nodal curve of genus 0;
2. $x_1, \ldots, x_n \in C$ are smooth marked points;
3. $\operatorname{Aut}(f)$ is finite.

The moduli space of stable maps is

$$\overline{\mathcal{M}}_{0,n}(X,\beta).$$

It is a Deligne–Mumford stack.

Definition 7.2. The *evaluation map* is

$$\mathrm{ev}_i : \overline{\mathcal{M}}_{0,n}(X,\beta) \to X$$

given by

$$\mathrm{ev}_i(f) = f(x_i).$$

Definition 7.3. The *Gromov–Witten invariant* is

$$\langle \phi_1, \ldots, \phi_n \rangle^X_{0,n,\beta} = \int_{[\overline{\mathcal{M}}_{0,n}(X,\beta)]^{\mathrm{vir}}} \prod_{i=1}^{n} \mathrm{ev}_i^*(\phi_i).$$

where the ϕ_i are cohomology classes on X.

Definition 7.4. For a graded basis (α_k) of $H^*(X)$, put

$$\tau = \sum_k T_k \alpha_k.$$

The *big quantum product* is

$$\phi_i \star_\tau \phi_j = \sum_{k,n,\beta} \langle \phi_i, \phi_j, \phi_k, \tau^n \rangle^X_{0,3+n,\beta} \frac{\phi^k}{n!} Q^\beta.$$

These are formal power series.

GKM spaces.

Flag varieties carry a torus action with enough fixed-point and one-dimensional orbit data to make GKM localization effective.

Flag varieties.

Definition 7.5. Let G be a connected complex reductive group and $B \subset G$ a Borel subgroup. The *flag variety* G/B parametrizes complete flags in the standard representation.

The T-action on G/B is induced by a maximal torus $T \subset B$. Its T-fixed points are indexed by the Weyl group W.

Remark 7.6. For a flag variety, the T-fixed points are isolated and the one-dimensional T-orbits are controlled by the roots of G. This is the GKM situation.

The point of flag varieties is that we can study them using the generalized eigenspace decomposition. In particular, we can study their GW invariants this way.

GKM data.

Definition 7.7. A T-space X is a *GKM space* if the fixed-point set X^T is finite and the one-dimensional orbits are isolated in the sense required by GKM theory.

For $X = G/B$, the combinatorial model is the moment graph: the vertices are W and the edges are given by reflections. Each edge is labeled by a root.

Remark 7.8. The moment graph encodes the Bruhat order, the root system, and the localization data needed for equivariant cohomology computations.

Equivariant cohomology.

Definition 7.9. The *T-equivariant cohomology* of X is

$$H_T^*(X) := H^*(ET \times_T X).$$

For GKM spaces, localization identifies $H_T^*(X)$ with tuples of polynomial functions on the fixed points satisfying edge-compatibility relations. For flag varieties, this gives a combinatorial description of Schubert classes.

Proposition 7.10. *The Schubert basis of $H_T^*(G/B)$ is compatible with the Weyl group parametrization of fixed points, and the restriction maps to X^T are determined by the GKM graph.*

Quantum part.

Let X be a smooth projective variety. We now switch to the small quantum cohomology.

Definition 7.11. Let $T_i, T_j \in H^*(X)$. The *small quantum product* is

$$T_i \star_{\text{small}} T_j = \sum_{a,b} \sum_{\beta \in \overline{\mathrm{NE}}(X,\mathbb{Z})} \langle T_i, T_j, T_a \rangle_{3,\beta} q^\beta g^{ab} T_b.$$

The ring

$$(QH^*(X), \star_{\text{small}})$$

is called the small quantum cohomology.

The point is that these are no longer series but finite sums, so we can do actual computations.

Flag varieties.

Definition 7.12. A *flag variety* is a complex homogeneous space

$$G^{\mathbb{C}}/P,$$

where $G^{\mathbb{C}}$ is a connected, simply connected semisimple complex Lie group and $P \subset G^{\mathbb{C}}$ is a parabolic subgroup, that is, a Lie subgroup containing a Borel subgroup.

Definition 7.13. Let G/B be a flag variety. The *Weyl group* W is the finite reflection group generated by the simple reflections S_α associated to the simple roots $\alpha \in \Pi$. It carries the length function

$$\ell : W \to \mathbb{Z}_{\geq 0},$$

where for $w \in W$ the value $\ell(w)$ is the length of any reduced expression of w as a product of simple reflections.

Example 7.14. Let

$$\mathrm{Fl}_3 = SL_3/B$$

be the space of full flags in $\mathbb{C}^3$:

$$\{0 \subset V_1 \subset V_2 \subset \mathbb{C}^3\}.$$

It can also be described as the projectivized tangent bundle of the projective plane:

$$\mathrm{Fl}_3 \cong \mathbb{P}(T_{\mathbb{P}^2}) \cong \mathbb{P}^1 \times \mathbb{P}^2.$$

The point is to use the root space decomposition to encode the geometry. The flag variety Fl_3 embeds into

$$\mathbb{P}^2 \times (\mathbb{P}^2)^*$$

as a smooth incidence hypersurface of bidegree $(1,1)$. It is cut out by the incidence relation

$$x_0y_0 + x_1y_1 + x_2y_2 = 0.$$

The set of positive roots is

$$R^+ = \{\alpha, \beta, \alpha+\beta\}.$$
$$W = \langle s_\alpha, s_\beta \rangle \cong S_3.$$

Schubert basis.

Given an element $w \in W$, define the Schubert cell of w in the flag variety G/B by

$$C_w = \omega_0 B \omega_0^{-1} \cdot wB/B.$$

The Schubert variety is

$$X_w = \overline{C_w}.$$

The Schubert class

$$\sigma_w = [X_w] \in H^{2\ell(w)}(X).$$

The set

$$\{\sigma_w\}_{w\in W}$$

is a homogeneous basis for $H^*(X)$. In particular,

$$H^*(X) = \sum_{w\in W} \mathbb{Q}\sigma_w.$$

Example 7.15. For Fl_2, we have the following basis for our cohomology:

$$\sigma_{\mathrm{id}}, \sigma_{s_\alpha}, \sigma_{s_\beta}, \sigma_{s_\beta s_\alpha} \in H^2,$$
$$\sigma_{s_\alpha s_\beta} \in H^3, \qquad \sigma_{s_\alpha s_\beta s_\alpha} \in H^6.$$

A nonzero linear functional

$$\alpha \in \mathfrak{h}^*$$

is called a root of $\mathfrak{g}$ with respect to $\mathfrak{h}$ if the corresponding root space

$$\mathfrak{g}_\alpha := \{X \in \mathfrak{g} : [H, X] = \alpha(H)X \text{ for all } H \in \mathfrak{h}\}$$

is nonzero.

The Lie algebra $\mathfrak{g}$ decomposes as a direct sum of weight spaces for the adjoint action of $\mathfrak{h}$:

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in R} \mathfrak{g}_\alpha.$$

Remark 7.16. The root data of $G^{\mathbb{C}}/P$ is what makes the GKM description effective.

Definition 7.17. Let

$$T \cong (\mathbb{C}^*)^r$$

be an algebraic torus. A smooth projective variety X with an effective $\mathbb{T}$-action is called *GKM*, after Goresky, Kottwitz, and MacPherson, if the following hold:

1. $X^{\mathbb{T}}$ is the fixed point set, and it is finite.
2. The one-dimensional orbits are isolated.

Definition 7.18. Let X be a GKM variety. The *GKM graph* of X is the graph

$$\Gamma = (V(\Gamma), E(\Gamma))$$

where:

(1) the vertices are the $\mathbb{T}$-fixed points;
(2) the edges are the closures of the one-dimensional orbits.

Remark 7.19. For a flag variety $X = G/B$ there is a natural action of a torus $T \subset G$. The fixed points are the points wB/B for $w \in W$.

Given $w \in W$ and a root $\alpha \in R$, the subgroup $G_\alpha \cong SL_2$ defined by α gives an invariant $\mathbb{P}^1$:

$$wG_\alpha B/B.$$

It connects wB/B and $ws_\alpha B/B$.

Remark 7.20. The GKM graph of our running example has six vertices.

Definition 7.21. If X is a GKM space with a T-action, there is an induced action on

$$\overline{\mathcal{M}}_{g,n}(X, \beta).$$

Given $t \in T$, define

$$t(C, p_1, \dots, p_n, f) = (C, p_1, \dots, p_n, t \circ f).$$

We denote by $\overline{\mathcal{M}}_{g,n}(X, \beta)^T$ the *fixed point set* of this action.

Definition 7.22. For $n \geq 0$, an *n-marked tree* γ consists of:

(1) a finite nonempty set $V(\gamma)$, the vertices;
(2) a subset $E(\gamma) \subset \binom{V(\gamma)}{2}$ such that the graph $(V(\gamma), E(\gamma))$ is a tree;
(3) a map $\psi : \{1, \dots, n\} \to V(\gamma)$;
(4) a map assigning an effective curve class to each vertex;
(5) a map assigning a positive degree to each edge.

Lemma 7.23. *The marked trees that occur are in bijection with the fixed points of* $\overline{\mathcal{M}}_{0,n}(X, \beta)$ *under the* $\mathbb{T}$*-action.*

Theorem 7.24. *Let* X *be a smooth projective GKM space, and fix a splitting*

$$s : H^*(X, \mathbb{Q}) \to H^*_{\mathbb{T}}(X, \mathbb{Q}).$$

Then the equivariant Gromov–Witten invariants are given by the localization formula, as a sum over the fixed marked trees.

Remark 7.25. For Fl_3, the equivariant Gromov– Witten invariants

$$\langle \gamma_i, \gamma_j \rangle_{2,\beta}$$

are encoded by a matrix. The point is that the localization theorem lets us compute these invariants.

Once GW invariants are computable, the quantum cohomology is computable as well. For Fl_3, this gives the matrix for small quantum multiplication. Hence we get six Hodge atoms of dimension 1 for Fl_3.

Detour.

The following map:

$$\mathrm{Fl}_3 \to \mathrm{Fl}_3, \qquad (V_1, V_2) \mapsto (V_2^\perp, V_1^\perp)$$

defines a $\mathbb{Z}_2$-action on Fl_3. It permutes the generators of the Picard group. This gives the following Hodge diamond:

To compute the G-Hodge atoms for this variety, we now look for the invariant Hodge class that gives us the maximum amount of eigenvalues. So now it is invariant classes, in contrast with the usual atoms.

The $\mathbb{Z}_2$ atoms for Fl_3 are:

ρ	$\rho_{\mathbb{Z}_2}$
λ_1	$2, 1$
λ_2	$2, 2$
λ_3	$1, 1$
λ_3	$1, 1$

The idea is to compute the matrix multiplication in quantum cohomology using the ambient space.

Theorem 7.26. *Let X be a smooth projective GKM space, and let $E \to X$ be a $\mathbb{T}$-equivariant vector bundle generated by global sections. Let $Y \subset X$ be a generic zero locus. Then the Gromov–Witten invariants of Y are computed by a localization formula. This is the same kind of computation as before, but now for complete intersections.*

In particular, one can compute the matrix for quantum multiplication.

Example 7.27. We construct a del Pezzo surface S directly from the canonical geometry of the flag variety Fl_3. We can compute the matrix multiplication.

Takeaway.

The Lie-theoretic structure of a flag variety turns the geometry into combinatorics, and GKM methods make equivariant calculations explicit.

8. Geometric aspects of conditional measures

Speaker.

Renata Possobon

Title.

Geometric aspects of conditional measures

Abstract.

From a logistical problem formulated in the 18th century emerged what we now know as Optimal Transport Theory, whose development has been particularly intense in recent decades. By providing a natural geometric framework for comparing measures, this theory establishes deep connections with several areas, including Geometry, Probability, Analysis, Dynamical Systems, Economics, and Data Science. In this talk, we will discuss the concept of disintegration of measures from the perspective of optimal transport problems, aiming to investigate the structure and geometric arrangement of the associated conditional measures. In particular, we will analyze rigidity conditions for disintegration maps, as well as propose an approach to characterize foliations induced by disintegration. This talk is based on joint work with Florentin Münch and Christian Rodrigues.

Optimal transport theory.

Let (X, d) be a metric space. We study the space $\mathcal{P}(X)$ of Borel probability measures on X.

Optimal transport began with a logistical problem posed by Caspard Monge in 1781.

The main idea is to transport mass from one point to another at minimal cost. Think of soil in a pile in the garden being moved to a hole made somewhere else.

The soil is represented by a distribution $\mu \in \mathcal{P}(X)$, and the hole, or destination, by another distribution $\nu \in \mathcal{P}(X)$.

Let X and Y be Radon spaces, and let $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$. Fix a Borel cost function $c : X \times Y \to [0, \infty]$. We minimize

$$\gamma \mapsto \int_{X\times Y} c(x,y)\, d\gamma(x,y)$$

among all transport plans with marginals μ and ν.

9. Rational Cubic Fourfolds and Their Relation to K3 Surfaces

Speaker.

Jérémy Guéré

Title.

Rational Cubic Fourfolds and Their Relation to K3 Surfaces

Abstract.

I will briefly review the construction of atoms, specifically addressing certain technical difficulties regarding the behavior of Hodge structures under Iritani's blow-up formula. Subsequently, I will introduce a new atomic invariant and provide a proof for the following theorem: if a smooth complex cubic fourfold is rational, then its primitive cohomology is isomorphic - as a rational Hodge structure - to the shifted middle cohomology of a projective K3 surface.

AI-generated summary.

The aim is to build a stronger birational obstruction for cubic fourfolds by mixing two kinds of data. This is the first dimension in which the rationality answer for cubics is not uniform: some are rational, and others are not.

First, classical Hodge theory says that blow-ups preserve the Hochschild degree $p-q$. Thus, if a cubic fourfold were rational, then some blow-up center in any weak factorization to $\mathbb{P}^4$ would have to contribute the missing $H^{(2)}$ piece. This already forces the center to be surface-like, and eventually K3-like.

Second, quantum cohomology gives the theory of atoms. Hodge theory alone gets one only to something like a surface with nef canonical class. To force the surface to be K3, one uses Iritani's quantum blow-up formula. The atoms are generalized eigenspaces of quantum multiplication by the Euler vector field.

The technical work is to make these atoms compatible across different blow-up factorizations. This is why one introduces evaluation maps into a common non-archimedean power-series field S: otherwise the completed coefficient rings depend on the chosen blow-up map.

The payoff is the following strategy. If X is rational, then weak factorization together with Iritani's theorem forces some blow-up center Z to fail a chosen property $(*)$. One then chooses $(*)$ so that the possible failing centers are geometrically meaningful. In KKPY, showing that no such center exists gives irrationality for very general cubic fourfolds. Here the property is refined so that rationality of a cubic fourfold forces the primitive rational Hodge structure of X to come from a projective K3 surface.

In slogan form:

$$\begin{aligned} \text{rational cubic fourfold} \quad &\Longrightarrow \quad \text{K3 center in weak factorization} \\ &\Longrightarrow \quad H^4(X,\mathbb{Q})_{\mathrm{prim}} \text{ is of K3 type.} \end{aligned}$$

Introduction.

Let

$$X_3 \subset \mathbb{P}^5$$

be a cubic fourfold. All varieties in this talk are smooth, projective, and over $\mathbb{C}$.

Theorem 9.1 (Weak factorization)**.** *Let X and Y be two birational varieties. Then there exists a weak factorization*

$$X = X_0 \dashrightarrow X_1 \dashrightarrow \cdots \dashrightarrow X_{n-1} \dashrightarrow X_n = Y,$$

where each dashed arrow is either a morphism or its inverse, and each morphism is the blow-up along a smooth center.

Remark 9.2. The reference is the weak factorization theorem of Abramovich–Karu–Matsuki–Wlodarczyk, also proved independently by Wlodarczyk. In these notes, the related blow-up technology appeared earlier in the discussion of Iritani's blow-up formula.

Now look at what happens for one blow-up.

The blow-up formula says that $D^b(Y)$ can be obtained from $D^b(X)$ by adding or removing semi-orthogonal pieces of the form $D^b(Z)$, where Z is the center of the blow-up.

Theorem 9.3. *Let*

$$\widetilde{X} = \mathrm{Bl}_Z X,$$

where $Z \hookrightarrow X$ has codimension r. Then

$$H^{p,q}(\widetilde{X}) \cong H^{p,q}(X) \oplus \bigoplus_{k=1}^{r-1} H^{p-k,q-k}(Z).$$

Remark 9.4. Not only is the bidegree (p,q) controlled, but the difference $p-q$ is preserved.

Definition 9.5. Define

$$H^{(k)}(X) = \bigoplus_{p-q=k} H^{p,q}(X).$$

We call k the *Hochschild degree.*

The point is that $H^{(n)}$ and $H^{(n-1)}$ are birational invariants, where $n = \dim X$:

Example 9.6. Let $n = \dim X = \dim Y$, and suppose X is birational to Y. Then

$$\dim H^{(n)}(X) = \dim H^{(n)}(Y).$$

The same holds for $H^{(n-1)}$. This follows directly from the blow-up formula and the definition.

For example, let

$$Q \subset \mathbb{P}^4$$

be a quintic threefold. Then

$$H^{3,0}(Q) = \mathbb{C}, \qquad H^{3,0}(\mathbb{P}^3) = 0.$$

Thus a quintic threefold is never rational.

Example 9.7. Let

$$X \subset \mathbb{P}^5$$

be a smooth cubic fourfold. Its Hodge diamond is

$$\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 0 & & 0 & & & \\
 & & 0 & & 1 & & 0 & & \\
 & 0 & & 0 & & 0 & & 0 & \\
0 & & 1 & & 21 & & 1 & & 0 \\
 & 0 & & 0 & & 0 & & 0 & \\
 & & 0 & & 1 & & 0 & & \\
 & & & 0 & & 0 & & & \\
 & & & & 1 & & & &
\end{array}$$

whereas the Hodge diamond of $\mathbb{P}^4$ is

$$\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 0 & & 0 & & & \\
 & & 0 & & 1 & & 0 & & \\
 & 0 & & 0 & & 0 & & 0 & \\
0 & & 0 & & 1 & & 0 & & 0 \\
 & 0 & & 0 & & 0 & & 0 & \\
 & & 0 & & 1 & & 0 & & \\
 & & & 0 & & 0 & & & \\
 & & & & 1 & & & &
\end{array}$$

For the cubic, the first nonzero off-center Hochschild degree is $H^{(2)}$, while for $\mathbb{P}^4$ all off-center Hochschild degrees vanish. Thus direct comparison with $\mathbb{P}^4$ forces nontrivial centers in any factorization.

The point is that this invariant is obtained by comparing the leftmost columns in the Hodge diamond.

If

$$X \dashrightarrow \mathbb{P}^4$$

is birational, then in every weak factorization there is a blow-up center Z such that

$$H^{(2)}(Z) \neq 0.$$

As a first observation, Z must be a surface. From the classification of surfaces, the minimal model Σ of such a surface has K_Σ nef.

A variety with K_X nef has only one atom. This shows that the nefness of K_X is important for Gromov–Witten theory.

Remark 9.8 (Hassett's conjecture)*.* A cubic fourfold X is rational if and only if

$$H^4(X, \mathbb{Z})_{\text{prim}} \xrightarrow{\sim} H^2(S, \mathbb{Z})(-1)$$

as integral Hodge structures, i.e. $\mathbb{Z}$-HS, for some projective K3 surface S.

Remark 9.9*.* The point is that Hassett's conjecture is an integral statement. It concerns integral Hodge structures, not only rational ones.

Theorem 9.10. *The direct implication of the conjecture holds with rational Hodge structures. That is, if X is rational, then*

$$H^4(X,\mathbb{Q})_{\mathrm{prim}} \cong H^2(S,\mathbb{Q})(-1)$$

as rational Hodge structures, for some projective K3 surface S.

Lemma 9.11 (KKPY)**.** *A very general cubic fourfold X is irrational.*

The obstruction is carried by the transcendental classes in

$$H^4(X,\mathbb{Q})_{\mathrm{prim}}.$$

Remark 9.12 (On the proof of the theorem)*.* If X is rational, then in every weak factorization there exists a blow-up center Z which is a K3 surface. In fact, to pass from K_Z being nef to Z being a K3 surface, one needs a little more than Hodge theory. This leads to the next topic.

Atom construction.

To pass from K_Σ nef to Σ being a K3 surface, we use a refined invariant composed with $H^{(2)}$.

The blow-up formula is an isomorphism of quantum rings.

Let us recall the basics of quantum cohomology.

$$\begin{tikzcd}\mathcal{C} \arrow[r, "f"] & X\end{tikzcd}$$

$\overline{\mathcal{M}}_{g,n}(X,\beta) = \{f : C \to X : C \text{ curve of genus } g, f_*([C]) = \beta \in H_2(X), x_1,\ldots,x_n \in C\}\,/\sim$

Here $\mathcal{C}$ is the universal curve.

$$\begin{tikzcd}\mathcal{C} \arrow[r, "f"] \arrow[d] & X \\ \overline{\mathcal{M}}_{g,n}(X,\beta) \arrow[u, bend left, "x_1,\ldots,x_n"] & \end{tikzcd}$$

There exists

$$[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}}$$

an algebraic cycle of the expected dimension. The virtual dimension is

$$\mathrm{vdim}_{\mathbb{C}} = (1-g)\big(\dim_{\mathbb{C}} X - 3\big) + n + \int_\beta c_1(X).$$

Definition 9.13. To compute the Gromov–Witten invariants, let

$$\alpha_1,\ldots,\alpha_n \in H^\bullet(X).$$

The *Gromov–Witten invariant* is something like counting curves:

$$\langle \alpha_1,\ldots,\alpha_n\rangle^X_{g,n,\beta} = \int_{[\overline{\mathcal{M}}_{g,n}(X,\beta)]^{\mathrm{vir}}} \prod_i \mathrm{ev}_i^*\big(\alpha_i\big) \in \mathbb{Q}.$$

Unless $g=0, \beta=0, n=2$, we have the string equation

$$\langle \alpha_1,\ldots,\alpha_n, \nVdash_X\rangle = 0.$$

And the divisor equation, if

$$\gamma \in H^1(X) \text{ or } H^2(X),$$

then

$$\langle \alpha_1, \dots, \alpha_n, \gamma \rangle = \int_\beta \gamma \, \langle \alpha_1, \dots, \alpha_n \rangle \, .$$

Definition 9.14. Let (ϕ_i) be a graded basis of $H(X, \mathbb{Q})$. Let (ϕ^j) be the dual basis, so that

$$\int_X \phi_i \vee \phi^j = \delta_{ij}.$$

For every graded vector subspace

$$V \subset H(X, \mathbb{Q}),$$

choose a graded basis (α_k) of V. For example, one can take

$$V = H^{\mathrm{even}}, \qquad V = H^{(0)}, \qquad V = H_{\mathrm{Hdg}}.$$

Put a formal variable T_k for each basis element α_k, and define

$$\tau = \sum_k T_k \alpha_k.$$

Then

$$\phi_i \star_\tau \phi_j = \sum_{k,n,\beta} \langle \phi_i, \phi_j, \phi_k, \tau^n \rangle^X_{0,3+n,\beta} \frac{\phi^k}{n!} Q^\beta.$$

Another invariant is the Euler vector field:

Definition 9.15. The *Euler vector field* is

$$\mathrm{Eu}_\tau = c_1(X) + \sum_k \frac{2 - \deg \alpha_k}{2} T_k \alpha_k.$$

One of the most important objects in theory is:

Definition 9.16.

$$\kappa_\tau(-) := \mathrm{Eu}_\tau \star_\tau (-) \in \mathrm{End}\big(H(X, \mathbb{Q})_R\big),$$

where

$$R = \mathbb{Q}[[Q, (T_k)]]$$

is the base-change ring introduced because of the formal variables above.

Properties.

- $$\kappa_{\tau + \lambda \, \mathrm{id}_X} = \kappa_\tau + \lambda \, \mathrm{id} \, .$$
  (string)
- κ_τ is a morphism of $\mathbb{Q}$-Hodge structures.

Iritani's formula is like the blow-up formula we had before but upgraded to the quantum level:

Theorem 9.17 (Iritani)**.** *Let*

$$\widetilde{X} = \mathrm{Bl}_Z X.$$

Then

$$\kappa^{\widetilde{X}}_{\widehat{\tau}}$$

is conjugate, as an endomorphism, to

$$\kappa^X_{\tau_0} \oplus \bigoplus_{k=1}^{r-1} \kappa^Z_{\tau_k}$$

in

$$\mathrm{End}_{\widehat{R}_{\mathbb{C}}}\big(H(\widetilde{X},\mathbb{C})_{\widehat{R}_{\mathbb{C}}}\big).$$

Here the parameters are related by a change of variables

$$\widehat{\tau} \longleftrightarrow (\tau_0, \tau_1, \ldots, \tau_{r-1}).$$

Here

$$R_{\mathbb{C}} = \mathbb{C}[[Q,(T_k)]], \qquad \widehat{R}_{\mathbb{C}} = \mathbb{C}[q^{\pm 1}][[Q,(T_k)]],$$

and

$$R_{\mathbb{C}} \hookrightarrow \widehat{R}_{\mathbb{C}}.$$

Remark 9.18. $\mathbb{C}$-HS are deformation invariants, while $\mathbb{Q}$-HS aren't.

These rings are written with respect to X. One could also ask for the corresponding rings with respect to $\widetilde{X}$, for example

$$R^{\widetilde{X}}_{\mathbb{C}} = \mathbb{C}[[\widetilde{Q},(\widetilde{T}_k)]].$$

There is a curve class $\ell_E \subset \mathrm{Exc}$ which is a line in the exceptional divisor. We do not write its precise definition here, but under the change of rings one has

$$\widetilde{Q}^{\ell_E} \longmapsto q^{2(r-1)}.$$

There is also a map

$$R^Z_{\mathbb{C}} \longrightarrow \widehat{R}_{\mathbb{C}},$$

which need not be injective, but at least gives a ring map.

Remark 9.19. Iritani's formula comes from a localization formula. Do not trust the change of variables: prove it.

Definition 9.20. An *atom* is a generalized eigenspace of $\kappa^{\widetilde{X}}_{\widetilde{\tau}}$, namely:

$$\mathbb{E}_{\tau,\lambda} = \mathrm{Ker}((\kappa_\tau - \lambda)^m), \qquad \gg 1.$$

Up to this point everything is deformation invariant, since we are in Gromov–Witten theory. If we want an invariant which is not deformation invariant, we need to add an object which is not deformation invariant: Hodge structures.

Refinements.

Replace $\mathbb{C}$ by a field K. Replace

$$V = H(X,\mathbb{Q})$$

by either

$$H(X)_{\mathrm{Hdg}} \qquad \text{or} \qquad H^{(0)}(X).$$

We now keep track of rational Hodge structures.

The operator can be written as

$$\kappa_\tau = \sum_i s_i K_i,$$

where (s_i) is a basis of $K/\mathbb{Q}$, and the K_i are morphisms of rational Hodge structures.

Big issue.

The ring $\widehat{R}$ inverts

$$\widetilde{Q}^{\ell_E},$$

and this depends on the morphism

$$\widetilde{X} \to X.$$

For example, in a diagram

$$\begin{tikzcd} & \widetilde{X} \arrow[dl] \arrow[dr] & \\ X_1 & & X_2 \end{tikzcd}$$

the resulting rings $\widehat{R}_1$ and $\widehat{R}_2$ need not be isomorphic.

Remark 9.21. One of the greatest difficulties of the theory if to compare the rings $\hat{R}_1$ and $\hat{R}_2$. Here comes Levi-Civita . . . [missing]

The solution is to look, for every R and every X, for a diagram

$$\begin{tikzcd} R^X \arrow[r] \arrow[d, hook] & S \\ \widehat{R} \arrow[ur, dashed] & \end{tikzcd}$$

where S is a common ring independent of X, of $\widetilde{X}$, and of the chosen map $\widetilde{X} \to X$.

The ring S must be:

- a ring of power series;
- large enough to contain roots and related elements needed by the change of variables.

In sum, take

$$S = \mathbb{Q}[a^{\mathbb{Q}}][b^{\pm 1}].$$

The factor $b^{\pm 1}$ keeps track of degree. We will ignore this point for now.

Here

$$\mathbb{Q}[a^{\mathbb{Q}}] \ni \sum_{n \in \mathbb{Q}} x_n a^n, \qquad x_n \in \mathbb{Q},$$

with the condition that, for every $m \in \mathbb{Q}$,

$$\#\{n : x_n \neq 0,\ n < m\} < +\infty.$$

Thus these are generalized power series with rational exponents and well-ordered support toward $-\infty$.

This is a field. On this field one defines a non-archimedean absolute value. For

$$0 \neq f = \sum_{n \in \mathbb{Q}} x_n a^n,$$

put

$$\rho(f) = \min\{n : x_n \neq 0\}$$

and define

$$|f| := 2^{-\rho(f)}.$$

Producing the ring map.

How do we produce the ring map

$$R^X \to S?$$

Since

$$R^X = \mathbb{Q}[[Q, T_k]],$$

one is tempted to define the map on the generators Q and T_k and then extend by linearity. This does not always work: there are convergence issues. The absolute value above is useful precisely because it gives a criterion for when such a map on generators can be extended.

Now let

$$\mathrm{ev} : \widehat{R}_K \to S_K$$

be an evaluation map. From

$$\mathrm{ev}(\kappa_\tau)$$

we compute the generalized eigenspaces

$$\mathbb{E}_{\mathrm{ev},\lambda} = \ker\bigl(\mathrm{ev}(\kappa_\tau) - \lambda\bigr)^m$$

for $m \gg 1$. These are the atoms.

For almost all evaluations ev and for all λ, if a property $(*)$ holds on $\mathbb{E}_{\mathrm{ev},\lambda}$, then we say that X satisfies $(*)$. Thus the point is to cook up a property $(*)$ of vector spaces.

Remark 9.22. The upshot is that eigenspaces are conjugation-invariant, so that we can apply Iritani's formula.

Proposition 9.23. *If Z satisfies $(*)$, then X satisfies $(*)$ if and only if*

$$\widetilde{X} = \mathrm{Bl}_Z X$$

satisfies $(*)$.

Proof. (Uses the string equation.) Z satisfies $(*)$, X not, exists τ_0, exists λ, $\mathbb{E}_{\tau_0,\lambda}$ not $(*)$. Choose τ_k such that $\mathbb{E}_{\tau_k} = 0 \rightsquigarrow$, $\mathbb{E}_{\tilde{\tau},\lambda}$ not $(*)$. ☐

Lemma 9.24 (Corollary)**.** *Assume that*

$$\begin{cases} X \text{ is birational to } Y, \\ X \text{ fails } (*), \\ Y \text{ satisfies } (*). \end{cases}$$

Then there exists a blow-up center Z which fails $(*)$.

Define

$$\rho_{\mathrm{ev},\lambda} = \dim\bigl(\mathbb{E}_{\mathrm{ev},\lambda} \cap H_{\mathrm{Hdg}}\bigr),$$
$$\nu^{(k)}_{\mathrm{ev},\lambda} = \dim\bigl(\mathbb{E}_{\mathrm{ev},\lambda} \cap H^{(k)}\bigr),$$

and

$$\gamma_{\mathrm{ev},\lambda} = \mathrm{rk}\,(\mathrm{ev}(\kappa_\tau) - \lambda)_{\mathbb{E}_{\mathrm{ev},\lambda}}\,.$$

Here $\mathbb{E}_{\mathrm{ev},\lambda}$ is the generalized tangent space.

The property from the KKPY paper is

$$(*)_{\mathrm{KKPY}} \quad \Longleftrightarrow \quad \nu^{(2)} = 0 \text{ or } \rho \geq 3.$$

i.e., that $H^{(2)} = 0$ or at least three algebraic cycles, because of point, H and fundamental class.

The property from the present paper is

$$(*)_{\text{new}} \quad \iff \quad \nu^{(2)} = 0 \text{ or } \nu^{(1)} \neq 0 \text{ or } \gamma \geq 2,$$

which is a modification of the former, such that the only surfaces that satisfy it are K3.

Thus, by the theorem, one gets a blow-up center Z failing $(*)$. The question becomes: which points, curves, and surfaces can fail this property?

For KKPY one shows that there is no center Z failing $(*)$. This gives the contradiction used to prove the desired non-birationality statement.

10. Birational Geometry of Hyperkähler Manifolds and the Hu–Yau Conjecture

Speaker.

Andrey Soldatenkov

Title.

Birational geometry of hyperkähler manifolds and the Hu–Yau conjecture

Abstract.

The talk will be based on a joint work with Katya Amerik and Misha Verbitsky. I will try to explain our proof of the Hu-Yau conjecture for compact hyperkähler manifolds. The conjecture predicts a description “in codimension two” of bimeromorphic maps between holomorphic symplectic manifolds. It states that any such map should be a composition of certain elementary transformations called Mukai flops, at least if we ignore some “bad locus” that can be of codimension three or higher. The conjecture has been proven by Wierzba and Wiśniewski in the case of symplectic fourfolds. Building on their proof, and using more recent results on the structure of the Kähler cone of hyperkähler manifolds, we extend the result to arbitrary dimension.

Definition 10.1. A *hyperkähler manifold* is a compact smooth manifold M with a metric g such that

$$\pi_1(M) = 1$$

and

$$\mathrm{Hol}(g) = \mathrm{Sp}(n).$$

Remark 10.2. Such an M admits a triple of complex structures I, J, K such that

$$IJ = -JI = K$$

and such that g is Kähler for all of them. We have the Kähler forms ω_I, ω_J, and ω_K.

Consider the 2-form

$$\sigma_I = \omega_J + \sqrt{-1}\,\omega_K.$$

Then σ_I is a holomorphic symplectic form of type $(2, 0)$. In fact, it is nondegenerate.

Let $\Omega^k_{M_I}$ be the bundle of I-holomorphic k-forms on M. Then σ_I^n is a nowhere vanishing section of the canonical bundle

$$K_{M_I} = \Omega^{2n}_{M_I}.$$

Thus M_I is an IHS.

Turning to variational geometry.

Example 10.3. Let S be a complex K3 surface, for example

$$S = \{x_0^4 + x_1^4 + x_2^4 + x_3^4 = 0\} \subset \mathbb{P}^3.$$

The metric is not immediate here: it is a consequence of the Calabi–Yau theorem. But it does exist: there is a hyperkähler metric making S a hyperkähler manifold.

Example 10.4. Let S be a complex K3 surface. The symmetric power $S^{(n)}$ is singular, but admits a resolution of singularities

$$r : S^{[n]} \to S^{(n)},$$

where $S^{[n]}$ is the Hilbert scheme of length n subschemes of S. The manifold $S^{[n]}$ is hyperkähler.

Example 10.5. Let

$$T = \mathbb{C}^2/\mathbb{Z}^4.$$

The Albanese morphism is

$$a : T^{[n+1]} \to T, \qquad (x_0, \ldots, x_n) \mapsto \sum_i x_i.$$

The generalized Kummer variety

$$K_n(T) = a^{-1}(0)$$

is hyperkähler.

Example 10.6. There are O'Grady's exceptional hyperkähler manifolds of dimensions 6 and 10.

Today we will be more interested in the Hilbert scheme example. In fact, these are all the examples we know.

Mukai flops.

Today we discuss birational maps of these manifolds. Birational maps of manifolds with trivial canonical bundle are similar to flops. They are modeled by so-called Mukai flops.

Mukai flops are local models of a birational transformation between symplectic manifolds.

Recall the cotangent space construction, which gives a symplectic manifold. Look at the cotangent bundle of $\mathbb{P}^n$. The idea is to contract the zero section of this manifold, and then blow it up in a different direction.

Let

$$\dim_{\mathbb{C}}(V) = k + 1$$

and

$$X = \operatorname{Tot}\big(\Omega^1_{\mathbb{P}(V)}\big).$$

To contract the zero section we use the Euler sequence

$$0 \to \Omega^1_{\mathbb{P}(V)} \to V^* \otimes \mathcal{O}_{\mathbb{P}(V)}(-1) \to \mathcal{O}_{\mathbb{P}(V)} \to 0.$$

Also

$$\mathcal{O}_{\mathbb{P}(V)}(-1) \hookrightarrow V \otimes \mathcal{O}_{\mathbb{P}(V)}.$$

Hence

$$\Omega^1_{\mathbb{P}(V)} \hookrightarrow \operatorname{End}(V) \otimes \mathcal{O}_{\mathbb{P}(V)}.$$

Remark 10.7. Here is the point of the sequence calculation. At a point $[u] \in \mathbb{P}(V)$, the cotangent space is

$$\Omega^1_{\mathbb{P}(V),[u]} \cong \mathrm{Hom}(V/\mathbb{C}u, \mathbb{C}u).$$

Such a map can be viewed as an endomorphism A of V by first quotienting by $\mathbb{C}u$ and then landing in $\mathbb{C}u$. Thus

$$\mathrm{im}(A) \subset \mathbb{C}u, \qquad A(u) = 0.$$

In particular $A^2 = 0$ and $\mathrm{rk}(A) \leq 1$. Conversely, an endomorphism satisfying these conditions gives a cotangent vector at $[u]$.

Therefore

$$X = \left\{([u], A) \in \mathbb{P}(V) \times \mathrm{End}(V) \middle| \mathrm{im}(A) \subset \mathbb{C}u,\ A^2 = 0\right\}.$$

The zero section is $\mathbb{P}(V) \subset X$. The projection

$$\pi : X \to \mathrm{End}(V)$$

contracts the zero section to the origin. Let

$$Y = \pi(X) = \{A \in \mathrm{End}(V) \mid \mathrm{rk}(A) \leq 1,\ A^2 = 0\}.$$

Analogously, put

$$X' = \left\{([u], B) \in \mathbb{P}(V^*) \times \mathrm{End}(V^*) \middle| \mathrm{im}(B) \subset \mathbb{C}u,\ B^2 = 0\right\}$$

and

$$Y' = \pi'(X') = \{B \in \mathrm{End}(V^*) \mid \mathrm{rk}(B) \leq 1,\ B^2 = 0\}.$$

Then $Y \cong Y'$ via

$$A \mapsto A^*.$$

Hence we get a diagram

$$\begin{tikzcd}
X = \mathrm{Tot}\big(\Omega^1_{\mathbb{P}(V)}\big) \arrow[rr, dashed, "\mu"] \arrow[dr, "\pi"'] & & X' = \mathrm{Tot}\big(\Omega^1_{\mathbb{P}(V^*)}\big) \arrow[dl, "\pi'"] \\
& Y \cong Y' &
\end{tikzcd}$$

where

$$\mu([u], A) = ([\ker(A)], A^*).$$

The upshot is this. We blow down the zero section, which is a Lagrangian subvariety, and then perform the other blow-up using the dual vector space V^*.

Use the above diagram to construct a local model of a birational map. Let

$$U = Y \cap B_\varepsilon(0),$$

where

$$B_\varepsilon(0) \subset \mathrm{End}(V)$$

is a small ball. Then

$$\begin{tikzcd}
\pi^{-1}(U) \times \Delta^d \arrow[rr, dashed, "\mu \times \mathrm{id}"] \arrow[dr, "\pi \times \mathrm{id}"'] & & (\pi')^{-1}(U) \times \Delta^d \arrow[dl, "\pi' \times \mathrm{id}"] \\
& U \times \Delta^d &
\end{tikzcd}$$

where

$$\Delta = \{z \in \mathbb{C} : |z| < 1\}.$$

Definition 10.8. Let

$$\varphi : M \dashrightarrow M'$$

be a bimeromorphic map between complex manifolds. We say that φ is *Mukai's elementary transformation in codimension k* if there exists a commutative diagram

$$\begin{tikzcd}
M \arrow[rr, dashed, "\varphi"] \arrow[dr, "p"'] & & M' \arrow[dl, "p'"] \\
& N &
\end{tikzcd}$$

where p and p' are proper small bimeromorphic contractions, and for every

$$x \in N_{\mathrm{sing}}$$

there exists an open neighborhood of x over which the diagram is isomorphic to the local model

$$U \times \Delta^d.$$

Remark 10.9. We prefer to describe the situation in terms of contractions, as in the diagram

$$\begin{tikzcd}
M \arrow[rr, dashed, "\varphi"] \arrow[dr, "p"'] & & M' \arrow[dl, "p'"] \\
& N &
\end{tikzcd}$$

When we contract something of codimension bigger than 1, the contracted space N will be singular. This is why the definition refers to N_{sing}. Further, N_{sing} is smooth. So N is singular, but not too badly.

rather than in terms of blow-ups, as in a diagram

$$\begin{tikzcd}
& \widetilde{M} \arrow[dl] \arrow[dr] & \\
M \arrow[rr, dashed] & & M'.
\end{tikzcd}$$

Hu–Yau conjecture.

Let

$$\varphi : M \dashrightarrow M'$$

be a bimeromorphic map between hyperkähler manifolds. Hu and Yau conjectured that, up to removing from M and M' some subvarieties of codimension strictly bigger than 2, the map φ is a composition of several Mukai elementary transformations.

Definition 10.10. Let

$$\psi : M \dashrightarrow M'$$

be a bimeromorphic map between two complex manifolds. We say that ψ is a *Hu–Yau transformation* if there exist closed subvarieties

$$Z \subset M, \qquad Z' \subset M',$$

of codimension strictly bigger than 2 such that

$$\psi : M \setminus Z \dashrightarrow M' \setminus Z'$$

is a Mukai elementary transformation in codimension two.

Theorem 10.11 (Wierzba–Wiśniewski)**.** *Assume that M is a symplectic manifold of dimension 4. Let*

$$\pi : M \to N$$

be a small proper bimeromorphic contraction such that N is quasiprojective. Then the exceptional locus of π is a disjoint union of several copies of $\mathbb{P}^2$, and π is locally analytically isomorphic to the contraction of the zero section in

$$\mathrm{Tot}\big(\Omega^1_{\mathbb{P}^2}\big).$$

This is the first step of the Mukai flop. It is the contraction of $\mathbb{P}^2$, which sits in its cotangent bundle as the zero section. Thus this theorem is exactly the picture explained above.

Theorem 10.12 (Wierzba–Wiśniewski)**.** *The Hu–Yau conjecture holds in dimension four. More precisely, if*

$$\varphi : M \dashrightarrow M'$$

is a birational map between projective hyperkähler fourfolds, then φ is a composition of several Mukai flops.

The approach to this conjecture is different. Instead of using the MMP, one uses the wall-and-chamber structure. Let us state the result first.

Theorem 10.13. *[Amerik–Soldatenkov–Verbitsky] Let M and M' be compact hyperkähler manifolds, and let*

$$\varphi : M \dashrightarrow M'$$

be a bimeromorphic map which is not regular in codimension two. Then φ is a composition of a finite number of Hu–Yau transformations.

Idea of the proof.

Wall crossing.

To describe birational transformations of hyperkähler manifolds, we look at Kähler cones. Look at $H^2(M)$ with its Beauville–Bogomolov–Fujiki form q. The restriction of q to $H^{1,1}(M)$ has signature

$$(1, b_2 - 3).$$

Let $\mathcal{K}_M$ be the Kähler cone of M, and let $\mathcal{C}_M$ be the connected component of the positive cone

$$\{x \in H^{1,1}(M) : q(x) > 0\}$$

which contains $\mathcal{K}_M$.

Theorem 10.14 (Amerik–Verbitsky)**.** *There exists a locally finite collection of hyperplanes*

$$W_i \subset \mathcal{C}_M$$

given by orthogonal complements to the MBM classes, such that for any bimeromorphic map

$$\varphi : M \dashrightarrow M'$$

with M' hyperkähler, the set

$$\varphi^* \mathcal{K}_{M'}$$

is a connected component of

$$\mathcal{C}_M \setminus \bigcup_i W_i.$$

The union of such chambers is the birational cone.

Definition 10.15. A *bimeromorphic map*

$$\varphi : M \dashrightarrow M'$$

between two hyperkähler manifolds is a *wall-crossing flop* if

$$\mathcal{K}_M \quad \text{and} \quad \varphi^* \mathcal{K}_{M'}$$

share a common wall.

For any wall-crossing flop there exists a diagram

$$\begin{tikzcd}
M \arrow[rr, dashed, "\varphi"] \arrow[dr, "p"'] & & M' \arrow[dl, "p'"] \\
& N &
\end{tikzcd}$$

where N is a singular symplectic variety. Now N may be more singular than before. The maps p and p' are proper small bimeromorphic contractions.

Since the collection of walls in $\mathcal{C}_M$ is locally finite, any bimeromorphic map

$$\varphi : M \dashrightarrow M'$$

between hyperkähler manifolds is a composition of a finite number of wall-crossing flops.

Example 10.16. Let

$$S \subset \mathbb{P}^3$$

be a quartic K3 surface containing a line C. Take

$$M = S^{[3]}.$$

Picard rank 1 is not very interesting from the point of view of wall-and-chamber structure. The pencil of planes passing through C defines an elliptic fibration

$$f : S \to \mathbb{P}^1$$

with fiber F. In the basis $[C], [F]$, the intersection form on $\mathrm{Pic}(S)$ is

$$\begin{pmatrix} -2 & 3 \\ 3 & 0 \end{pmatrix}.$$

Moreover

$$\mathrm{Pic}(M) \cong \mathrm{Pic}(S) \oplus \mathbb{Z}\delta,$$

where

$$\delta = [\Delta]/2,$$

and Δ is the exceptional divisor of the contraction

$$S^{[3]} \to S^{(3)}.$$

Thus δ defines a wall in the wall-and-chamber structure.

There is a picture of the birational ample cone of $S^{[3]}$, as seen in the Poincare hyperbolic plane.

Some MBM classes on M are:

- δ, the exceptional divisor of the contraction $S^{[3]} \to S^{(3)}$;
- $\alpha = [C]$, the divisor of subschemes that intersect C;

- $\epsilon = 2[F] - \delta$, the closure of the image of
$$\mathrm{Hilb}^2(S \setminus F) \times S \dashrightarrow S^{(3)};$$
- $\gamma = 2[C] - \delta$, the set of subschemes supported on C, which is isomorphic to
$$C^{(3)} \cong \mathbb{P}^3.$$

We have decomposed everything into a sequence of wall-crossing flops. We need to show that those wall-crossing flops are in fact Hu–Yau transformations.

Consider a wall-crossing flop

$$\begin{tikzcd}
M \arrow[rr, dashed, "\varphi"] \arrow[dr, "p"'] & & M' \arrow[dl, "p'"] \\
& N. &
\end{tikzcd}$$

Theorem 10.17 (Bakker–Lehn)**.** *There exists a small locally trivial deformation of the above diagram which makes M, M', and N projective.*

Theorem 10.18 (Kaledin)**.** *The symplectic variety N admits a finite locally closed stratification such that the strata are symplectic manifolds.*

Remark 10.19. The open stratum of the Kaledin stratification is the smooth locus of N. The symplectic forms on the strata glue to a nice Poisson structure.

Consider a wall-crossing flop

$$\begin{tikzcd}
M \arrow[rr, dashed, "\varphi"] \arrow[dr, "p"'] & & M' \arrow[dl, "p'"] \\
& N &
\end{tikzcd}$$

such that the indeterminacy locus of φ has a component of codimension 2. Let
$$N' \subset N_{\mathrm{sing}}$$
be a stratum of codimension 4, and let $x \in N'$. Use the Poisson structure on N to construct a slice
$$S \subset N$$
passing through x, transverse to N', and symplectic.

Then
$$p_S : p^{-1}(S) \to S$$
is a small contraction of symplectic fourfolds. Applying the theorem of Wierzba–Wiśniewski, p_S is the contraction of a copy of $\mathbb{P}^2$. Thus over N' we have a $\mathbb{P}^2$-bundle for both p and p'. It follows that φ is a Hu–Yau transformation. This completes the proof.

11. Holonomy of the Obata connection on Joyce hypercomplex manifolds

Speaker.

Beatrice Brienza

Title.

Holonomy of the Obata connection on Joyce hypercomplex manifolds

Abstract.

The Obata connection on a hypercomplex manifold is the unique torsion-free connection that preserves the hypercomplex structure. Up to taking the product with a torus, Joyce constructed left-invariant hypercomplex structures on compact semisimple Lie groups and certain related homogeneous spaces. Soldatenkov showed in 2011 that every Joyce hypercomplex structure on SU(3) has Obata holonomy equal to the quaternionic general linear group and it was expected that the same should hold for all Joyce hypercomplex manifolds. For all such group manifolds except for SU(2n+1), we will show that the holonomy group is strictly contained in the quaternionic general linear group. The case of SU(2n+1) is more subtle, however for every n > 1, there still exist infinitely many Joyce hypercomplex structures with Obata holonomy strictly contained in the quaternionic general linear group and at least one with full holonomy on SU(5). This talk is based on a joint work with Udhav Fowdar, Giovanni Gentili and Luigi Vezzoni.

Notes.

Theorem 11.1 (Berger)**.** *Let M be an oriented simply connected Riemannian manifold of dimension n. Assume that M is irreducible and not locally symmetric. Then its Riemannian holonomy group must be one of the following:*

$$\mathrm{SO}(n), \quad \mathrm{U}(m), \quad \mathrm{SU}(m), \quad \mathrm{Sp}(m),$$

$$\mathrm{Sp}(m)\,\mathrm{Sp}(1), \quad G_2, \quad \mathrm{Spin}(7).$$

Here $n = 2m$ for the unitary cases, $n = 4m$ for the symplectic cases, $n = 7$ for G_2, and $n = 8$ for Spin(7)*.*

At the time this list was a list of possible irreducible Riemannian holonomies. Examples of Riemannian manifolds with holonomy G_2 and Spin(7) were found by Bryant–Salamon in 1989.

Possible generalization.

Consider torsion-free connections which are not metric. In fact, this idea was discussed in the same paper by Berger. The theorem actually still makes sense when the signature of the metric is not Riemannian.

Merkulov–Schwachhöfer gave in 1999 the complete list of all possible irreducible holonomy groups for torsion-free connections.

What about the Obata connection? Its holonomy group fits this scenario.

Hypercomplex geometry in a nutshell.

Let M be a manifold. A *hypercomplex structure* on M is a triple (I, J, K) of complex structures on M satisfying the quaternionic relations:

$$I^2 = J^2 = K^2 = IJK = -1.$$

Then

$$\dim_{\mathbb{R}} M = 4n.$$

We call (M, I, J, K) a *hypercomplex manifold*.

Example 11.2. The manifold $\mathbb{H}^n$, with the complex structures given by left multiplication by i, j, and k, is a hypercomplex manifold.

Question.

Is any hypercomplex manifold locally isomorphic to

$$(\mathbb{H}^n, i, j, k)?$$

For each

$$L \in \{I, J, K\},$$

we have

$$(M, L) \cong_{\mathrm{loc}} (\mathbb{C}^{2n}, i).$$

That is, each complex structure is locally standard by the Newlander–Nirenberg theorem. However, in general

$$(M, I, J, K) \not\cong_{\mathrm{loc}} (\mathbb{H}^n, i, j, k).$$

Thus there is no direct hypercomplex version of the Newlander–Nirenberg theorem.

Theorem 11.3 (Obata)**.** *On TM there exists a unique torsion-free connection ∇ which preserves I, J, and K.*

Furthermore,

$$(M, I, J, K) \cong_{\mathrm{loc}} (\mathbb{H}^n, i, j, k)$$

if and only if

$$\mathrm{Hol}^0(\nabla) = \{e\}.$$

Examples of Obata-flat manifolds.

1. $(\mathbb{H}^n, i, j, k)$.
2. Many hypercomplex nilmanifolds. There are many examples among the 2-step ones; see for instance DF, ABB, and BD.
3. Hopf surfaces

   $$\mathbb{H} \setminus \{0\} / \langle q \rangle,$$

   where

   $$q \in \mathbb{H}^* \quad \text{and} \quad |q| > 1.$$

The holonomy of the Obata connection satisfies

$$\mathrm{Hol}(\nabla) \subseteq \mathrm{GL}(n, \mathbb{H}).$$

Only three subgroups of $\mathrm{GL}(n, \mathbb{H})$ appear in the Merkulov–Schwachhöfer list.

1. $\mathrm{Sp}(n)$. We have

   $$\mathrm{Hol}(\nabla) \subseteq \mathrm{Sp}(n)$$

   if and only if the Obata connection is the Levi-Civita connection of a metric. Equivalently, (g, I, J, K) is hyperkähler.
2. $$\mathrm{SL}(n, \mathbb{H}) = [\mathrm{GL}(n, \mathbb{H}), \mathrm{GL}(n, \mathbb{H})].$$

   Hypercomplex nilmanifolds have Obata holonomy contained in $\mathrm{SL}(n, \mathbb{H})$; see BDV.
3. The most mysterious case is $\mathrm{GL}(n, \mathbb{H})$. Soldatenkov showed that $\mathrm{SU}(3)$, with any compact invariant hypercomplex structure, has Obata holonomy

   $$\mathrm{GL}(2, \mathbb{H}).$$

   In fact, this is our first example of a Joyce hypercomplex manifold.

Joyce's decomposition.

Let G be a compact semisimple Lie group. Let

$$T \subseteq G$$

be a maximal torus, and put

$$\mathfrak{g} = \mathrm{Lie}(G), \qquad \mathfrak{t} = \mathrm{Lie}(T).$$

Let $r = \mathrm{rk}(G)$, and let Δ be a system of ordered roots with respect to $\mathfrak{t}^{\mathbb{C}}$.

Step 1.

Pick a maximal positive root

$$\alpha \in \Delta.$$

Set

$$\widetilde{\partial}_1 = \mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha} \oplus [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \cong \mathfrak{sl}(2, \mathbb{C}),$$

and

$$\partial_1 = \mathfrak{g} \cap \widetilde{\partial}_1 \cong \mathfrak{su}(2).$$

Then

$$\mathfrak{g}_0 = \mathfrak{g} = \partial_1 \oplus \mathfrak{c}_1 \oplus \mathfrak{f}_1,$$

where $\mathfrak{c}_1$ is the centralizer of ∂_1, and $\mathfrak{f}_1$ is the orthogonal complement of

$$\partial_1 \oplus \mathfrak{c}_1$$

with respect to the Killing form B.

Step 2.

Write

$$\mathfrak{c}_1 = \mathfrak{g}_1 \oplus \mathfrak{b}_1.$$

Apply Step 1 to $\mathfrak{g}_1$.

Step $m + 1$.

At the final step the centralizer is abelian:

$$\mathfrak{c}_{m+1} = \mathfrak{b}_{m+1}.$$

Thus the inductive process stops when the centralizer is only abelian.

We obtain

$$\mathfrak{g} = \mathfrak{b} \oplus \bigoplus_{j=1}^{m} \partial_j \oplus \bigoplus_{j=1}^{m} \mathfrak{f}_j.$$

Here

$$\mathfrak{b} = \mathfrak{b}_1 \oplus \cdots \oplus \mathfrak{b}_{m+1}$$

is an abelian subalgebra of dimension $r - m$. Each ∂_j is a subalgebra isomorphic to $\mathfrak{su}(2)$. Moreover,

$$[\mathfrak{b}, \partial_j] = 0, \qquad [\partial_i, \partial_j] = 0 \quad (i \neq j).$$

The spaces $\mathfrak{f}_j$ have dimension $4k_j$, and

$$[\partial_i, \mathfrak{f}_j] = 0 \quad (i < j),$$

while

$$[\partial_i, \mathfrak{f}_i] \subseteq \mathfrak{f}_i.$$

Thus $\partial_i \cong \mathfrak{su}(2)$ acts on $\mathfrak{f}_i$ through the standard action on $\mathbb{C}^{2k_i}$. This is the Joyce decomposition of $\mathfrak{g}$.

Some examples.

Consider Sp(2). We write

$$\mathfrak{sp}(2) = \{A \in \mathfrak{gl}(2,\mathbb{H}) \mid A + \overline{A}^t = 0\}.$$

Thus an element of $\mathfrak{sp}(2)$ has the form

$$A = \begin{pmatrix} x & y \\ -\overline{y} & z \end{pmatrix}, \qquad x, z \in \mathrm{Im}(\mathbb{H}).$$

In this case one has

$$\mathfrak{sp}(2) = \partial_1 \oplus \partial_2 \oplus \mathfrak{f}_1.$$

Here

$$\partial_1 = \left\{ \begin{pmatrix} x & 0 \\ 0 & 0 \end{pmatrix} \right\},$$

with the remaining pieces to be filled in.

One may also consider colored Dynkin diagrams. In OP, 1998, the colors account for the Joyce decomposition. Using this other technique one obtains the same decomposition, namely

$$\mathfrak{sp}(k) = \partial_1 \oplus \mathfrak{c}_1 \oplus \mathfrak{f}_1.$$

One can also obtain

$$\mathfrak{sp}(k) = \partial_1 \oplus \mathfrak{sp}(k-1) \oplus (\text{missing term}).$$

Hypercomplex structures.

Let G be compact and semisimple, with

$$\mathrm{rk}(G) = r.$$

Consider

$$\mathbb{T}^{2m-r} \times G.$$

Its Lie algebra is

$$(2m-r)\mathfrak{u}(1) \oplus \mathfrak{g} = (2m-r)\mathfrak{u}(1) \oplus \mathfrak{b} \oplus \bigoplus_{j=1}^{m} \partial_j \oplus \bigoplus_{j=1}^{m} \mathfrak{f}_j.$$

Using the Joyce decomposition and additional formulas, Joyce constructs a hypercomplex structure (I, J, K) on this Lie algebra. This is a hypercomplex structure in the sense of Joyce, 1992.

Theorem 11.4 (DT, BGP)**.** *Joyce hypercomplex manifolds are in one-to-one correspondence with compact Lie groups with left-invariant hypercomplex structures.*

Let G be simple. The dimension of the torus can be checked from the list of Spindel and collaborators.

dim	example
4	$\mathrm{SU}(2) \times S^1$, Obata-flat, simplest case
8	$\mathrm{SU}(3), \mathrm{Hol}(\nabla) = \mathrm{GL}(2,\mathbb{H})$
$4n$	Joyce hypercomplex manifolds

Folklore conjecture.

Any Joyce hypercomplex manifold in the list of Spindel and collaborators, except for

$$S^1 \times \mathrm{SU}(2),$$

has full Obata holonomy, that is,

$$\mathrm{Hol}(\nabla) = \mathrm{GL}(n, \mathbb{H}).$$

Theorem 11.5 (BFGV)**.** *For any Joyce hypercomplex manifold in the list of Spindel and collaborators, with the exception of* $\mathrm{SU}(2k+1)$*, one has*

$$\mathrm{Hol}(\nabla) \subset \mathrm{GL}(n, \mathbb{H}).$$

Moreover, this inclusion is strict.

Idea of proof.

We may assume that $\dim M \geq 8$. Let

$$M = \mathbb{T}^{2m-r} \times G.$$

Then

$$(2m-r)\mathfrak{u}(1) \oplus \mathfrak{g}$$

decomposes as

$$(2m-r)\mathfrak{u}(1) \oplus \mathfrak{b} \oplus \bigoplus_{j=1}^{m} \partial_j \oplus \bigoplus_{j=1}^{m} \mathfrak{f}_j.$$

There are further formulas describing the corresponding decomposition of the Lie algebra.

Step 1.

Lemma 11.6 (BFGV)**.** *Let*

$$M = \mathbb{T}^{2m-r} \times G,$$

and let (I, J, K) *be a Joyce hypercomplex structure on* M*. Equivalently,* (I, J, K) *is a left-invariant hypercomplex structure. Then*

$$\nabla_X e_1 = \begin{cases} -X, & X \in \mathfrak{h}_j \oplus \mathfrak{f}_j, \\ 0, & \textit{otherwise.} \end{cases}$$

More details of the formula remain to be filled in.

Step 2.

By Step 1, the holonomy representation is reducible. Therefore the holonomy group cannot be $\mathrm{GL}(n, \mathbb{H})$. In fact, if $p \in M$ and γ is a loop based at p, then the parallel transport P_γ has the form

$$P_\gamma = \big(\text{missing}\big).$$

Step 3.

Look at every simple Lie group and check whether it has a trivial $\mathfrak{g}_j$ summand. All of them except $\mathrm{SU}(2k+1)$ have at least one such summand.

We have already seen that

$$\mathfrak{sp}(k) = \partial_1 \oplus \mathfrak{sp}(k-1) \oplus \mathfrak{f}_1.$$

By induction, in the case $k-1$ we start from

$$\mathfrak{sp}(2) = \partial_1 \oplus \partial_2 \oplus \mathfrak{f}_1, \qquad \mathfrak{f}_2 = \{0\}.$$

Assume that $\mathfrak{sp}(k-1)$ has one trivial $\mathfrak{g}_j$ summand. Since $\mathfrak{sp}(k-1)$ appears in the decomposition of $\mathfrak{sp}(k)$, we look for the $\mathfrak{f}_j$ component inside the $\mathfrak{sp}(k-1)$ factor.

The $\mathrm{SU}(2k+1)$ **case.**

For $k = 1$, this is the Soldatenkov case. What happens when $k > 1$?

For $k = 2$, the Joyce decomposition of $\mathfrak{su}(5)$ is

$$\mathfrak{su}(5) = \partial_1 \oplus \underbrace{\mathfrak{su}(3)}_{\langle e_1\rangle \oplus \partial_1 \oplus \mathfrak{f}_2} \oplus \underbrace{\mathfrak{u}(1)}_{\langle \delta_1 \rangle} \oplus \mathfrak{f}_1.$$

The point is that we can find a copy of $\mathfrak{su}(3)$ inside.

Theorem 11.7 (BFGV)**.** *If the Joyce hypercomplex structure has block form*

$$(I, J, K) = \begin{pmatrix} a & 0 \\ b & d \end{pmatrix},$$

then

$$\mathrm{Hol}(\nabla) \subset \mathrm{GL}(n, \mathbb{H}).$$

Moreover, this inclusion is strict.

There is a decomposition of the form

$$\mathfrak{su}(5) = \mathfrak{b}_1 \oplus \mathfrak{f}_1 \oplus \mathfrak{b}_2 \oplus \mathfrak{f}_2 \oplus (\text{missing term}).$$

The point is that $\mathfrak{su}(3)$ is not only a subalgebra, but a hypercomplex subalgebra. It is preserved by the Obata connection. Thus we get a reduction of the holonomy. The same procedure works for other $\mathrm{SU}(2k+1)$, but not for $2k + 1 < 5$.

12. Bogomolov–Guan precursors and quasi-diagonals in a product of elliptic curves

Speaker.

Misha Verbitsky

Title.

Bogomolov–Guan precursors and quasi-diagonals in a product of elliptic curves

Joint work.

Leila Abubakarova and Alexandra Kuznetsova.

The main point of the talk is that quasidiagonals in products of elliptic curves control projective subvarieties of the Bogomolov–Guan precursor. This gives a concrete way to study the precursor through the geometry of elliptic curves and their products.

Definition 12.1. Let L be an ample line bundle on an elliptic curve E. Let

$$p_i : E^n \to E$$

denote the projection to the ith component. A *quasidiagonal* is a curve

$$S \subset E^n$$

such that, for all i and j, the line bundle

$$V_{i,j} = p_i^* L\big|_S \otimes \left(p_j^* L\big|_S\right)^{-1}$$

is torsion. In other words, there exists $k > 0$ such that

$$V_{i,j}^{\otimes k} = \mathcal{O}_S.$$

Remark 12.2. For $n = 2$, a curve

$$S \subset E \times E$$

is a correspondence from E to E. Since both factors are the same elliptic curve E, the two projections restrict to maps

$$p_1|_S, p_2|_S : S \to E.$$

Thus both pullbacks of L live on the same curve S, and can be compared. The quasidiagonal condition does not use a pushforward; it only says that the ratio of these two pullbacks is torsion.

The theorem says there are at most countably many quasidiagonals for any given pair (E, L).

Example 12.3. Let

$$\nu : E \to E$$

be an automorphism of order d. Given a line bundle L_1 on E, set

$$L = \bigotimes_{i=0}^{d-1} (\nu^i)^* L_1.$$

Then L is ν-invariant. Therefore the graph

$$\Gamma_\nu \subset E^2$$

of ν is a quasidiagonal for (E, L).

Open questions.

(1) Can a quasidiagonal have geometric genus $p_g(C) > 1$? There are examples which are elliptic and singular.

(2) Call a line bundle $L \in \mathrm{Pic}^n(E)$ commensurable with $\mathcal{O}(n[e])$ if

$$L \otimes \mathcal{O}(n[e])^{-1}$$

is torsion. Is it true that there are no restricted quasidiagonals on (E, L) when L is not commensurable with $\mathcal{O}(n[e])$?

(3) Consider E^n, where n is odd and E does not admit complex multiplication. Are there any restricted quasidiagonals in E^n, for some ample line bundle L on E?

(4) In the case $n = 3$ and

$$E = \mathbb{C}/\mathbb{Z}[\zeta],$$

where ζ is a primitive third root of unity, there is an example of a restricted quasidiagonal in E^3. Does E^3 contain any restricted quasidiagonal except this one?

Remark 12.4. These questions may be accessible projects for students learning algebraic geometry.

Hilbert schemes.

Definition 12.5. Let M be a complex surface. The *Hilbert scheme of points* $M^{[n]}$ is a natural smooth resolution of the nth symmetric power $M^{(n)}$.

Remark 12.6. If M is holomorphically symplectic, then $M^{[n]}$ is also holomorphically symplectic. This follows from Serre duality.

Remark 12.7. The Hilbert scheme of a K3 surface is simply connected and holomorphically symplectic. The Hilbert scheme of a torus T is not simply connected, but the fiber of its Albanese map

$$T^{[n]} \to T$$

has finite fundamental group. The universal cover of this fiber is called a generalized Kummer variety. In this way one obtains the two main examples of simply connected holomorphically symplectic manifolds.

Todorov conjectured that any compact simply connected holomorphically symplectic manifold should be Kähler. Guan produced counterexamples: compact simply connected holomorphically symplectic manifolds which are not Kähler. Bogomolov then used Guan's example to produce what are now called Bogomolov–Guan manifolds.

Definition 12.8. Let L be a line bundle on an elliptic curve E with

$$c_1(L) \neq 0.$$

Let $\widetilde{S}$ be the $\mathbb{C}^*$-bundle on E obtained by removing the zero section:

$$\widetilde{S} = \mathrm{Tot}(L) \setminus 0.$$

Fix a complex number λ with $|\lambda| > 1$, and let

$$h_\lambda : \widetilde{S} \to \widetilde{S}$$

be the corresponding homothety. The quotient

$$\widetilde{S}/\langle h_\lambda \rangle$$

is called a *primary Kodaira surface*, or *Kodaira–Thurston surface*, or *Kodaira surface*.

Remark 12.9. The Kodaira surface is a principal elliptic fibration

$$p : \mathrm{Kod} \to E,$$

with fiber another elliptic curve E_1. It is holomorphically symplectic. Therefore

$$\mathrm{Kod}^{[n]}$$

is also holomorphically symplectic, but it is not simply connected.

Applying p to each component gives a map

$$\mathrm{Kod}^{(n)} \to E^{(n)}.$$

Composing with the Hilbert–Chow resolution

$$\mathrm{Kod}^{[n]} \to \mathrm{Kod}^{(n)}$$

and with addition on the elliptic curve gives a holomorphic map

$$\mathrm{Kod}^{[n]} \to E.$$

Let

$$\mathcal{D} \subset \mathrm{Kod}^{[n]}$$

be the fiber over $0 \in E$. If Ω is the holomorphic symplectic form on $\mathrm{Kod}^{[n]}$, then

$$T\mathcal{D}^{\perp\Omega} \subset T\mathcal{D}.$$

The corresponding rank-one distribution is the characteristic foliation.

The leaf space of this foliation is holomorphically symplectic, but may have orbifold singularities. Equivalently, in the precursor model one considers

$$D \subset \mathrm{Kod}^{n+1}$$

defined by

$$D = \left\{ (z_1, \ldots, z_{n+1}) \mid \sum_i p(z_i) = 0 \right\}.$$

The diagonal E_1-action on D is by leaves of the characteristic foliation. The quotient

$$D/E_1$$

is the restricted *Bogomolov–Guan precursor.*

Theorem 12.10. *Let*

$$P : X \to B$$

be the Lagrangian fibration on the Bogomolov–Guan precursor, and let

$$Z \subset X$$

be an irreducible subvariety. Then Z is projective if and only if

$$P(Z) \subset B \subset E^{n+1}$$

is either a point or a quasidiagonal.

Definition 12.11. A *nilmanifold* is a smooth manifold equipped with a transitive action of a nilpotent Lie group.

Remark 12.12. All nilmanifolds are obtained as quotients

$$M = G/\Gamma,$$

where G is a nilpotent Lie group and $\Gamma \subset G$ is a discrete subgroup. The Bogomolov–Guan precursor is particularly easy to study because it is a complex nilmanifold.

Definition 12.13. Let $\mathfrak{g}$ be a real Lie algebra. An *integrable complex structure* on $\mathfrak{g}$ is a subalgebra

$$\mathfrak{g}^{1,0} \subset \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$$

such that

$$\mathfrak{g}^{1,0} \oplus \overline{\mathfrak{g}^{1,0}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}.$$

Remark 12.14. Such a decomposition is exactly the same thing as a complex structure on the Lie algebra. Indeed, one lets the complex structure act by $\sqrt{-1}$ on $\mathfrak{g}^{1,0}$ and by $-\sqrt{-1}$ on

$$\overline{\mathfrak{g}^{1,0}}.$$

The condition that $\mathfrak{g}^{1,0}$ is a Lie subalgebra is precisely the integrability condition.

Definition 12.15. A *complex nilmanifold* is a nilmanifold

$$M = G/\Gamma$$

equipped with a complex structure such that the projection

$$(G, I) \to M$$

is holomorphic, where I is a left-invariant complex structure on G.

Theorem 12.16 (Bishop)**.** *Let (M, I) be a compact complex manifold, and let ω be a Hermitian form. Fix $a \in \mathbb{R}$, and let*

$$\mathscr{Z}_a$$

be the set of k-cycles

$$Z \subset M$$

such that

$$\int_Z \omega^k \leq a.$$

Then $\mathscr{Z}_a$ is compact.

Remark 12.17. It is not clear at this point why this is relevant.

Definition 12.18. A Hermitian form ω on a complex manifold is *SKT* or *pluriclosed* if

$$dd^c\omega = 0.$$

Theorem 12.19 (Bishop-Gromov)**.** *Let $p \in M$ and $0 \leq t \leq i(p)$. Denote by $B_t(p)$ the ball of radius t centered at p, and by $B_{t,k}$ the ball of radius t in the space of constant curvature k. If $\mathrm{Ric} \geq k$, then*

$$\frac{\mathrm{Vol}(B_t(p))}{\mathrm{Vol}(B_{t,k})}$$

is a nonincreasing function of t.

Remark 12.20. This is the Bishop-Gromov volume comparison theorem, not Bishop's compactness theorem for cycles.

Definition 12.21. A compact complex variety X is *Moishezon* if it is bimeromorphic to a projective variety.

Campana's work is foundational in the birational study of rational curves and rational connectedness. The guiding idea is to connect points of a manifold using rational curves, or more generally families of curves of varying type. The notes also use a theorem about a holomorphic map

$$M \to B$$

with Moishezon fibers and Moishezon base. The precise relation between that theorem and rational connectedness is left open here.

Definition 12.22. A Hermitian form ω on a complex manifold is *SKT* or *pluriclosed* if

$$dd^c\omega = 0.$$

Definition 12.23. A *nilmanifold* is a smooth manifold equipped with a transitive action of a nilpotent Lie group.

Remark 12.24. All nilmanifolds are obtained as quotients

$$M = G/\Gamma,$$

where G is a nilpotent Lie group and $\Gamma \subset G$ is a discrete subgroup. The Bogomolov–Guan precursor is particularly easy to study because it is a complex nilmanifold.

Definition 12.25. Let $\mathfrak{g}$ be a real Lie algebra. An *integrable complex structure* on $\mathfrak{g}$ is a subalgebra

$$\mathfrak{g}^{1,0} \subset \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$$

such that

$$\mathfrak{g}^{1,0} \oplus \overline{\mathfrak{g}^{1,0}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}.$$

Remark 12.26. Such a decomposition is exactly the same thing as a complex structure on the Lie algebra. Indeed, one lets the complex structure act by $\sqrt{-1}$ on $\mathfrak{g}^{1,0}$ and by $-\sqrt{-1}$ on $\overline{\mathfrak{g}^{1,0}}$. The condition that $\mathfrak{g}^{1,0}$ is a Lie subalgebra is precisely the integrability condition.

Definition 12.27. A *complex nilmanifold* is a nilmanifold

$$M = G/\Gamma$$

equipped with a complex structure such that the projection

$$(G, I) \to M$$

is holomorphic, where I is a left-invariant complex structure on G.

Theorem 12.28. *There are at most countably many quasidiagonals for any given pair (E, L).*

13. Quantum cohomology and irrationality of Gushel-Mukai fourfolds

Speaker.

Vladimiro Benedetti

Title.

Quantum cohomology and irrationality of Gushel-Mukai fourfolds

Abstract.

Gushel-Mukai fourfolds behave very similarly to cubic fourfolds: they are Fano manifolds of K3-type, one can associate to them a (general) IHS manifold, their rationality is conjecturally controlled by their cohomology. In this talk, I will explain how one can get the irrationality of very general Gushel-Mukai fourfolds through the theory of atoms. This will be done, as for cubics, by computing the small quantum cohomology of Gushel-Mukai fourfolds. Moreover, thanks to a suitable deformation of the small quantum cohomology ring, we will also deduce that a rational Gushel-Mukai fourfold has the same rational cohomology as some K3 surface. This is a joint work with L. Manivel and N. Perrin.

AI-generated summary.

The talk studies rationality of Gushel–Mukai fourfolds through the same circle of ideas that appears for cubic fourfolds: Hodge theory, period domains, associated K3 surfaces, and atoms coming from quantum cohomology. The Hodge-theoretic part describes the vanishing lattice, the period map, and Noether–Lefschetz divisors $\mathcal{D}_d$; Hodge-special fourfolds are precisely those whose periods acquire extra algebraic classes. This leads to the Hodge rationality expectation: rational Gushel–Mukai fourfolds should have associated K3 surfaces.

The proof strategy uses small quantum cohomology. Quantum multiplication by the Euler vector field decomposes into generalized eigenspaces, and these pieces behave like atoms under birational modifications. For a rational variety, weak factorization forces such atoms to appear either from projective space or from the centers of blow-ups. Computing the quantum cohomology of a Gushel–Mukai fourfold and a suitable deformation of it shows that the relevant Jordan blocks cannot be accounted for unless a K3-type piece appears. Consequently a Hodge general, in particular very general, Gushel–Mukai fourfold is irrational; more precisely, rationality forces the vanishing cohomology to be the rational Hodge structure of a projective K3 surface, up to the usual shift.

Notes.

Definition 13.1. A variety X is *rational* if there is a birational map

$$X \dashrightarrow \mathbb{P}^n.$$

Equivalently, there are Zariski open subsets

$$U \subset X, \qquad V \subset \mathbb{P}^n$$

which are isomorphic.

The first question is how rationality behaves under deformations. We want to deform the complex structure and ask whether rationality is preserved. Equivalently, one asks for the locus of rational varieties inside the moduli space.

Motivation.

Conjecture (now theorem) [KKPY]. A very general cubic fourfold is irrational.

Let us study Fano fourfolds with

$$H^{(2)} \neq 0.$$

Some examples are:

- Birational to cubics.
- Birational to Gushel–Mukai fourfolds.
- Blow-ups with K3 center.
- Quartic bundles.
- Kücük̆̈cü’s examples.

Cubic fourfolds.

For cubic threefolds the situation is rigid in the opposite direction: all of them are irrational. For cubic fourfolds

$$C \subset \mathbb{P}^5$$

the moduli space $\mathcal{M}$ has dimension 20. One can ask for the rational locus

$$\mathrm{Rat}(C) \subset \mathcal{M}.$$

This locus is a countable union of divisors.

The interesting point is that the Hodge structure controls the complex structure. More precisely, the Hodge decomposition

$$H^k(C) = \bigoplus_{p+q=k} H^{p,q}(C)$$

determines the complex structure. Katzarkov–Kontsevich–Pantev–Yu used this idea to prove that a very general cubic fourfold is irrational.

Gushel–Mukai fourfolds.

Gushel–Mukai fourfolds are very similar to cubic fourfolds. They are the main objects of this talk.

Definition 13.2. A *Gushel–Mukai variety* is a Fano n-fold X with

$$\operatorname{Pic}(X) = \mathbb{Z}H, \qquad \operatorname{index}(X) = n-2, \qquad H^n = 10.$$

In the ordinary case one can write

$$X = H \cap Q \cap G(2, V_5),$$

or equivalently

$$X = H \cap Q \subset G(2, V_5).$$

The Hodge structure decomposes as

$$H^\bullet(X) = H^\bullet(X)_{\text{amb}} \oplus H^4(X)_{\text{van}}.$$

Correspondingly, the Hodge diamond of X is the sum of the ambient and vanishing Hodge diamonds. Note that the Hodge diamond resembles that of a K3.

$$\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 0 & & 0 & & & \\
 & & 0 & & 1 & & 0 & & \\
 & 0 & & 0 & & 0 & & 0 & \\
0 & & 1 & & 22 & & 1 & & 0 \\
 & 0 & & 0 & & 0 & & 0 & \\
 & & 0 & & 1 & & 0 & & \\
 & & & 0 & & 0 & & & \\
 & & & & 1 & & & &
\end{array}$$

$$= \begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 0 & & 0 & & & \\
 & & 0 & & 1 & & 0 & & \\
 & 0 & & 0 & & 0 & & 0 & \\
0 & & 0 & & 2 & & 0 & & 0 \\
 & 0 & & 0 & & 0 & & 0 & \\
 & & 0 & & 1 & & 0 & & \\
 & & & 0 & & 0 & & & \\
 & & & & 1 & & & &
\end{array} \oplus \begin{array}{ccccccccc}
 & & & & 0 & & & & \\
 & & & 0 & & 0 & & & \\
 & & 0 & & 0 & & 0 & & \\
 & 0 & & 0 & & 0 & & 0 & \\
0 & & 1 & & 20 & & 1 & & 0 \\
 & 0 & & 0 & & 0 & & 0 & \\
 & & 0 & & 0 & & 0 & & \\
 & & & 0 & & 0 & & & \\
 & & & & 0 & & & &
\end{array}.$$

Periods of Gushel–Mukai fourfolds.

Recall the lattice

$$\Lambda = (H^4(X, \mathbb{Z})_{\text{van}}, \cup),$$

which is deformation invariant. The period map is

$$p : \mathcal{M}_{\text{GM}} \longrightarrow \mathcal{P}, \qquad [X] \longmapsto [H^{3,1}(X)].$$

Here

$$\mathcal{P} = \{[\omega] \in \mathbb{P}(\Lambda \otimes_{\mathbb{Z}} \mathbb{C}) \mid \omega \cup \omega = 0, \quad \omega \cup \overline{\omega} > 0\} / \widetilde{O}(\Lambda)$$

is the period domain.

Remark 13.3. We have

$$\dim(\mathcal{M}_{\text{GM}}) = 24, \qquad \dim(\mathcal{P}) = 20.$$

Remark 13.4 (Debarre-Kuznetzov). GM in fibers of p are birational.

Hodge classes using period domain.

The Hodge classes in degree 4 are

$$H^4(X)_{\mathrm{Hdg}} = H^4(X)_{\mathrm{amb}} \oplus \left(\Lambda \cap (H^{3,1}(X) \oplus \overline{H^{3,1}(X)})^{\perp}\right),$$

where the ambient cohomology is always algebraic. The idea is that for very general GM,

$$H^{\bullet}(X)^{\mathrm{Hdg}} = H^{\bullet}(X)^{\mathrm{amb}}$$

That is, for a very general Gushel–Mukai fourfold, the Hodge classes are exactly the ambient classes.

If $\alpha \in \Lambda$, then it defines a divisor

$$\mathcal{D}_d = \{[\omega] \in \mathcal{P} \mid \omega \cup \alpha = 0\} \subset \mathcal{P}.$$

Here d is the discriminant of

$$\alpha^{\perp} \subset \Lambda.$$

Definition 13.5. A Gushel–Mukai fourfold is *Hodge-special* if

$$p([X]) \in \bigcup_d \mathcal{D}_d.$$

It looks like the next step is to analyse different values of d:

- For which d, $\mathcal{D}_d \neq 0$.
- For which d, there exists an associated K3 surface: $\alpha^{\perp} \simeq H^2(K3, \mathbb{Z})_{\mathrm{prim}}$.

Remark 13.6. **Rationality, Hodge version.** Conjecture 1: Gushel–Mukai fourfold is rational if and only if it has a Hodge associated K3 surface.

Evidence.

The conjecture is known in the cases

$$\mathcal{D}_{10} \quad \text{and} \quad \mathcal{D}_{20}.$$

The first case is due to Debarre–Iliev–Manivel, and the second to Hoff and Sagliáno.

More evidence is the analogy with cubics: a cubic is rational if and only if there exists a Hodge K3 associated.

Remark 13.7. If the associated K3 surface is projective, then there exists a non-ambient Hodge class. Thus X is not Hodge general.

Beware.

For Gushel–Mukai fourfolds there are two nonequivalent rationality conjectures.

Remark 13.8. **Rationality, derived version.** Conjecture 2: a Gushel–Mukai fourfold X is rational if and only if there exists a derived associated K3 surface, that is,

$$\mathcal{K}_X \simeq D^b(\mathrm{K3}),$$

where

$$D^b(X) = \langle \mathcal{K}_X, \mathcal{O}_X, \mathcal{U}_X^{\vee}, \mathcal{O}_X(1), \mathcal{U}_X^{\vee}(1) \rangle.$$

The upshot is that the first conjecture is a divisorial condition, whereas the second conjecture is a higher-codimension condition.

Theorem 13.9 (Debarre–Iliev–Manivel)**.** *The divisor $\mathcal{D}_d$ is nonempty if and only if the following conditions hold:*

- *If* $d \equiv 0 \pmod 3$*, then [missing].*

missing

Theorem 13.10. *There exists a projective K3 surface* S *such that*

$$\alpha^{\perp} \cong (H^2(S, \mathbb{Z})_{\mathrm{prim}}, -\cup)$$

if and only if either

$$d \equiv 2 \pmod 8 \qquad \text{or} \qquad d \equiv 4 \pmod 8)$$

and every odd prime dividing d *is congruent to* 1 *modulo* 4*.*

Remark 13.11*.* There was a small additional detail at the end of the statement which was missed in the notes.

Theorem 13.12 (Benedetti–Manivel– Perrin)**.** *A Hodge general Gushel–Mukai fourfold is not rational. More precisely, if a Gushel–Mukai fourfold* X *is rational, then there exists a projective K3 surface* Σ *and an isomorphism of rational Hodge structures*

$$H^4(X, \mathbb{Q})_{\mathrm{van}} \cong H^2(\Sigma, \mathbb{Q})(-1).$$

Remark 13.13*.* There are also partial results on the birational geometry of singular (nodal) Gushel–Mukai fourfolds due to Benedetti–Guéré–Manivel– Perrin.

Remark 13.14*.* We can also state the theorem by using some number field (we don't know exactly which) instead of $\mathbb{Q}$ (and that (-1) term, I suppose) in Theorem 13.12.

Strategy via atoms.

Consider the small quantum cohomology ring

$$(QH(X), \star, \mathrm{Eu}_0), \qquad \mathrm{Eu}_0 = c_1(X).$$

One computes the matrix of

$$\mathrm{Eu}_0 \star$$

using periods, GKM theory, and geometry.

Let E_λ be the generalized eigenspace with eigenvalue λ which contains $H^{3,1}(X)$. Set

$$\begin{aligned} \rho_\lambda &= \dim(E_\lambda \cap QH^{\bullet}(X)_{\mathrm{Hdg}}), \\ \nu_\lambda &= \dim(E_\lambda \cap QH^{(2)}(X)), \\ \nu'_\lambda &= \dim(E_\lambda \cap QH^{(1)}(X)), \end{aligned}$$

and

$$\gamma_\lambda = \mathrm{rank}\ (\mathrm{Eu}_0 - \lambda)|_{E_\lambda}.$$

One can think of small quantum cohomology just as a ring attached to the variety. It decomposes as

$$QH(X) = \bigoplus_{\lambda} E_\lambda$$

into generalized eigenspaces.

These generalized eigenspaces are preserved under the blow-up and blow-down processes which link X to $\mathbb{P}^4$. So if X was rational, we have something of the kind

$$\begin{tikzcd}
 & \widetilde{X}_0 \arrow[dl, "\pi_0"'] \arrow[dr, "\pi_0'"] & & \widetilde{X}_1 \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_1'"] & & \cdots \arrow[dl] \arrow[dr] & & \widetilde{X}_k \arrow[dl, "\pi_k"'] \arrow[dr, "\pi_k'"] & \\
X = X_0 \arrow[rr, dashed, "\varphi_0"] & & X_1 \arrow[rr, dashed, "\varphi_1"] & & X_2 & & \cdots \arrow[rr, dashed, "\varphi_k"] & & X_{k+1} = \mathbb{P}^4.
\end{tikzcd}$$

The generalized eigenspace appears as an atom of some blow-up center.

Thus the idea is that these atoms appear for rational varieties and are preserved by rational manipulations. More precisely, by Iritani, $\mathbb{E}_0$ should appear in the quantum cohomology of $\mathbb{P}^4$ or in a smooth center. Studying them can therefore rule out rationality for some varieties.

Quantum cohomology setup.

Let X be a Gushel–Mukai fourfold with

$$\operatorname{Pic}(X) = \mathbb{Z}[\mathcal{O}_X(1)].$$

We start from

$$H(X) = (H^\bullet(X, \mathbb{Q}), \cup)$$

and deform the cup product to obtain the small quantum cohomology ring

$$QH(X) = (H^\bullet(X)[q], \star).$$

Here one may use small quantum cohomology, but in general one has to work with series.

The product has the form

$$\sigma \star \sigma' = \sigma \cup \sigma' + \sum_{d \geq 1} q^d \sum_{\tau \in H(X)} \langle \sigma, \sigma', \tau \rangle_d \, \tau^\vee.$$

Let $M_{3,d}(X)$ be the moduli space of stable maps

$$f : C \to X$$

from a nodal genus zero curve with three marked points

$$p_1, p_2, p_3,$$

with finite automorphism group and

$$f_*[C] = d[\mathcal{O}_X(1)]^\vee.$$

For $i = 1, 2, 3$ there are evaluation maps

$$\operatorname{ev}_i : M_{3,d}(X) \to X, \qquad \operatorname{ev}_i([f]) = f(p_i).$$

The Gromov–Witten invariant is

$$\langle \sigma, \sigma', \tau \rangle_d = \int_{[M_{3,d}(X)]^{\mathrm{vir}}} \operatorname{ev}_1^* \sigma \cup \operatorname{ev}_2^* \sigma' \cup \operatorname{ev}_3^* \tau.$$

Roughly, this counts degree d curves passing through the chosen cycles.

Remark 13.15. There is a missing point here: one has to prove that this rough counting interpretation applies in the needed case.

Monodromy action.

Let

$$\mathcal{F} = \{\text{smooth } X \hookrightarrow G(2, V_5)\}$$

be the family of hypersurfaces. The monodromy action gives an action of

$$\pi_1(\mathcal{F})$$

on

$$H^4(X, \mathbb{Z})_{\mathrm{van}},$$

and this action is irreducible.

Gromov–Witten invariants are monodromy, hence deformation, invariants.

Lemma 13.16. *Let*

$$\sigma_1, \dots, \sigma_k \in H^\bullet(X)_{\mathrm{amb}}$$

and let

$$\tau \in H^4(X)_{\mathrm{van}}.$$

Then

$$\langle \sigma_1, \dots, \sigma_k, \tau \rangle_d = 0.$$

As consequences, if

$$\alpha \in H^\bullet(X)_{\mathrm{amb}},$$

then

$$\alpha \star QH^\bullet(X)_{\mathrm{amb}} \subset QH^\bullet(X)_{\mathrm{amb}},$$
$$\alpha \star QH^4(X)_{\mathrm{van}} \subset QH^4(X)_{\mathrm{van}},$$

and

$$(\alpha\star)|_{QH^4(X)_{\mathrm{van}}} = \lambda\,\mathrm{id}.$$

These statements hold for any ambient deformation of $QH(X)$. The same statements hold for the Euler vector field. Thus, with respect to the decomposition into ambient and vanishing parts, one has a block form

$$\mathrm{Eu}_0 \star = \begin{pmatrix} * & 0 \\ 0 & \lambda\,\mathrm{id} \end{pmatrix}.$$

where the first column corresponds to the ambient cohomology, while the right column corresponds to the vanishing cohomology.

For a deformation parameter t_α, the corresponding quantum Euler vector field has the form

$$\mathrm{Eu}_{t_\alpha} = c_1(X) - \left(1 - \frac{\deg(\alpha)}{2}\right) t_\alpha.$$

Enumerativity.

For the atom calculation one computes

$$\mathrm{Eu}_{t_\alpha} \star.$$

At $\alpha = 0$ one has

$$\mathrm{Eu}_0 = 2h.$$

Moreover

$$\ker(\mathrm{Eu}_0 \star)|_{QH^4(X)_{\mathrm{van}}} = 0,$$

because the index is at least 2.

Proposition 13.17 (Enumerativity)**.** *The varieties $F_1(X)$ and $G_2(X)$ are smooth and irreducible. For $d = 1$ and $d = 2$, after using the* PGL_5*-action, the Gromov–Witten invariant*

$$\langle \alpha, \dots, \alpha \rangle_d$$

is the degree of a suitable enumerative cycle.

Remark 13.18. This proposition is very incomplete in these notes: the hypotheses and the degree formula still have to be filled in.

Computing invariants.

A basis for $H^\bullet(X)_{\mathrm{amb}}$ is

$$1,\ h,\ \sigma_2,\ \sigma_{11},\ I = h^\vee,\ \mathrm{pt}.$$

Here

$$\sigma_2 = [\text{missing}], \qquad \sigma_{11} = [\text{missing}], \qquad I = h^\vee = [\text{missing}].$$

The explicit expressions were missed in the notes.

Eigenspace decomposition.

The point is to compute

$$\mathrm{Eu}_0 \star|_{H^4(X)_{\mathrm{amb}}}.$$

This gives a matrix whose characteristic polynomial is

$$P(T) = T^2(T^2 - 44qT - 16q^2).$$

Therefore

$$\ker(\mathrm{Eu}_0 \star) = \ker(\mathrm{Eu}_0 \star) \cap QH^\bullet(X)_{\mathrm{amb}} \oplus H^4(X)_{\mathrm{van}}.$$

Equivalently,

$$\ker(\mathrm{Eu}_0 \star) = \langle \alpha, \beta \rangle \oplus H^4(X)_{\mathrm{van}}.$$

A deformation.

By deforming $QH(X)$, one finds that the ambient part becomes a Jordan block in the matrix of quantum multiplication.

Remark 13.19. The details of this deformation were missed in the notes.

Strategy via atoms, continued.

- Consider the small quantum cohomology ring with its Euler vector field.
- Compute the matrix of $\mathrm{Eu}_0 \star$.
- Study the generalized eigenspaces.
- By KKPY and Guéré, following weak factorization, E_λ should appear as a generalized eigenspace of either $QH_t(\mathbb{P}^4)$ or QH_t of a smooth center.
- By KKPY, if $\rho_\lambda \leq 2$ and $\nu_\lambda \neq 0$, then E_λ is not a generalized eigenspace of the quantum cohomology of a variety of dimension at most 2, or of $\mathbb{P}^4$. This implies irrationality.
- By Guéré, rationality implies that, if $\nu_\lambda \neq 0$, $\nu'_\lambda = 0$, and $\gamma_\lambda \leq 1$, then there exists a K3 small center and a map

  $$H^\bullet(\mathrm{K3}) \longrightarrow H^4(X)_{\mathrm{van}}.$$

  That is,

  $$QH_t^\bullet(K3) \xrightarrow{\sim} \mathbb{E}_0$$

- If $\gamma_\lambda = 1$, then separating Jordan blocks gives a Hodge-theoretic relation from $H^2(\mathrm{K3})$ to $H^4(C)_{\mathrm{van}}$. Even if you deform, you get only one eigenvalue.

The upshot is that, under deformation, one must send a Jordan block to a Jordan block.

14. ISOMETRIC SOLUTIONS TO THE HETEROTIC G2-SYSTEM

Speaker.
Viviana del Barco

Title.
Isometric solutions to the heterotic G2-system

Abstract.
I will report on new solutions to the heterotic G-system with non-abelian gauge group, both compact and non-compact, over certain 3-Sasakian manifolds and 2-step nilmanifolds. Moreover, we construct families of solutions on torus bundles over manifolds carrying SU(3)-structures with special torsion forms. Our methods allow us to produce solutions to the heterotic G_2-system on a fixed Riemannian manifold endowed with non-equivalent G_2-structures. This is achieved by modifying either the gauge group of the corresponding principal bundle or a torsion class of the G_2-structure. In particular, we obtain solutions within the same isometry class of a G_2-structure exhibiting both vanishing and non-vanishing scalar torsion. This is a joint work with Andrés J. Moreno and Udhav Fowdar.

AI-generated summary.
The talk constructs new solutions to the heterotic G_2-system, whose data consist of an integrable G_2-structure, a gauge connection, and a Bianchi identity relating the torsion of the characteristic connection to the curvature of the gauge field. The first part recalls the basic G_2-geometry: torsion forms, integrability, the characteristic connection, and the instanton and Bianchi equations.

The main constructions come from two ansätze. The first uses manifolds with three orthogonal vector fields, producing coclosed G_2-structures on nilmanifolds and on certain 3-Sasakian spaces, with compact and non-compact gauge groups. The second uses principal S^1-bundles over 6-manifolds with SU(3)-structures, giving families

$$\varphi_t$$

of integrable, generally non-coclosed G_2-structures. In both methods, the key point is that one can obtain different heterotic solutions on the same underlying Riemannian manifold, often with the same connection but different torsion forms. This explains the phrase "isometric solutions" in the title.

Outline.
1. Preliminaries on G_2-structures and the heterotic G_2-system.
2. Previous solutions to the system.
3. New solutions by two different methods.

References.
- *G_2-instantons on 2-step nilpotent Lie groups.*
- *Isometric solutions to the heterotic G_2-system.*

Preliminaries on G_2-structures.

Definition 14.1. A *G_2-manifold* is a 7-dimensional manifold M equipped with a G_2-structure. Such a structure is a 3-form

$$\varphi \in \Omega^3(M)$$

such that, at every point $p \in M$,

$$\iota_v\varphi \wedge \iota_v\varphi \wedge \varphi \neq 0$$

for every nonzero

$$v \in T_pM.$$

Such a structure defines a Riemannian metric g_φ and an orientation by the formula

$$g_\varphi(v, w)\ \mathrm{vol}_\varphi = \frac{1}{6}\iota_v\varphi \wedge \iota_w\varphi \wedge \varphi.$$

G_2-module decompositions.

A G_2-structure gives decompositions

$$\Lambda^2(M) = \Lambda^2_7 \oplus \Lambda^2_{14},$$

and

$$\Lambda^3(M) = \langle\varphi\rangle \oplus \Lambda^3_7 \oplus \Lambda^3_{27}.$$

Each summand has an explicit description.

Torsion forms.

The decomposition allows one to define the torsion forms

$$\tau_0 \in C^\infty(M), \qquad \tau_1 \in \Lambda^1(M),$$

$$\tau_2 \in \Lambda^2_{14}, \qquad \tau_3 \in \Lambda^3_{27}.$$

They are defined by the components of $d\varphi$ and $d\psi$, where

$$\psi = *_\varphi\varphi,$$

with respect to the G_2-module decompositions:

$$d\varphi = \tau_0\psi + 3\tau_1 \wedge \varphi + *_\varphi\tau_3,$$

and

$$d\psi = 4\tau_1 \wedge \psi + \tau_2 \wedge \varphi.$$

The G_2-structure φ is called:

- integrable if

$$d\psi = 4\tau_1 \wedge \psi;$$

- coclosed if [missing];
- purely coclosed if [missing];
- nearly parallel if [missing];
- parallel if [missing].

Theorem 14.2 (Friedrich–Ivanov, 2002)**.** *Being integrable is equivalent to the existence of a metric connection on M with skew-symmetric torsion such that*

$$\nabla^c\varphi = 0.$$

Heterotic G_2-system.

Let φ be an integrable G_2-structure on a differentiable manifold M. Consider:

- a principal G-bundle

$$P \to M;$$

- a connection 1-form A;

- an $\mathrm{Ad}(G)$-invariant nondegenerate symmetric bilinear form
$$\langle \, , \, \rangle_{\mathfrak{g}}$$
on
$$\mathfrak{g} = \mathrm{Lie}(G).$$

Definition 14.3. We say that (φ, A) satisfies the *heterotic G_2-system* in the sense of da Silva et al. 2025 if the following system of equations holds:

$$d\psi = 4\tau_1 \wedge \psi,$$
$$F_A \wedge \psi = 0,$$

and

$$dT_\varphi = \langle F_A \wedge F_A \rangle_{\mathfrak{g}}.$$

The first equation is the integrability condition, the second is the instanton condition, and the third is the heterotic Bianchi identity.

Here
$$T_\varphi = \frac{1}{6}\tau_0\varphi + *_\varphi(\tau_1 \wedge \varphi) - \tau_3$$
is the torsion of the canonical connection.

Remark 14.4. This is a particular case of a heterotic system. There is a more general way to define such systems.

Previous approaches.

Given a manifold with an integrable G_2-structure (M, φ), it is not clear in general which bundles, connections, and Ad-invariant metrics define solutions to the heterotic G_2-system. The following list is non-exhaustive.

- Harland–Nölle, 2012: the canonical connection ∇^c is an instanton when φ is nearly parallel.
- Ivanov–Ivanov, 2005: solutions on
$$M = S^1 \times \Gamma/N_{3,2}.$$
- Fernández et al., 2011: solutions on nilmanifolds
$$\Gamma/\mathbb{H}_7,$$
where $\mathbb{H}_7$ is the 7-dimensional Heisenberg group.
- Moroianu et al., 2025: solutions on 2-step nilmanifolds.
- Lotay–Sa Earp, 2023: approximate solutions to the system on contact Calabi–Yau manifolds.
- Galdeano–Stecker, 2025: approximate solutions on 3-(α, δ)-Sasakian manifolds, with an exact solution occurring in a degenerate case.

Part 2: nilmanifold solutions.

In joint work with Clarke and Moreno, we considered the following setup:

- $$M = \Gamma \backslash N,$$
where N is a 2-step nilpotent Lie group of dimension 7 and $\Gamma \subset N$ is a discrete cocompact subgroup;
- φ is a coclosed left-invariant G_2-structure on N;
- $A = \nabla^c$ is the characteristic connection.

First we aimed at solving the instanton condition for ∇^c. We obtained the following theorem.

Theorem 14.5. *There exists a G_2-structure φ on*

$$M = \Gamma\backslash N$$

for which the characteristic connection is a G_2-instanton if and only if the Lie algebra $\mathfrak{n}$ is one of the following:

$$\mathfrak{n} = \mathfrak{h}_7, \qquad \mathfrak{n} = \mathbb{R} \oplus \mathfrak{n}_{3,2}, \qquad \mathfrak{n} = \mathfrak{h}_{\mathbb{H}}.$$

In each case one can check that ∇^c also solves the heterotic Bianchi identity. Thus each of the above gives a solution to the heterotic G_2-system. The gauge groups are

$$\mathrm{SU}(3) \subset G_2.$$

The Riemannian Lie groups obtained are naturally reductive, so

$$\nabla^c T = 0, \qquad \nabla^c R = 0.$$

The torsion is not closed.

Remark 14.6. These solutions differ from the ones studied by Moroianu et al. in 2025.

Remark 14.7. The solution corresponding to the first Lie algebra was already known from Fernández et al. in 2011.

Remark 14.8. The solution corresponding to the second Lie algebra is different from the one in Ivanov–Ivanov.

Part 3: new solutions by two methods.

First ansatz.

- This applies to manifolds with a triple of orthogonal vector fields.
- It produces examples with both
$$\tau_0 = 0 \qquad \text{and} \qquad \tau_0 \neq 0.$$
- It applies to nilmanifolds and 3-Sasakian manifolds.
- It gives coclosed G_2-structures, so
$$\tau_1 = \tau_2 = 0.$$
- It provides compact and non-compact structure groups G.

Second ansatz.

- This applies to manifolds which are S^1-bundles over manifolds with SU(3)-structures.
- It gives integrable solutions, which are non-coclosed in general:
$$\tau_1 \neq 0, \qquad \tau_2 = 0.$$
- Again one obtains both cases
$$\tau_0 = 0 \qquad \text{and} \qquad \tau_0 \neq 0.$$
- There are 1-parameter families of solutions over the same Riemannian manifold, with nonequivalent G_2-structures and fixed connection.

The main feature in both cases is that the solutions are isometric.

First ansatz: the manifold and the G_2-structure.

Let (M,g) be an oriented Riemannian spin manifold. Assume that (M,g) admits a triple of orthogonal vector fields

$$e_5, e_6, e_7.$$

Then

$$\varphi = \sigma_1^+ \wedge e^5 + \sigma_2^+ \wedge e^6 + \sigma_3^+ \wedge e^7 + e^{567}$$

is an SO(3)-family of G_2-structures, where

$$\sigma_1^+, \sigma_2^+, \sigma_3^+$$

are self-dual 2-forms on the distribution [missing].

First ansatz: the gauge group and connections.

Let G be a 3-dimensional Lie group. [missing]

First ansatz: solutions on nilmanifolds.

Assume

$$M = \Gamma \backslash N, \qquad \mathfrak{n} = \mathrm{Lie}(N),$$

with a global coframe

$$\{e^i\}_{i=1}^7$$

satisfying the structure equation [missing].

Any Lie algebra satisfying that equation with $\delta \neq 0$ is 2-step nilpotent and isomorphic to one of

$$\mathbb{R} \oplus \mathfrak{n}_{3,2}, \qquad \mathfrak{n}_{7,3,A},$$

$$\mathfrak{n}_{7,3,B_1}, \qquad \mathfrak{n}_{7,3,D}, \qquad \mathfrak{h}_{\mathbb{H}}.$$

First ansatz: results.

Theorem 14.9. *Let*

$$M = \Gamma \backslash N$$

be a nilmanifold, where Γ *is a cocompact lattice. There exists a coclosed* G_2*-structure* φ *on* M *and a* G*-connection* A *solving the heterotic* G_2*-system in the following cases:*

(1)

$$G = \mathbb{T}^3$$

and

$$\mathfrak{n} \cong \mathbb{R}^2 \oplus \mathfrak{h}_5, \quad \mathbb{R} \oplus \mathfrak{h}_3^{\mathbb{C}}, \quad \mathfrak{h}_{\mathbb{H}}.$$

(2)

$$G = \mathrm{SU}(2).$$

[missing]

(3)

$$G = \mathrm{SL}(2,\mathbb{R}).$$

[missing]

Moreover, in the nonabelian cases, both the underlying metric and volume are the same, but the G_2*-structures* φ *are distinct.*

Let (M,g) be a 3-Sasakian manifold. We denote by φ_{ts} the 3-Sasakian nearly parallel G_2-structure, by φ_{np} the squashed nearly parallel G_2-structure, and by $\widehat{\varphi}_{\mathrm{ts}}$ a coclosed, but not nearly parallel, G_2-structure related to φ_{ts}.

Theorem 14.10. *Let M be either S^7 or the Aloff–Wallach space $N^{1,1}$. For the G_2-structure φ_{ts}, there exist G-connections solving the heterotic G_2-system in the following cases:*

(1)
$$G = \mathrm{SL}(2,\mathbb{R}) \times \mathrm{SU}(3) \times G_2$$
and $M = S^7$.

(2)
$$G = \mathrm{SL}(2,\mathbb{R}) \times \mathrm{U}(1)$$
and $M = N^{1,1}$.

[something else missing]

Second ansatz: the manifold and G_2-structure.

Let M be a principal S^1-bundle over a 6-manifold Q with an SU(3)-structure
$$(Q, g_\omega, J, \omega, \gamma).$$
Let η be a connection 1-form on M. Consider the family of G_2-structures on M
$$\varphi_t = \eta \wedge \omega + \mathrm{Re}\left(e^{it}(\gamma_+ + i\gamma_-)\right).$$

Analogously to the G_2 case, the spaces $\Lambda^\bullet(Q)$ decompose as SU(3)-modules. For example,
$$\Lambda^2(Q) = \langle\omega\rangle \oplus \Lambda^2_6 \oplus \Lambda^2_8 \oplus \cdots .$$
More of this decomposition is missing from the notes.

Proposition 14.11. *The G_2-structure φ_t is integrable for all*
$$t \in [0, 2\pi)$$
if and only if
$$\sigma_2 = \pi_2 = 0, \qquad \pi_1 = 2\nu_1,$$
and $d\eta$ is of type $(1,1)$. In this case one has formulas for
$$T_{\varphi_t}, \qquad \tau_0(t), \qquad \tau_1(t),$$
which were missed in the notes.

Here $(d\eta)_0$ denotes the ω-component of $d\eta$, and T_ω is the torsion form of the Bismut connection of
$$(\omega_+, \gamma_+).$$

Remark 14.12. More details of this proposition are missing.

Theorem 14.13. *Let M be a principal S^1-bundle over*
$$(Q, \omega, \gamma_+).$$
Assume that $d\eta$ and the torsion forms of the SU(3)- *structure are such that T_{φ_t} is constant. Assume also that*
$$(\omega, \gamma_+, A)$$
solves the SU(3) *heterotic Bianchi identity, and [missing]. Then the pullback connection*
$$\eta \oplus A$$
is a solution of the heterotic G_2-system for every
$$t \in [0, 2\pi).$$

Example 14.14. Let

$$\mathfrak{n}_{3,2} = \langle e_2, \dots, e_7 \rangle$$

with structure equations

$$de^i = 0 \quad \text{for } i = 2, 3, 4,$$

and [missing].

There is a left-invariant SU(3)-structure induced on $N_{3,2}$, given by

$$\omega = \text{[missing]}, \qquad \gamma_+ = \text{[missing]}.$$

On the manifold

$$M = S^1 \times \Gamma \backslash N_{3,2}$$

we take

$$\eta = e^1$$

and [missing]. The theorem above gives a 1-parameter family of solutions to the heterotic G_2-system on M, with the same connection A but with varying τ_0 and τ_1. In particular, these structures are not coclosed in general.

The family φ_t contains both a coclosed G_2-structure, for $t = 0$, and integrable but not coclosed G_2-structures for other values of t.

15. Moduli spaces of pointed genus one curves and logarithms of algebraic numbers

Speaker.

Tiago Fonseca

Title.

Moduli spaces of pointed genus one curves and logarithms of algebraic numbers

Abstract.

This is joint work with Francis Brown. I will explain the relation between motives of moduli spaces of genus one curves with marked points and the so-called Gross-Zagier conjecture, according to which certain special values of automorphic Green's functions are logarithms of algebraic numbers.

Goals of the talk.

- Recall the Gross–Zagier log-algebraicity conjecture.
- Relate it to the moduli spaces

$$\mathcal{M}_{1,n}.$$

- Discuss geometric and motivic aspects.

Notation.

- $\mathbb{H}$ denotes the upper half-plane. Recall that $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathbb{H}$ by Möbius transformations.
- Klein's J-invariant is a map

$$J : \mathbb{H} \to \mathbb{C}$$

which induces an isomorphism

$$\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H} \cong \mathbb{C}.$$

- The moduli space $\mathcal{M}_{1,1}$ parametrizes genus 1 curves with one marked point, and the moduli interpretation is
$$\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H} \cong \mathcal{M}_{1,1},$$
via
$$z \longmapsto (E_z = \mathbb{C}/(\mathbb{Z}+\mathbb{Z}z), [0]).$$

AI-generated note on J.

Klein's J-function is the classical modular function which classifies elliptic curves over $\mathbb{C}$ up to isomorphism. It is invariant under the action of
$$\mathrm{SL}_2(\mathbb{Z})$$
on the upper half-plane, so two points $z, z' \in \mathbb{H}$ give the same value of J exactly when the corresponding marked complex tori are isomorphic. Thus J gives a coordinate on the coarse moduli space of elliptic curves. In the more standard normalization, one often writes
$$j(z) = q^{-1} + 744 + 196884q + \cdots, \qquad q = e^{2\pi i z},$$
and Klein's function is
$$J = j/1728.$$

Arithmetic of J.

The function J satisfies many interesting arithmetic properties. One of them is that this highly transcendental function sends quadratic numbers to algebraic numbers.

Theorem 15.1 (Kronecker)**.** *If $z \in \mathbb{H}$ is a quadratic number, then*
$$J(z) \in \overline{\mathbb{Q}}.$$

Example 15.2 (Weber)**.** One has
$$J(\sqrt{-14}) = 6^{-3}\left(323 + 228\sqrt{2} + (231 + 161\sqrt{2})\sqrt{2\sqrt{2}-1}\right)^3.$$

Geometrically, this says that an elliptic curve with too many endomorphisms, meaning
$$\mathrm{End}(E) \supset \mathbb{Z}, \quad \mathrm{End}(E) \neq \mathbb{Z},$$
is defined over $\overline{\mathbb{Q}}$. We say that such a point $z \in \mathbb{H}$ is a *CM point*, where CM stands for complex multiplication.

> The theory of complex multiplication, which forms a powerful link between number theory and analysis, is not only the most beautiful part of mathematics but also of all science.
>
> D. Hilbert

Gross–Zagier Green functions.

In the 1980s, Gross and Zagier published *Heegner points and derivatives of L-series.* This proves a key case of the Birch–Swinnerton-Dyer conjecture. The proof is interesting here because it involves a long explicit calculation with the higher Green functions
$$G_s(z,w) = -s \sum_{\gamma \in \mathrm{SL}_2(\mathbb{Z})} Q_{s-1}\left(1 + \frac{|z-\gamma w|^2}{2\,\mathrm{Im}(z)\,\mathrm{Im}(\gamma w)}\right),$$

where $z, w \in \mathbb{H}$ are not in the same orbit, and $s \geq 1$ is a parameter.

These are higher versions of

$$G_1(z,w) = \log|J(z) - J(w)|^2,$$

but their geometric interpretation is less clear. Thus the function appears in the proof, but it is not immediately clear what it means geometrically.

When $s \geq 1$ is an integer, G_s has remarkable arithmetic properties.

- The Gross–Zagier conjecture is known for
  $$s \in \{1,2,3,4,5,7\}.$$
  If z and w are CM points of discriminants d_z and d_w, then
  $$G_s(z,w) = (d_z d_w)^{(s-1)/2} \log|\alpha|$$
  for some
  $$\alpha \in \overline{\mathbb{Q}}.$$
  There is a more complicated version for other values of s.
- For
  $$G_1(z,w) = \log|J(z) - J(w)|^2,$$
  this follows from Kronecker's theorem.
- Mellit computed the example
  $$G_2\left(\frac{1+\sqrt{-7}}{2}, i\right) = \frac{8}{\sqrt{7}} \log(8 - 3\sqrt{7}).$$
- Previous work includes Gross–Kohnen–Zagier 1987, Zhang 1997, Mellit 2008, Viazovska 2011, Li 2023, and Bruinier–Li–Yang 2025.

The goal is to relate the Gross–Zagier conjecture to algebraic geometry.

Differential forms on $M_{1,n}$.

The case $s = 1$ is already suggestive. Since

$$G_1(z,w) = \log|J(z) - J(w)|^2$$

is a logarithm, it can be expressed in terms of integrals. Equivalently, with one variable fixed, one may think of it as a primitive

$$\nu + \overline{\nu},$$

where ν is a 1-form on $M_{1,1}$ with poles along the corresponding value of J.

Let $\overline{M}_{1,n}$ denote the Deligne–Mumford compactification, whose points are stable curves of genus one with n marked points. There is an action of S_n on $\overline{M}_{1,n}$ by permuting the marked points.

Theorem 15.3 (Brown–Fonseca). *There are unique real-analytic $(n-1)$-forms $g^{p,q}$ on*

$$\overline{M}_{1,n} \setminus D_w$$

such that

$$dg^{p,q} = \nu^{p,q} + \overline{\nu^{q,p}}.$$

The forms $\nu^{p,q}$ define algebraic n-forms on

$$\overline{M}_{1,n} \setminus D_w.$$

Their cohomology classes are S_n-alternating and satisfy

$$H^n(\overline{M}_{1,n} \setminus D_w)_{\mathrm{sgn}} = H^n(\overline{M}_{1,n})_{\mathrm{sgn}} \oplus \bigoplus_{p+q=n-1} \mathbb{C}[\nu^{p,q}].$$

To summarize, for $n \geq 3$ one gets an $n \times n$ matrix of Green functions. Its entries are coefficients of $(n-1)$-forms $g^{p,q}$ such that

$$dg^{p,q} = \nu^{p,q} + \overline{\nu^{q,p}},$$

where the $\nu^{p,q}$ are algebraic n-forms.

Motives.

Let $K \subset \mathbb{C}$ be a number field. Many cohomology theories with constant coefficients for algebraic varieties over K include:

Betti	de Rham	ℓ-adic
$H^\bullet_{\mathrm{B}}(X)$	$H^\bullet_{\mathrm{dR}}(X)$	$H^\bullet_\ell(X)$
$H^\bullet_{\mathrm{sing}}(X(\mathbb{C});\mathbb{Q})$	$H^\bullet_{\mathrm{Zar}}(X;\Omega^\bullet_{X/K})$	$\varprojlim_n H^\bullet_{\mathrm{et}}(X,\mathbb{Z}/\ell^n) \otimes \mathbb{Q}_\ell$
$\mathbb{Q}$	K	$\mathbb{Q}_\ell$
complex conj.	Hodge filtration	Galois rep.

Grothendieck's comparison theorem shows that the coefficients are periods:

$$H^\bullet_{\mathrm{dR}}(X) \otimes_K \mathbb{C} \xrightarrow{\sim} H^\bullet_{\mathrm{B}}(X) \otimes_{\mathbb{Q}} \mathbb{C},$$

$$\omega \mapsto \left([\sigma] \mapsto \int_\sigma \omega \right).$$

These integrals are called periods. The motive $H(X)$ is a linear algebra object containing all cohomological information of X for all Weil cohomology theories.

Voevodsky motives.

- $DM(K)$ is Voevodsky's triangulated category of motives over K. Morally, it is the bounded derived category of the true abelian category of motives over K.
- $DM(K)$ is $\mathbb{Q}$-linear, triangulated, pseudo-abelian, and rigid tensor.
- A subvariety $Y \subset X$ determines a relative cohomology motive

  $$H(X,Y) \in DM(K).$$

  We write $H(X,\emptyset) = H(X)$.
- There is a distinguished triangle

  $$H(Y)[-1] \to H(X,Y) \to H(X) \xrightarrow{+1}$$

  This is the relative cohomology long exact sequence.
- There are realization functors

  $$R_{\mathrm{B}} : DM(K) \to D\mathrm{Vect}_{\mathbb{Q}}, \qquad R_{\mathrm{dR}} : DM(K) \to D\mathrm{Vect}_K.$$

  For example,

  $$R_{\mathrm{B}}(X,Y) = H^\bullet_{\mathrm{B}}(X,Y).$$
- Grothendieck's comparison theorem can be rephrased as

  $$R_{\mathrm{dR}} \otimes_K \mathbb{C} \cong R_{\mathrm{B}} \otimes_{\mathbb{Q}} \mathbb{C}.$$

The basic motives are the building blocks for all such cohomological realizations.

Basic motives.

- The trivial motive is
$$\mathbb{Q}(0) = H(\mathrm{pt}).$$
- The Lefschetz motive is
$$\mathbb{Q}(-1) = H(\mathbb{A}^1 \setminus \{0\}, \{1\})[1].$$
- More generally,
$$\mathbb{Q}(n) = \mathbb{Q}(-1)^{\otimes -n}, \qquad n \in \mathbb{Z}.$$
- The Tate twist is
$$M(n) = M \otimes \mathbb{Q}(n).$$
- For $\alpha \in K^\times$, the Kummer motive is
$$\mathcal{K}_\alpha = H(\mathbb{A}^1 \setminus \{0\}, \{1, \alpha\})[1].$$

There is a distinguished triangle
$$\mathbb{Q}(0) \to \mathcal{K}_\alpha \to \mathbb{Q}(-1) \xrightarrow{+1}$$
The periods of these Kummer motives are logarithms. This is one way to think about logarithms in algebraic geometry.

If you want to know whether a number is a logarithm, you attach a motive to it and look for the corresponding extension of Tate motives. In that way, the logarithm is read off from the motive's mixed structure.

Theorem 15.4 (Brown–Fonseca)**.** *Let $n \geq 3$.*

1. *The functions $G^{p;q}_{r,s}(z,w)$ are single-valued periods of a motive $M_{n,z,w}$.*
2. *The motive $F_{n,z}$ is a subquotient of $\mathrm{Sym}^{n-1} H^1(E_z)$. If z is a CM point, then*
$$F_{n,z} = \mathbb{Q}\left(\frac{n-1}{2}\right) \oplus \widetilde{F}_{n,z}.$$
3. *If z and w are CM points and*
$$H(\overline{M}_{1,n})_{\mathrm{sgn}}[1] = 0,$$
then one gets a subquotient
$$\mathbb{Q}\left(\frac{n-1}{2}\right) \to \mathcal{GZ}_{n,z,w} \to \mathbb{Q}\left(\frac{n-3}{2}\right) \xrightarrow{+1}$$

If $n = 2s-1$, then $G^{p;q}_{r,s}(z,w)$ is the single-valued period of $\mathcal{GZ}_{n,z,w}$. The Gross–Zagier conjecture holds for the corresponding CM points.

We know that
$$R_{\mathrm{B}} H(\overline{M}_{1,n})_{\mathrm{sgn}}[n] = 0$$
for
$$s \in \{1,2,3,4,5,6,7\}.$$

Remark 15.5. **Conservativity.** The functor
$$R_{\mathrm{B}} : DM(K) \to D\mathrm{Vect}_{\mathbb{Q}}$$
is conservative.

This would imply the Gross–Zagier conjecture. We do not know how to prove this in general. The vanishing of the motives above is not out of reach. For instance, conservativity is known on the subcategory of mixed Tate motives. So this is a completely geometric proof of the conjecture in that case. It works because mixed Tate motives are conservative.

Theorem 15.6 (Brown–Fonseca)**.** *The motive of* $\overline{M}_{1,3}$ *is mixed Tate. In particular, the above argument proves the Gross–Zagier conjecture for* $n = 3$*, that is, for* $G_2(z, w)$*.*

We also expect the mixed Tate property for all $n \leq 10$. Compare with Belorousski's rationality result.

16. ATOMIC SEMI-ORTHOGONAL DECOMPOSITIONS FOR DERIVED CATEGORIES OF SURFACES

Speaker.

Alexey Elagin

Title.

Atomic Semi-Orthogonal Decompositions for Derived Categories of Surfaces.

Abstract.

I will talk about our recent work with Julia Schneider and Evgeny Shinder. We construct, for any surface with a group action, a canonical "atomic" G-invariant semi-orthogonal decomposition of the derived category, so that these decompositions are compatible with geometric operations such as blow-ups and conic bundles. Non-trivial "atoms" of these decompositions are birational invariants, this enables a uniform approach to birational classification of G-surfaces in terms of atoms. The construction of atomic decompositions is based on the G-minimal model program in dimension two and on the Sarkisov link decomposition. Our work is inspired by the atoms by Katzarkov-Kontsevich-Pantev-Yu, although the approach is more elementary and makes no use of quantum cohomology.

AI-generated note.

Elagin uses Sarkisov's program in the same structural way as the Cremona notes do in Section **??**: factor a birational map into elementary links, then read off the effect of each link on the derived category. For surfaces with group action, this lets him build the atomic semi-orthogonal decomposition compatibly with blowups and conic bundles, and then show that the nontrivial atoms are birational invariants.

Let X be a variety over a field $\daleth$. We write

$$D^b(X) := D^b(\mathrm{Coh}(X)).$$

Then $D^b(X)$ is a k-linear triangulated category.

Triangulated categories are the derived-category version of exact categories. Triangles replace extensions, kernels, and cokernels.

If X and Y are isomorphic, then $D^b(X) \simeq D^b(Y)$. The converse is false in general. By Bondal–Orlov, if X and Y are smooth projective and K_X or $-K_X$ is ample, then an equivalence $D^b(X) \simeq D^b(Y)$ implies $X \simeq Y$.

Definition 16.1. A semi-orthogonal decomposition $D^b(X) = \langle A_1, \ldots, A_n \rangle$ means that $\mathrm{Hom}(A_j, A_i) = 0$ for $j > i$, and the A_i generate $D^b(X)$.

Example 16.2 (Beilinson)**.** There is an SOD

$$D^b(\mathbb{P}^n) = \langle \mathcal{O}(-n), \dots, \mathcal{O}(-1), \mathcal{O} \rangle.$$

Definition 16.3. An object E in $D^b(X)$ is exceptional if

$$\operatorname{Ext}^i(E,E) = 0 \quad \text{for } i \neq 0, \qquad \operatorname{Hom}(E,E) = \daleth.$$

A collection of objects is exceptional if each object is exceptional and they are semi-orthogonal.

Theorem 16.4 (Orlov)**.** *If $\widetilde{X} = \operatorname{Bl}_Z X$ is the blow up of a smooth center Z, then*

$$D^b(\widetilde{X}) = \langle \underbrace{D^b(Z), \dots, D^b(Z)}_{\operatorname{codim}(Z,X)-1}, p^* D^b(X) \rangle.$$

Theorem 16.5 (Bondal–Orlov)**.** *For a standard flop $X \dashrightarrow Y$ there is an equivalence of derived categories. For a standard flip $X \dashrightarrow Y$ the category $D^b(Y)$ is a semi-orthogonal component of $D^b(X)$.*

The idea is to interpret the MMP in the language of derived categories.

Definition 16.6. Let X and Y be birational varieties. We say that X and Y are K-equivalent if there is a birational diagram

$$\begin{tikzcd}
 & Z \arrow[dl, "p"'] \arrow[dr, "q"] & \\
X \arrow[rr, dashed, "\varphi"] & & Y
\end{tikzcd}$$

with $p^* K_X = q^* K_Y$.

Remark 16.7 (Kawamata). The slogan is that reducing the canonical class K_X is reflected by reducing the derived category $D^b(X)$ by a semi-orthogonal component. If X and Y are K-equivalent, then $D^b(X) \simeq D^b(Y)$. Conversely, if $D^b(X) \simeq D^b(Y)$, then X and Y are K-equivalent.

Example 16.8. For $X \subset \mathbb{P}^5$ a smooth cubic hypersurface, there is an SOD

$$D^b(X) = \langle \mathcal{A}_X, \mathcal{O}(-2), \mathcal{O}(-1), \mathcal{O} \rangle.$$

The category $\mathcal{A}_X$ is triangulated and 2-Calabi–Yau: the Serre functor is isomorphic to the shift functor.

Remark 16.9 (Kuznetsov). X is rational if and only if $\mathcal{A}_X$ is equivalent to $D^b(S)$ for a K3 surface S.

Remark 16.10. A natural approach is to decompose semi-orthogonally as far as possible. One could try to call an SOD maximal if all its components are semi-orthogonally indecomposable. Then one might define atoms of X as the components of a maximal SOD of $D^b(X)$. A nontrivial atom would be one that does not appear as an atom of $D^b(Z)$ for $\dim Z \leq \dim X - 2$. The blow-up formula would then imply that nontrivial atoms are birational invariants. This would give the DK conjecture in the globally generated case, and one direction of Kuznetsov's conjecture.

Unfortunately this does not work. Atoms are not well defined this way, because SODs are not unique. For a blow-up of $\mathbb{P}^2$ at ten general points, one gets an SOD with a phantom component. This shows that the same category may admit different maximal SODs. The point of Kontsevich's idea is that atomic decompositions should be extra data, not a natural phenomenon.

Definition 16.11. Let X be a smooth projective variety. We say that $D^b(X)$ admits an atomic decomposition if

$$D^b(X) = \langle A_1, \dots, A_n \rangle$$

where the A_i are atoms. An atom is one of the components in such a decomposition.

Definition 16.12. A G-atomic theory is given by

- a class of G-varieties and G-morphisms $\mathcal{V}$;
- a class of G-invariant atomic SODs of $D^b(X)$ for $X \in \mathcal{V}$.

It is subject to:

1. as before;
2. as before.

The point is that the atomic decomposition is part of the data.

Remark 16.13. A prototypical case is

$$D^b(X) = \langle \operatorname{Ker} Rf_*, f^* D^b(Y) \rangle, \qquad Rf_* \mathcal{O}_X = \mathcal{O}_Y.$$

Definition 16.14. For an SOD

$$D^b(X) = \langle A_1, \dots, A_i, \dots, A_n \rangle$$

and $1 \leq i \leq n-1$, the ith right mutation is the unique SOD

$$D^b(X) = \langle A_1, \dots, A_{i-1}, A_{i+1}, \widetilde{A}_i, A_{i+2}, \dots, A_n \rangle.$$

Left mutation is defined similarly.

Remark 16.15. The subcategories $\widetilde{A}_i$ and A_i are equivalent. Mutations define an action of the braid group Br_n on the set of SODs.

Definition 16.16. Let

$$\mathcal{V}_{k,2}^{\mathrm{Bir},G} = \{G\text{-surfaces over } k,\ G\text{-equivariant birational morphisms}\}.$$

Theorem 16.17 (Elagin–Schneider–Shinder)**.** *Let $k = \overline{k}$ and let G be a group. Then there exists a G-atomic theory for $\mathcal{V}_{k,2}^{\mathrm{Bir},G}$.*

Remark 16.18. Let k be perfect. Then there is an atomic theory for $\mathcal{V}_{k,2}^{\mathrm{Bir}}$.

The main operation is the blow up or blow down of

$$\widetilde{X} \to X$$

along a finite G-orbit $Z = \{p_1, \dots, p_n\} \subset X$. The exceptional divisor is a disjoint union $E = E_1 \cup \cdots \cup E_n$ of (-1)- curves.

$$\begin{tikzcd}
\widetilde{X} \arrow[d] & E = E_1 \cup \cdots \cup E_n \arrow[l, hook'] \\
X & Z = \{p_1, \dots, p_n\} \arrow[l, hook']
\end{tikzcd}$$

Definition 16.19. If such a contraction exists, we say that X is G-minimal.

Theorem 16.20 (…,Iskovskikh-Manin,…)**.** *Given a G-surface Y, there exists a chain of blow ups*

$$Y = X_0 \to X_1 \to \cdots \to X_n = X$$

where X is G-minimal. Such X is one of:

(1) *a surface with* K_X *nef;*
(2) *a del Pezzo surface with* $\mathrm{Pic}^G(X) \cong \mathbb{Z}$*;*
(3) *a conic bundle* $X \to C$ *over a curve with* $\mathrm{Pic}^G(X/C) \cong \mathbb{Z}$*.*

Remark 16.21. If X is of type 1, then X and the chain $Y \to \cdots \to X$ are unique. If X is of type 2 or 3, they may not be unique. A Mori fibre space over a point, that is, a G-minimal del Pezzo surface $X \to \mathrm{pt}$, has degree $d = K_X^2$ in $\{1, \ldots, 9\}$. A Mori fibre space over a curve, that is, a relatively G-minimal conic bundle $X \to C$, has degree $d = K_X^2$ in $\{0, \ldots, 8\}$.

Remark 16.22. Let $f : \widetilde{Y} \to Y$ be the blow up of a G-orbit. An atomic SOD is

$$D^b(\widetilde{Y}) = \langle \mathcal{O}_{E_1}(-1), \ldots, \mathcal{O}_{E_d}(-1), f^*B_1, \ldots, f^*B_m \rangle,$$

where

$$D^b(Y) = \langle B_1, \ldots, B_m \rangle$$

is an atomic SOD.

Definition 16.23. An exceptional collection $E_1, \ldots, E_n$ is called a block if

$$\mathrm{Ext}^i(E_j, E_k) = 0 \quad \text{for all } i \text{ and } j \neq k.$$

Remark 16.24. For a G-minimal surface X/B we write $\mathcal{B}(X/B)$ for the standard atomic decomposition. It is defined by cases.

- If K_X is nef, then
$$\mathcal{B}(X) = \langle D^b(X) \rangle.$$
- If $\pi : X \to C$ is a G-conic bundle and C is irrational, then
$$\mathcal{B}(X/C) = \langle \ker \pi^*,\ \pi^* D^b(C) \rangle.$$
- If X is a G-minimal del Pezzo surface of degree $d \leq 4$, then
$$\mathcal{B}(X) = \langle \mathcal{O}_X^{\perp},\ \langle \mathcal{O}_X \rangle \rangle.$$
- If $\pi : X \to \mathbb{P}^1$ is a G-Mori conic bundle of degree $d \leq 4$, then
$$\mathcal{B}(X/P^1) = \langle \ker \pi^*,\ \langle \pi^* \mathcal{O}_{P^1}(-1) \rangle, \langle \mathcal{O}_X \rangle \rangle.$$
- If X is birationally rich, then the standard decompositions are
$$\mathcal{B}(\mathbb{P}^2) = \langle \langle \mathcal{O}(-2H) \rangle, \langle \mathcal{O}(-H) \rangle, \langle \mathcal{O} \rangle \rangle,$$
$$\mathcal{B}(\mathbb{P}^1 \times \mathbb{P}^1) = \langle \langle \mathcal{O}(-h_1 - h_2) \rangle, \langle \mathcal{O}(-h_1), \mathcal{O}(-h_2) \rangle, \langle \mathcal{O} \rangle \rangle,$$
$$\mathcal{B}(X_6) = \langle \langle \mathcal{O}(-H_1), \mathcal{O}(-H_2) \rangle, \langle \mathcal{O}(-h_1), \mathcal{O}(-h_2), \mathcal{O}(-h_3) \rangle, \langle \mathcal{O} \rangle \rangle,$$
and
$$\mathcal{B}(X_5) = \langle \langle \mathcal{E} \rangle, \langle \mathcal{O}(-h_1), \ldots, \mathcal{O}(-h_5) \rangle, \langle \mathcal{O} \rangle \rangle.$$

Theorem 16.25. *Any birational isomorphism between* G*-Mori fibre spaces* $X_1/B_1 \dashrightarrow X_2/B_2$ *decomposes into a composition of* G*-links, or is an isomorphism.*

Definition 16.26. A Sarkisov G-link is a birational diagram of Mori fibre spaces obtained by one of the standard Sarkisov types.

Theorem 16.27. *There is a classification of links.*

Example 16.28. A link of type II with $(\deg X_1, \deg Z, \deg X_2) = (9, 7, 8)$:

$$X_1 = \mathbb{P}^2, \qquad X_2 = F_0.$$

Here Z is the blow up of X_1 at an orbit of order 2 and of X_2 at a fixed point. The mutations are as follows.

Step 3. Classification of links.

Step 4. There is nothing to do due to the smart definition given in Step 1.

Plan.

1. Define SODs for nonminimal surfaces via minimal ones. 2. Define atomic SODs for minimal surfaces by considering cases. 3. Check that the definitions do not depend on the choices. 4. Check the axioms of an atomic theory.

Notes from now on are from Elagin's second talk. Some ideas may be repeated.

These atoms are different from the atoms discussed earlier in the notes. They play the same role in detecting obstructions to birational morphisms.

Definition 16.29 (Atoms in the sense of derived categories)**.** (Incomplete definition. . .) Let X be a smooth projective variety. We say that $D^b(X)$ admits an atomic decomposition if

$$D^b(X) = \langle A_1, \ldots, A_n \rangle$$

where the A_i are atoms. An atom is one of the components in such a decomposition.

The Brauer class

$$a = [A] \in \mathrm{Br}(\urcorner)$$

has index 3.

$$\mathrm{SB}(A) \cong \mathrm{SB}(A')$$

if and only if

$$A \cong A'$$

if and only if

$$a = a'.$$

Theorem 16.30 (Amitsur)**.**

$$\mathrm{SB}(A) \sim \mathrm{SB}(A')$$

if and only if

$$a = \pm a'$$

if and only if

$$\langle a \rangle = \langle a' \rangle$$

as subgroups in $\mathrm{Br}(\urcorner)$*.*

Theorem 16.31 (Kollár)**.** *Let C_1 and C_2 be conics over $\urcorner$, with*

$$C_i = \mathrm{SB}(A_i),$$

where A_i are central simple algebras of degree 2*. Then*

$$C_1 \times C_2 \sim C_1' \times C_2'$$

if and only if

$$\langle a_1, a_2 \rangle = \langle a_1', a_2' \rangle$$

as subgroups in $\mathrm{Br}(\daleth)$.

Example 16.32. Let X be a minimal del Pezzo surface of degree 6. Let

$$\overline{X} = X \times_{\daleth} \overline{\daleth} \cong \mathrm{Bl}_{p_1,p_2,p_3}(\mathbb{P}^2),$$

and let

$$G = \mathrm{Gal}(\overline{\daleth}/\daleth).$$

The invariant (L_2, a_3):

- $H_1, H_2 \in \mathrm{Pic}(\overline{X})$ are two classes of contractions $\overline{X} \to \mathbb{P}^2$. They are permuted by the Galois action.
- $L_2/\daleth$ is the splitting field of H_1 or H_2. Then

  $$[L_2 : \daleth] = 2.$$

- $$G_2 = \mathrm{Gal}(\overline{\daleth}/L_2) = \mathrm{Stab}(H_1) = \mathrm{Stab}(H_2) \subset G.$$
- The G_2-equivariant map

  $$|H_1| : \overline{X} \to \mathbb{P}^2$$

  descends to a birational map

  $$X_{L_2} \to \mathrm{SB}(A_3)$$

  for some central simple algebra A_3 of degree 3 over L_2.
- $$a_3 := [A_3] \in \mathrm{Br}(L_2).$$

The invariant (L_3, a_2) is introduced similarly using three classes $h_1, h_2, h_3 \in \mathrm{Pic}(\overline{X})$ that define conic bundles

$$\overline{X} \to \mathbb{P}^1.$$

Example 16.33. Atoms of a Severi–Brauer surface:

$$X = \mathrm{SB}(A), \qquad a = [A] \in \mathrm{Br}(\daleth), \qquad 3a = 0.$$

The atoms are

$$(\daleth, 2a), \quad (\daleth, a), \quad (\daleth, 0).$$

The subgroup is

$$\langle a \rangle = \{0, a, 2a\}.$$

Example 16.34. Let $X = C_1 \times C_2$, where

$$C_1 = \mathrm{SB}(A_1), \qquad C_2 = \mathrm{SB}(A_2),$$

and A_1, A_2 are central simple algebras of degree 2.

Over $\overline{\daleth}$ one has

$$D^b(\mathbb{P}^1 \times \mathbb{P}^1) = \langle \mathcal{O}(-1,-1), \mathcal{O}(-1,0), \mathcal{O}(0,-1), \mathcal{O} \rangle.$$

Over $\daleth$,

$$D^b(C_1 \times C_2) = \langle \mathcal{E}_{-1,-1}, \mathcal{E}_{-1,0}, \mathcal{E}_{0,-1}, \mathcal{O} \rangle.$$

The atoms are

$$(\daleth, a_1 + a_2), \quad (\daleth, a_1), \quad (\daleth, a_2), \quad (\daleth, 0).$$

Definition 16.35. Let X be a smooth projective variety. We say that $D^b(X)$ admits an atomic decomposition if

$$D^b(X) = \langle A_1, \dots, A_n \rangle$$

where the A_i are atoms. An atom is one of the components in such a decomposition.

Definition 16.36. An atom T is *small* if

$$T \cong D^b(\Lambda),$$

where Λ is a division $\daleth$-algebra. It may be noncommutative.

Definition 16.37. An atom T is *trivial* if

$$T \cong D^b(F),$$

where F is a field extension of $\daleth$.

Remark 16.38. A small atom $D^b(\Lambda)$ is given by a pair

$$(F, a),$$

where

$$F = Z(\Lambda)$$

and

$$a = [\Lambda] \in \mathrm{Br}(F).$$

Definition 16.39. A surface X is *birationally rich* if X is birational to a MFS of degree at least 5.

Remark 16.40. A MGS surface X is birationally rich if and only if

$$\deg(X) \geq 5.$$

Theorem 16.41. *Let X be a surface over a perfect field.*

- *X is birationally rich if and only if all atoms of X are small.*
- *X is rational if and only if all atoms of X are trivial.*

Example 16.42. Del Pezzo surface of degree 6.

Over $\bar{\daleth}$,

$$D^b(\mathrm{Bl}_3(\mathbb{P}^2)) = \langle \langle \mathcal{O}(-H_1), \mathcal{O}(-H_2) \rangle, \langle \mathcal{O}(-h_1), \mathcal{O}(-h_2), \mathcal{O}(-h_3) \rangle, \langle \mathcal{O} \rangle \rangle.$$

Over $\daleth$,

$$D^b(X) = \langle \mathcal{E}_2, \mathcal{E}_2, \mathcal{O} \rangle.$$

The atoms are

$$(L_2, a_3), \quad (L_3, a_2), \quad (\daleth, 0).$$

If a_3 or a_2 is not zero, then the corresponding atom is a birational invariant.

Example 16.43. For X a minimal del Pezzo surface of degree 4, the atomic decomposition is

$$D^b(X) = \langle \mathcal{A}_X, \mathcal{O} \rangle.$$

The category $\mathcal{A}_X$ is called the Kuznetsov component.

Theorem 16.44 (ESS, Shramov–Trepalin, 2025)**.** *Let X and X' be minimal del Pezzo surfaces of degree* 4*. Then the following conditions are equivalent:*

1. *X and X' are isomorphic.*
2. *X and X' are birational.*
3. *$\mathcal{A}_X \cong \mathcal{A}_{X'}$.*

17. Enriched Inflection and Secant Planes for Linear Series on Algebraic Curves

Speaker.
Ethan Cotterill

Title.
Enriched Inflection and Secant Planes for Linear Series on Algebraic Curves.

Abstract.
The fundamental local invariant of a linear series on an algebraic curve is its inflection, as measured by the vanishing of local derivatives in a point. Secant planes, which emerge from the rank behavior of tuples of points on the curve, may be regarded as (the manifestations of) a global analogue of inflection. Here we report on joint work with Darago, Garay López, Han, Shaska and Zhang in which we compute the classes of inflection points and of tuples of points spanning secant planes in the Grothendieck–Witt ring of an arbitrary perfect base field of characteristic not equal to 2.

Rank Loci in AG.
Enumerative algebraic geometry classically deals with the following paradigm. We want to count the number of points in the degeneracy locus of a map of vector bundles.

Given a map of vector bundles $\varphi : E \to F$ over a complex algebraic variety X, we would like to count the number of points $x \in X$ over which φ has rank at most k, for a fixed $k \in \mathbb{N}$.

One feature of this paradigm over $\mathbb{C}$ is that the number of solutions, counted with multiplicities, is uniquely determined. This is the principle of conservation of number.

For other choices of base field $\mathbb{F}$, the number of $\mathbb{F}$-rational solutions is not typically unique. For example, the number of lines on a cubic over $\mathbb{C}$ is 27, but over $\mathbb{R}$ it depends on the topology of the real locus.

Enriched AG.
Enrichments, or weights, are used to address the loss of conservation of number.

1. GW theory. Over $\mathbb{C}$, one counts holomorphic maps $\mathbb{P}^1 \to X$ subject to generic incidence and contact conditions with linear subvarieties of X, for example with a point of X. Trying to replicate this over $\mathbb{C}$ leads to Welschinger invariants, that is, GW invariants with signs.

2. Quadratic-enriched AG. Over an arbitrary field $\mathbb{F}$, solutions are weighted by isomorphism classes of symmetric bilinear forms over $\mathbb{F}$. Welschinger invariants are a special case. Conservation of number is achieved in GW($\mathbb{F}$), the Grothendieck–Witt group of $\mathbb{F}$.

3. Random AG. Compute expectations for cardinalities of zero-dimensional solution spaces with respect to distinguished probability distributions, for example over $\mathbb{R}$ or $\mathbb{Q}_p$.

Topological Degrees over $\mathbb{R}$.

Consider a polynomial $f(x)$ on the real line. For every zero, check the sign of the derivative of f at that point. The topological degree of f is

$$\sum_{f(p)=0} \operatorname{sign}(f'(p)).$$

More generally, any continuously differentiable map between compact oriented n-manifolds has a degree, which generalizes to the induced map in homology

$$f_* : H_*(M) \to H_*(N).$$

Local ring.

In the 1970s, Eisenbud–Levine and independently Khimshiashvili discovered how to compute local degrees at not necessarily regular values using algebraic geometry.

For a germ of a map

$$f = (f_1, \dots, f_n) : (\mathbb{R}^n, 0) \to (\mathbb{R}^n, 0),$$

define the local algebra of f as

$$\mathcal{Q}(f) = \mathbb{R}[[x_1, \dots, x_n]]/\langle f_1, \dots, f_n \rangle.$$

Assuming the zero is isolated, this is finite-dimensional, that is, an Artinian algebra.

Much more is true. By the early 1960s, it was noted that $\mathcal{Q}(f)$ is a Gorenstein ring, with one-dimensional socle spanned by the residue class of

$$\det J_f.$$

It also comes with a distinguished symmetric nondegenerate bilinear form, called the Scheja–Storch form. Eisenbud–Levine and independently Khimshiashvili showed that

$$\deg_0(f) = \operatorname{sgn} \phi.$$

Theorem 17.1 (EKL). *The signature of the form is the topological degree.*

The talk continued.

18. Atom applications and MMP

Speaker.

Ludmil Katzarkov

Title.

Atom applications and MMP.

Abstract.

In this talk we introduce the theory of Atoms. Applications and connection with MMP will be discussed.

These are notes of the second instance of Ludmil's talk, which was different than the first.

Plan

(1) Example
(2) Categorical approach
(3) CFT to the rescue

(4) Where do we go now

Consdier a 3-dimensional cubic

$$x_0^3 + \cdots + x_4^3.$$

If you degenerate this cubic, create one singular point, and, as was known to Arquimides(?) it becomes rational. The multiplicity-2 line intersections give us a map to $\mathbb{P}^3$, which is what makes it rational.

Let's have a look at this example from the point of view of Hodge theory. The hodge diamond of this cubic is:

[Put the hodge diamong]

If you introduce one singular point, then you are actually going to reduce by ... and after a smooth resultion, the hodge diamond of the resolved cubic looks as follows:

[Hodge diamond with 4's instead of 5's]

It's well known that the variety is blow up of $\mathbb{P}^3$ [along a quadric?] and we blow down the quadric.

Now let's discuss LG models. In this smooth 3d cubic,

$$W = \frac{(x+y+z)^3}{xyz} + z$$

So the LG model is $(\mathbb{C}^*, W)$. By HMS, .. is equivalent to the category of Lagrangians, but we won't use this in this talk.

In this picture, we have now atoms interpretation in the cohomology of our smooth cubic, which is split as direct sum :

$$H = \bigoplus H_\lambda = H_{\lambda_0} + H_{\lambda_1} + H_{\lambda_2}$$

In this picture, there is a very singular fiber... *figure denoting fibers...*.

This is all very classical. Other classical constructions like degenerations and birational geometry actually fit quite well with LG models and the behaviours of atoms.

Galois spectrum

Initially introduced by Rafael Rouqet. Later Orlov developed this introducing Orlov spectrum. When playing with the LG of the cubic we noticed there are two Lagrangian models — we never published this.

Let α be a generator of a category— that is, you can get to any object in this category by sums, summands, shifts and triangles. Cf. Alexander's talk.

Associate a number to each of the generators by forgetting about this operation. The generation time of alpha is the minimum number of triangles that we need to get from α to any other object in the category. Denote this number by $t(\alpha)$.

The smallest of the $t(\alpha)$ is called the Rouquet dimension of the category, which was conjectured by Rouquet to be the dimension of X.

Orlov introduced the notion of Orlov spectrum. This is old generation times:

$$\mathrm{OSpec}(\varphi) = t(\alpha_1), \ldots, t(\alpha_n).$$

Example 18.1.

$$\begin{aligned} D^b(\mathbb{P}^1) &= \langle 1, 2 \rangle \\ D^b(E) &= \langle 1, 2, 3, 4 \rangle \,. \end{aligned}$$

Using mirror symmetry a lot was done using generation time of Fukaya categories and [missing]. As a result of that, the following conjecture was made: there is a gap which appears, the gap of the Orlov spectrum, then the conjecture says that the maximum gap of the Orlov spectrum is bigger than $\dim(X) - 2$, then X is not rational.

These numbers are very hard to compute.

The maker of CFT was Mostow in the 80s (confirm). Other important names are Knizhnik-Zamwbolchicov, among others. At the time theory of singularity was proliferating in moscow. Steen-Brink spectrum, as part of CFT, who stated several conjectures, many of which have been proved by Arnolds and others.

This correction was not observed in Moscow, before mirror symmetry, but by Vata-Cecov, leading to Ditti equations. A zero dimensional singularity.

In a paper we wrote with M Balland, D Esvero, and (another?) by Y. Lin.

Vafa understood the connection with the asymptotics of the [missing name] equation.

Conifold transition. The properties of this spectrum, the CFT spectrum, …, all this could lead to birational invariants. A lot of work is done when CFT is not related to geometry.

But when geometry helps you, you can do a lot. In a way that's what [name] will be doing from now on.

We will be using quantum differential equation.

This connection appeared already in the toric case, where the group of birational transformations of a toric variety was expressible of the behvd at 00 of the correpondng Lg model the proptis of the crma grp where exppressed of the 1dim 2 dim circuits of the toric combinatorics

1 dimensional circuits corresponds to generators of the the bir trnasformation 2 circuits to sarkisov links and ?? dim ciurcits correspon to relations in the cremona gp.

The new components to all this is the differential equations — that's what I plan to tell you about.

I want to bring to examples here.

Consider the cubic cross $\mathbb{P}^1$, i.e. $X_3 \times \mathbb{P}^1$. And also a quartic multiplied by itself, $X_4 \times X_4$.

A 5-dimensional Fano: the intersection of quadric and cubic in $\mathbb{P}^7$, i.e. $Q_2 \cap C_3 \subset \mathbb{P}^7$. Here we have an interesting Hodge diamond:

$$\begin{array}{ccccc} & & 1 & & \\ & & 1 & & \\ 1 & 83 & & 83 & 1 \\ & & 1 & & \\ & & 1 & & \end{array}$$

the mirror, as observed by Candelas corresponds to [missing, related to a CY].

Now for the 7-dimensional cubic $C_3^7 \subset \mathbb{P}^8$

$$\begin{tikzcd} C_3^7 \arrow[d] \\ C_{3,0}^7 \end{tikzcd}$$

$$\begin{array}{ccccc} & & 1 & & \\ & & 1 & & \\ 1 & 84 & & 84 & 1 \\ & & 1 & & \\ & & 1 & & \end{array}$$

Here we have a 3-dimensional CY associated.

The idea is that if we know a lot about $Q_2 \cap C_3$ we can study C_3^7.

Consider the quantum differential equation

$$\left(\frac{\partial}{\partial u} - \frac{K}{u^2} + \frac{G}{u}\right)\psi(u) = 0.$$

This equation has subexponential growth solutions

$$\psi(u) = u^{\delta i}\log(\dots)(r + \dots \exp \dots)$$

We all $u^{\delta i}$ the *asymptotics*, which appeared in Leonardo's talk.

$$\delta = 2\text{maximum asymptotic} = 2\text{max}|D_i|$$

Theorem 18.2. *Let X be a Fano hypersurface of degree d in $\mathbb{P}^N$. Then*

(1) $\delta = \dim X - 2\frac{N+1-d}{d}$

(2) *if $\delta > \dim X - 2$ then X is not rational.*

In the case of C_3^3, the 3-dimensional cubic, $\delta = 3 - 2^{\frac{5-3}{3}} = \frac{5}{3}$. In the case of $X_3 \times \mathbb{P}^1$, we can use the corresponding LG model, which for $\mathbb{P}^1$ is $z + \frac{1}{z} = W$. There is no 2-dimensional atom which looks like that [figure of singular fibers]. This proves that it is not rational! Which of course is not something new. This would also work for many Fanos times $\mathbb{P}^1$.

The example of $X_4 \times X_4$ was proven that if you take birationally rigid Fanos of big dimension and big degrees then the product is of degree as well. For X_4, $\delta = 3 - 2^{\frac{5-4}{4}} = \frac{5}{2}$. So for $X_4 \times X_4$ we have $\delta(X_4 \times X_4) = 5$, which is bigger than $\dim X - 2$, and so according to the theorem we get that it's not rational.

We can apply this for 4-dimensional quintics: we compute that $\delta = 4 - 2\frac{6-5}{5} = \frac{18}{5}$, so $\delta(X_5 \times X_5) = \frac{18}{5} \cdot 2 > 6$.

Now for the intersection of a quadric and a cubic in $\mathbb{P}^7$, we cannot use asympotics only. Denote it by Y. This is a 5-dimensional Fano. We have four atoms:

$$H(Y) = H_{\lambda_0} + H_{\lambda_1} + H_{\lambda_2} + H_{\lambda_3}.$$

The way this is done is:

$$\delta = \dim X - 2^{\frac{8-3-2}{3}}$$

(we substract the degrees of the hypersurfaces and divide by the maximal one).

Let Z be the minimal model of algebraic variety in dimension 3, CY, general type, there exists a resolution of Z which has only curves as fibers [Corti, Kologhiros].

Then the quantum differential equation splits. And this way, as combination of atoms and birational geometry we see get non rationality.

To combine atoms and MMP, use this roadmap for nonrationality: start with the splitting of atoms $H = \bigoplus H_{\lambda_i}$ and restrict the quantum differential equation. We have $\Delta_{-k}, \ldots, \Delta_k$ If $k < \dim X$, $\Delta_{-k} > \dim X - 2$, $\Delta_k = \dim X - 2$. And use your knowledge on mirror symmetry to rule out the example. Apply MMP and atoms to rule out this case.

19. Degeneracy loci of Poisson fivefolds

Speaker.
Jorge Vitório Pereira

Institution.
Instituto de Matemática Pura e Aplicada.

Abstract.
Let X be a Fano manifold of dimension five and let

$$\sigma \in H^0(X, \wedge^2 T_X)$$

be a holomorphic Poisson structure. We prove that the rank zero locus $D_0(\sigma)$ has an irreducible component of dimension at least 1 and that the rank at most two locus $D_2(\sigma)$ has an irreducible component of dimension at least 3. This confirms a conjecture by Bondal in dimension five. The proof uses: (i) an algebraic integrability criterion for codimension one foliations on weak Fano manifolds; (ii) a lemma of Gualtieri–Pym on degeneracy loci of ample Poisson modules; and (iii) a generalization of a result by Esteves-Kleiman on the zero loci of Pfaff fields along invariant subvarieties.

20. Degeneracy loci of Poisson fivefolds

Speaker.
Jorge Vitório Pereira

Institution.
Instituto de Matemática Pura e Aplicada

Abstract.
Let X be a complex manifold.

Definition 20.1. A Poisson structure is a global bivector

$$\sigma \in H^0(X, \wedge^2 T_X)$$

such that

$$[\sigma, \sigma] = 0,$$

where the bracket is the Schouten bracket.

This allows us to define a Hamiltonian map

$$\mathcal{O}_X \to T_X,$$

given by

$$f \mapsto \langle \sigma, df \rangle.$$

We also get a morphism

$$\sigma^\sharp : \Omega^1_X \to T_X,$$

and the map

$$j : \mathcal{O}_X \to \Omega^1_X.$$

Definition 20.2. The $2k$-th degeneracy locus of σ is

$$D_{2k}(\sigma) = \{x \in X : \mathrm{rk}(\sigma^\sharp_x) \leq 2k\}.$$

This is the vanishing of a Pfaffian and is thus an algebraic variety. Moreover it is a strong Poisson subvariety.

Conjecture (Bondal).

Suppose that X is a Fano manifold, that is, K_X is antiample, and that $\sigma \in \wedge^2 T_X$ is a Poisson structure. Then

$$\dim D_{2k}(\sigma) \geq 2k+1.$$

We expect it to have dimension $2k$, so the conjecture says it is in fact bigger than expected.

Known cases.

- Max degeneracy: D_{2n-2} in odd dimension (Polishchuk, 1997). Using index and Baum–Bott theory for foliations.
- Generalization: Beauville observed the same argument works for D_{2r-2} for any maximal rank $2r$.
- Even dimension: D_{2n-2} follows from nontriviality of the anticanonical bundle.
- Submaximal, even dimension: D_{2n-4} was proved by Gualtieri–Pym in 2013.

Theorem 20.3 (Druel–Pereira–Pym– Touzet)**.** *Let X be a Fano manifold of dimension $2n+1$, with $n > 1$, and let σ have rank $2n$. Then*

$$D_{2n-4}(\sigma)$$

has an irreducible component of dimension at least $2n-3$.

This gives Bondal's conjecture in dimension 5.

Strategy of proof.

1. Consider a saturated foliation $\mathcal{F}_\sigma \subset T_X$.
2. Two main cases according to $\mathrm{codim}(\mathrm{Sing}(\mathcal{F}_\sigma))$:
    - codim ≥ 3: algebraic integrability implies reduction to even dimension.
    - codim $= 2$: detailed analysis via modular residues and cohomology (obstructions).

Three main techniques.

Technique 1: algebraic integrability.

Theorem 20.4 (Druel–Pereira–Pym– Touzet)**.** *Let X be weak Fano and let $\mathcal{F}$ be codimension 1 on X with*

$$\mathrm{codim}(\mathrm{Sing}(\mathcal{F})) \geq 3.$$

Then $\mathcal{F}$ is defined by a morphism

$$f : X \to \mathbb{P}^1$$

with irreducible fibers.

- A foliation $\mathcal{F}$ of codim q on X is a saturated involutive subsheaf $T\mathcal{F} \subset T_X$ of corank q.
- Normal sheaf: the dual of $\operatorname{Ann}(T\mathcal{F}) \subset \Omega^1_X$.
- Singular locus: $\operatorname{Sing}(\mathcal{F})$ is where $T_X/T\mathcal{F}$ is not locally free.

Key facts.

- $N_{\mathcal{F}} \cdot C = 2$ (Baum–Bott for tangent curves).
- If C is not tangent, then $N_{\mathcal{F}} \cdot C \geq 2$.

Definition 20.5. A Pfaff field of degree p on X is a pair (L, η) where L is a line bundle and

$$\eta : \Omega^p_X \to L$$

is $\mathcal{O}_X$-linear. Equivalently,

$$\eta \in H^0(X, \wedge^p T_X \otimes L),$$

by contraction with η.

Singular locus. [missing]

Invariant subvarieties. [missing]

Poisson structures give Pfaff fields.

Proposition.

Let X be a compact Kähler manifold of dimension n, and let

$$\eta : \Omega^p_X \to L$$

be a Pfaff field. Let $S \subset Y \subset X$ be closed subvarieties with:

(1) Y is η-invariant, irreducible, and
$$\dim Y = p + r.$$
(2) $\dim S \leq p - 2$.
(3) $Y^\circ := Y \setminus S$ is smooth.
(4) $\eta|_T$ does not vanish in codimension 1, and defines a foliation $\mathcal{F}^\circ$ of dimension p.
(5) $\mathcal{F}^\circ$ admits a closed strongly directed positive current T°.

Then the morphism

$$H^\eta : H^p(X, \Omega^p_X) \to H^p(X, L)$$

is nonzero.

Remark.

Condition 5 is automatic when $r = 0$. The current of integration along Y° works.

Connection with the work of Esteves–Kleiman.

- Work on algebraic varieties over arbitrary fields.
- If Y is η-invariant and $\dim \operatorname{Sing}(Y, \sigma|_Y) \leq p - 2$, then the cohomological maps vanish.
- Etc.

Proposition.

Let X be a compact Kähler Poisson manifold. Suppose $Y \subset X$ is closed and Poisson.

(1) $\sigma|_Y$ has generic rank k.

(2) $\dim \mathrm{Sing}(Y, \sigma|_Y) \leq 2k - 2$.
(3) The symplectic foliation supports a strongly directed positive current.

Then

$$h^{0,2k}(X) > 0.$$

In other words, we deduce the existence of a $(0, 2k)$-form.

Modular residues.

Gualtieri–Pym introduced modular residues:

$$\mathrm{Res}_k^{\mathrm{mod}}(\sigma) \in H^0\left(D_{2k}(\sigma), \mathfrak{X}^{2k+1}_{D_{2k}(\sigma)}\right).$$

Here $\mathfrak{X}_{D_{2k}(\sigma)}$ is the singular version of the tangent sheaf.

The local description is omitted here; it involves a Poisson connection and a holomorphic connection.

Lemma (Gualtieri–Pym).

If $Y \subset D_{2k}(\sigma)$ is irreducible, then

$$\mathrm{Sing}(Y, \sigma|_Y)$$

has a component of dimension at least $2k - 1$.

Proof of the main theorem.

Case 1.

$\mathrm{codim}(\mathrm{Sing}(\mathcal{F})) \geq 3$.

- By the integrability theorem, σ defines a morphism

$$f : X \to \mathbb{P}^1.$$

- A general fiber F is a Fano manifold of dimension $2n$.
- F is a Poisson subvariety.
- Apply the known even-dimensional result of Gualtieri–Pym:

$$D_{2n-4}(\cdots).$$

Done. This is the easy case.

Case 2.

$\mathrm{codim}(\mathrm{Sing}(\mathcal{F})) = 2$.

Setup.

Let Y be an irreducible component of

$$\mathrm{Sing}(\mathcal{F}),$$

with

$$\dim Y = 2k - 1.$$

Set

$$S := \mathrm{Sing}(Y, \sigma|_Y).$$

Then S and Y are strong Poisson subvarieties of

$$D_{2k-2}(\sigma),$$

and

$$\dim S \leq 2k - 2.$$

Three subcases.

- $\dim S = 2k - 3$.

- $\dim S = 2k-2$: use the previous proposition and other considerations.
- $\dim S \leq 2k-4$: the most delicate case. Set

$$\rho = \mathrm{Res}^{\mathrm{mod}}_{k-1}(\sigma)|_{Y^\circ}.$$

Subcase 2.1.

If ρ vanishes somewhere, then the lemma gives a component.

Subcase 2.2.

The remaining case is omitted here.

Summary.

- Bondal's conjecture is confirmed for the submaximal degeneracy locus of odd-dimensional Fano manifolds.
- This proves the conjecture in dimension five.
- The proof combines:
  - algebraic integrability of codimension 1 foliations on weak Fano manifolds;
  - modular residues of Poisson structures;
  - Esteves–Kleiman type cohomological constraints via positive currents on compact Kähler manifolds.

21. Spectral decomposition of nc-Hodge structures and F-bundles

Speaker.

Shaowu Zhang

Title.

Spectral decomposition of nc-Hodge structures and F-bundles.

Abstract.

Non-commutative Hodge structures generalize classical Hodge structures for non-commutative spaces. They consist of de Rham and Betti data, encoded as Stokes structures. These structures naturally arise in mirror symmetry in the context of the Gamma conjectures. In this talk, I will first discuss the behavior of Betti data under the topological Laplace transform, relating this to the spectral decomposition of non-commutative Hodge structures. I will then talk about the formal and non-Archimedean analogues of nc-Hodge structures, known as F-bundles, focusing on the spectral decomposition theorem for maximal F-bundles and its connection to Hodge atoms.

22. Blowups, Gale duality and moduli spaces

Speaker.

Carolina Araujo

Title.

Blowups, Gale duality and moduli spaces.

Abstract.

We discuss the birational geometry of blowups of projective spaces at points in general position. For this we explore Gale duality, a correspondence between sets of $n = r + s + 2$ points in $\mathbb{P}^s$ and $\mathbb{P}^r$. For small values of s, this duality has a

geometric manifestation: the blowup of $\mathbb{P}^r$ at n points can be realized as a moduli space of vector bundles on the blowup of $\mathbb{P}^s$ at the Gale dual points.

Joint work with Castravet, Kaur, Martinelli.

Notes

Setup. Consider a collection of points in "very general position". Consider X, the blow up of $\mathbb{P}^n$ in at $P_1, \dots, P_k$. Problem: describe all the "rational contraction" that variety admits. If the number of points is small the geometry is simple. As the number grows, the geometry becomes increasingly more complicated and eventually becomes infinite.

Consider the example of $Bl_P\mathbb{P}^2$. It's a $\mathbb{P}^1$ bundle over $\mathbb{P}^1$. Now increase the number of points, say,m $Bl_{P_1,P_2}\mathbb{P}^2$ now we have two exceptional divisors. Consider the strict transform of a line $\ell \subset \mathbb{P}^2$, $\tilde{\ell}$ which is isomorphic to $\mathbb{P}^1$ and has $\ell^2 = -1$, which by Castelnuovo's thoerem we know we can blow down, yielding a quadric. These are all the rational contractions of this blow-up.

Finally consider three points, $Bl_{P_1,P_2,P_3}\mathbb{P}^2$ this is del Pezzo 3. Consdier the stric transform of the three lines passing therough this points. Blowing down leads to the standard Cremona transformation! If we lift it to X, we obtain an automorphism of X.

Denote $H = \pi^*\mathcal{O}_{\mathbb{P}^2}(1)$ and E_i the exceptional divisor over P_i. $\mathrm{Pic}(X)$ and $N^1 = \mathrm{Pic}(X) \otimes_{\mathbb{Z}} \mathbb{R} cong \mathbb{R}^4$. A reflection q^* across a hyperplane, $q : (x : y : z) \mapsto (???)$ which preserves $Nef(X) \subset Eff(X) \subset N^1$. We can construct such transformations in similar ways, and extending to reflections in PicL $q^*_{ijl} : N^1(X) \to N^1(X)$ isometry. We can do this for any collection of 3 points, and we obtain a Weyl subgroup

$$W := \langle q_{ijl} \rangle \subset O(N^1)$$

whose action preserves (-1)-curves and the cones $Nef(X) \subset Eff(X) \subset N^1(X)$.

W is finite $\iff k \leq 8$.

Theorem 22.1. *blowup of* $\mathbb{P}^2$ *??*

- for $k \leq 8$ X has finitely many (-1) curves ($-K_X$ is ample.
- for $k \geq 9$, X has inifintely many (-1)-curves

$$\begin{cases} k = 9 & -K_X \text{ is nef} \\ k > 9 & -K_X \text{ is not nef} \end{cases}$$

- $k = 9$ [Cone diagram]
- $k \geq 10$ [Cone diagram]

Theorem 22.2 (D'urso, impa!)**.** $Nef(X_{K_X \leq 0})$ *has a finite polyhedral fundamental domain for the action of* W*.*

Example 22.3. $X = Bl_{P_1,P_2}\mathbb{P}^n \xrightarrow{\pi} \mathbb{P}^n$, $n \geq 3$.

- $H = \pi^*\mathcal{O}_{\mathbb{P}^n}(1)$.
- E_i exceptional divisors
- H_i strict trfm of hyperplane
- H_{12} strict transform of hyperplane through P_1 and P_2.

$$Nef(X) \subset Mov(X) \subset Eff(X) \subset N^1(X).$$

[Diagram of a triangle of the Nef cone see cremona-transformations.tex]

$$\begin{tikzcd}
& Y \supset F = \mathbb{P}^1 \times \mathbb{P}^{n-2} \arrow[dl, "Bl_{\tilde{\ell}_{12}}"'] \arrow[dr, "\mathrm{Cont}_F"] & \\
\mathbb{P}^1 \cong \ell_{12} \subset X \arrow[rr, dashed, "f\ell_p"] \arrow[dr, "\mathrm{contr}_{\tilde{\ell}_{12}}"'] & & X' \supset Z \cong \mathbb{P}^{n-2} \arrow[dl] \\
& W &
\end{tikzcd}$$

Finally, the blow up of $\mathbb{P}^n$ at $k \geq n+1$ very generap points.

For $I \subset \{1, \ldots, k\}$ $|I| = n+1$ we obtain reflections $(q_l)^* : N_1(X_l) \cong N^1(X_l)$.

$$\begin{tikzcd}
X_l \arrow[r, dashed, "\cong \text{ pseudo}"] \arrow[d] & X_l \arrow[d] \\
\mathbb{P}^n \arrow[r, dashed, "q_l"] & \mathbb{P}^n
\end{tikzcd}$$

Coble pairing on $\mathrm{Pic}(X)$:

- $H^2 = n - 1$
- ??aaaa
- ??

Mori dream spaces

Definition 22.4.

$$\mathrm{Cox}(X) = \sum_{D \in \mathrm{Pic}(X)} H^0(X, \mathcal{O}_X(D)).$$

A variety is a *Mori dream space* is the Cox ring above is finitely generated.

Theorem 22.5 (Batryev-Popov, 2004)**.** *For Bl of $\mathbb{P}^2$ at k points in very general postn X is MDS iff $k \leq 0$ iff $-K_X$ is ample.*

Definition 22.6. [Missed it, picture]

Theorem 22.7 (Hu-Keel, 2000)**.** *Let X be a Mori Dream space, then*

- *The effective cone $Eff(X)$*
  - *is generated by finitely many prime divisors*
  - *decomposes into subcones $\leftrightsquigarrow$ rational contractions of X*
- *The nef cone is generate by finitely many semiample divisors.*
- *The movable cone decomposes into subcones.*

Upshot: The birational geometry of MDS is finite! And... we know exactly which MDS are out there.

Theorem 22.8 (Mukari, 2001, Castravel-Tevelev, 2005)**.** *Tje blowup of $\mathbb{P}^n$ at k points is a MDS iff **conditions***

Problems.

- Mori chamber decomposition of $Eff(X)$ in the MDS cases.
- What can be said in the non-MDS cases? (we shall focus on this one today!) in the above theorem, $k = n+2$ and $n \geq 5$.

Gale duality.
Let $n, s \in \mathbb{Z}_{\geq 0}$ and $k = n + s + 2$

Gale correspondence is:

$$\{\{P_1, \dots, P_k\} \subset \mathbb{P}^n\} / \sim \quad \leftsquigarrow\!\!\rightsquigarrow \quad \{\{Q_1, \dots, Q_n\} \subset \mathbb{P}^s\} / \sim$$

where the bijection is up to projective equivalence.

Example 22.9. Take $n+3$ points in $\mathbb{P}^1$ here's how I give $n+3$ points in $\mathbb{P}^n$: there's a unique rational curve passing through... [check, this is for $s = 1$ and is very simple].

More generally, we would use a matrix, its transpose, and complete with the identity matrix. This is due to Gale in the 50's, when he studied convex geometry, polytopes, etc. But is also related to the work of Coble (more recently).

This Gale duality is compatible with Cremona transformations. Let

$$P_1, \dots, P_{n+1}, P_{n+2}, \dots, P_{n+s+2} \leftsquigarrow\!\!\rightsquigarrow Q_1, \dots, Q_{n+1}, Q_{n+2}, \dots, Q_{n+s+2}$$

$iP_1, \dots, P_{n+1}, \dots, P_{n+1}, \hat{P}_{n+2}, \dots, \hat{P}_{n+s+2} \leftsquigarrow\!\!\rightsquigarrow \hat{Q}_1, \dots, \hat{Q}_{n+1}, Q_{n+1}, \dots, Q_{n+s+2}$

And in fact, this has related Dynkyn diagrams. [Obviously an explanation of why this has to do with Cremona transformations is missing]

Now, there exists a linear isomorphism

$$\varphi : N^1(X) \to N^1(X)$$

which

- sends the canonical to the canoinca
- and restricts to an isometry $K_X^\perp \to K_Y^\perp$.

This is due to the relation if Gale duality with the Cremona transformations explained above.

Theorem 22.10 (Mukai 2001,Baue r 1991)**.** *Description of the Mori chamber decomposition of the effective cone when X is the blowup of $\mathbb{P}^n$ at $n+3$ very general points.*

(This is the case we highlighted above as "the interesting for us" based on the description of the MDS there exist theorem.)

Theorem 22.11 (Mukai 2001, Casagrande- Codogni-Fanelli 2019)**.** *Description of mori chamber when X is bloup of $\mathbb{P}^4$ at 8 v general pts.*

Theorem 22.12 (A-Castravet-Kaur-Matrinelli)**.** *PArtiall extended this description for??*

Idea of proof.
$P_1, \dots, P_{n+4} \in \mathbb{P}^n$ and dual Gale points $Q_1, \dots, Q_{n+4} \in \mathbb{P}^2$.

$X =$ blowup of $\mathbb{P}^n$ at $P_1, \dots, P_{n+4}$, $S =$ blowup of $\mathbb{P}^2$ at $Q_1, \dots, Q_{n+4}$.

On S we fix invariants $r = 2, c_1 = -K_S$ and $c_2 = 2$.

$L \in Amp(S) \rightsquigarrow$ Moduli space of Gieseker L-semistable torsion free sheaves, $\overline{\mathcal{M}_L}$.

Variation of $\overline{\mathcal{M}_L}$ as L varies. Friedman-Qin (95), Göttsche (96), Matsuki-Wentworth (97).

- Chamber decomposition on $Amp(S)$ ($L \equiv L' \iff \overline{\mathcal{M}}_L = \overline{\mathcal{M}}_{L'}$.

- (In the 90s, what people where doing. . .) birational maps between models on $K_S < 0$.

Theorem 22.13 (A-Castravet-Kaur-Martinelli)**.**
- *$\exists L_0 \in Amp(S)$ such that $X \cong \overline{\mathcal{M}}_{L_0}$.*
- *Mori chamber decomposition on $Eff(X) \cap (K_X < 0)$.*

23. Lino

Speaker.
Lino Anderson da Silva Grama (IMECC)

Title.
SYZ for almost abelian Lie groups.

Abstract.

In this talk, we discuss SYZ mirror symmetry in the non-Kähler context, in the sense of Lau–Tseng–Yau. We construct SYZ dual pairs for a family of solvmanifolds called almost abelian, and we explore applications related to dualities between Bott–Chern and Tseng–Yau cohomologies. This is joint work in progress with L. Cavenaghi, L. Katzarkov, and P. Muniz.

Definition 23.1. An SU(3) structure (X, ω, Ω) is supersymmetric of type IIB if Ω defines an honest complex structure and ω defines a balanced metric. That is,

$$d\Omega = 0$$

and

$$d(\omega^2) = 0.$$

For the conformal factor F, there is ρ_B, the RR flux source current. It is a $(4, 4)$-form.

Remark 23.2. There is a system of equations. When ω is symplectic and F is constant, X is a CY and

$$\rho_B = 0.$$

In this case X has SU(3) holonomy.

Remark 23.3. We also have a system of three equations which determine the structure. If $d\Omega = 0$, we again get a CY manifold.

Definition 23.4. An SU(3) structure is supersymmetric of type IIA if

$$d\omega = 0$$

and

$$d(\operatorname{Re}\Omega) = 0.$$

Now ρ_A is an exact 3-form.

Definition 23.5. An SU(n) structure (X, ω, Ω) is semi-flat if X admits a Lagrangian torus bundle

$$\mu : X \to B$$

and

$$\omega, \Omega \in \Omega^*_B(X, \mathbb{C}).$$

That is, they are base dependent forms.

Definition 23.6. The polarization switch operator $\mathcal{P}$ acts on $\Omega^{p,q}_B(X)$ locally as

$$\alpha = \sum f_{I,J}\, dz_I \wedge d\bar{z}_J \mapsto \mathcal{P}(\alpha) = \sum f_{I,J}\, dz_I \wedge dz_J.$$

Fiber product setup.

Let

$$X \times_B \widehat{X}$$

be the fiber product of dual torus bundles over B. It is the correspondence space for the Fourier–Mukai transform.

Fourier–Mukai transform.

The transform is a push-pull operator with a correction term given by a wedge product with an exponential.

Theorem 23.7 (Lau–Tseng–Yau, 2015)**.** *Let ω be a torus invariant real (1,1)-form on X. Let*

$$\widehat{\Omega} = \mathrm{FT}(e^{2\omega}).$$

Then

- (X, Ω, ω) *is* $\mathrm{SU}(n)$ *if and only if*
- $(\widehat{X}, \widehat{\omega}, \widehat{\Omega})$ *is the dual* $\mathrm{SU}(n)$ *structure.*

SYZ for 3-d solvmanifolds.

Bedulli–Vannini (2024) study SYZ mirror symmetry in the setting of 3-dimensional solvmanifolds. Their result fits the semi-flat and Fourier–Mukai picture above.

Up to isomorphism, there are four simply connected unimodular solvable Lie groups of dimension 3.

- $\mathbb{R}^3$.
- Nil^3.
- Sol^3.
- G_α, the remaining non-nilpotent unimodular solvable case.

Theorem 23.8 (Bedulli–Vannini, 2024)**.** *For dual torus bundles*

$$\widehat{X} \to B \qquad \text{and} \qquad X \to B$$

- $\widehat{X}$ *is a solvmanifold for the semidirect product of G with* $(\mathbb{R}^n)^*$*.*
- X *is a solvmanifold for the semidirect product of G with* $\mathbb{R}^n$*.*
- *An invariant type IIA structure on X corresponds to an invariant type IIB structure on $\widehat{X}$.*

Here

$$\rho : G \to \mathrm{GL}(n, \mathbb{R})$$

is the linear part of the affine representation.

Definition 23.9. A Lie group is called almost abelian if its Lie algebra $\mathfrak{g}$ has an abelian ideal $\mathfrak{h}$ of codimension 1. That is, choosing $e_0 \notin \mathfrak{h}$, $\mathfrak{g}$ decomposes as the sum of $\mathbb{R}e_0$ and $\mathfrak{h}$. Such a Lie algebra is also called almost abelian.

Fix a basis $\mathcal{B}$ of $\mathfrak{h}$, and let A be the matrix representing $\mathrm{ad}_{e_0}|_{\mathfrak{h}}$ with respect to $\mathcal{B}$. If $\dim \mathfrak{h} = d$, then $\mathfrak{g}$ is determined by

$$[e_0, v] = Av.$$

There are two sides, the complex side and the symplectic side.

Theorem 23.10. *The pairs constructed above on each side are supersymmetric of types IIB and IIA. The duality is given by the Fourier–Mukai transform.*

Remark 23.11. Idea of the proof.

Type IIB for (X, Ω, ω):

- $d\Omega = 0$ by construction.
- To get $d(\omega^{n-1}) = 0$, use $\operatorname{tr} A = 0$.
- Compute $d\omega$.
- Then
$$d(\omega^{n-1}) = \cdots$$

Type IIA:

- Show
$$\mathrm{FT}(e^{2\omega}) = 0?$$
- Use the duality to get the symplectic side.

Theorem 23.12 (Andrada–Orfila, 2016)**.** *Let*

$$G$$

be the semidirect product of $\mathbb{R}$ *and* $\mathbb{R}^{2n+1}$ *via* φ*. Then* G *admits a lattice if and only if there exists* $t_0 \neq 0$ *such that* $\varphi(t_0)$ *is conjugate to an integer matrix. In that case, the lattice is the group* $t_0\mathbb{Z} \times \Gamma$*, with multiplication induced by* $\varphi(t_0)$*, where* $\Gamma \subset \mathbb{R}^{2n+1}$ *is a lattice.*

Theorem 23.13. *There is another lattice existence theorem for solvable Lie groups.*

Cohomology theories.

Definition 23.14. Let X be a real $2n$-fold and let ω be a nondegenerate 2-form. A real polarization with respect to ω is an integrable Lagrangian distribution

$$\{\Lambda_x \subset T_xX : x \in U\}$$

over an open dense subset $U \subset X$.

Given a real polarization (Λ, U) on (X, ω) and a metric on X, there is a splitting of the tangent bundle

$$TU = \Lambda \oplus \Lambda^{\perp}$$

and a splitting of forms

$$\Omega^k(U) = \bigoplus_{p+q=k} \Omega_{\Lambda}^{(p,q)}(U).$$

Here $\Omega_{\Lambda}^{(p,q)}(U)$ is the subspace of forms of type (p, q) with respect to Λ.

Definition 23.15. The Bott–Chern cohomology of a complex manifold X is

$$H^{p,q}_{BC}(X) = \frac{\ker d \cap \Omega^{p,q}(X)}{\operatorname{im} \partial\bar{\partial}}.$$

It is the complex-side analogue of a polarized cohomology theory.

Definition 23.16. Let $(\widehat{X}, \widehat{\omega})$ be a Lagrangian bundle over B with real polarization Δ. The Tseng–Yau cohomology is

$$H^{(p,q)^{\Delta}}_{B,TY}(\widehat{X}) = \frac{\ker(d + d^{\Lambda}) \cap \Omega_{\Delta}^{(p,q)}(\widehat{X})}{\operatorname{im} dd^{\Lambda}}.$$

The point is that it is similar to Bott–Chern cohomology, but using d^{Δ} on the symplectic side.

Theorem 23.17 (Lau–Tseng–Yau, 2015)**.** *If Δ is a real polarization on $\widehat{X}$ coming from the bundle structure, then the Fourier–Mukai transform gives an isomorphism between the Bott–Chern and Tseng–Yau cohomologies.*

The problem of computing Bott–Chern cohomology for solvmanifolds was studied by Angella–Kasuya.

Theorem 23.18 (Angella–Kasuya)**.** *Let G be a connected solvable Lie group, and let Γ be a lattice in G. Then the Bott–Chern cohomology of G/Γ can be computed from invariant forms under suitable assumptions.*

We use this theorem for

$$G \times \mathbb{R}^n \cong \mathbb{C} \times \mathbb{C}^{n-1},$$

where

$$\phi(x_1 + iy_1) = \operatorname{diag}\big(e^{x_1\lambda_2}, \ldots, e^{x_1\lambda_n}\big).$$

Theorem 23.19. *This theorem gives an expression for the Bott–Chern cohomology*

$$H^{p,q}_{BC}\big(G \times \mathbb{R}^n/\Gamma \times \Xi\big).$$

Lemma 23.20. *For*

$$h^{p,q}_{BC} = \dim H^{p,q}_{BC}$$

and

$$h^{p,q}_{TY} = \dim H^{p,q}_{TY},$$

we have

$$h^{p,q}_{BC} = N(p,q)$$

and

$$h^{n-p,q}_{TY} = N(n-p,q).$$

Lemma 23.21. *The Hodge numbers satisfy*

$$h^{p,q}_{BC} = h^{q,p}_{BC}$$

and

$$h^{p,q}_{BC} = h^{n-p,n-q}_{BC}.$$

Definition 23.22. The symplectic Hodge star operator is the operator

$$\star_s : \Omega^k(\widehat{X}) \to \Omega^{2n-k}(\widehat{X})$$

defined by

$$\alpha \wedge (\star_s \beta) \ = \ \langle \alpha, \beta \rangle \frac{\widehat{\omega}^n}{n!}$$

for forms α, β of degree k.

Theorem 23.23 (Brylinski)**.** *If $(\widehat{X}, \widehat{\omega})$ is a symplectic $2n$-manifold, then there is an isomorphism*

$$H^{2n-k}_{d^\Lambda}(\widehat{X}) \simeq H^k_{dR}(\widehat{X})$$

given by the symplectic Hodge star operator.

Definition 23.24. Tseng and Yau proposed the symplectic cohomology

$$H^k_{d+d^\Lambda}(\widehat{X}) = \frac{\ker(d + d^\Lambda) \cap \Omega^k(\widehat{X})}{\operatorname{im} d \cap \Omega^k(\widehat{X}) + \operatorname{im} d^\Lambda \cap \Omega^k(\widehat{X})}.$$

Definition 23.25. Tseng and Yau also proposed

$$H^k_{dd^\Lambda}(\widehat{X}) = \frac{\ker(dd^\Lambda) \cap \Omega^k(\widehat{X})}{\operatorname{im} d \cap \Omega^k(\widehat{X}) + \operatorname{im} d^\Lambda \cap \Omega^k(\widehat{X})}.$$

Properties.

- If X is compact, both cohomologies are finite.
- One can do Hodge theory: harmonic forms and Hodge decomposition.
- They are not isomorphic to de Rham cohomology.
- They are invariant under symplectomorphisms and Hamiltonian isotopy.

Theorem 23.26 (Tseng–Yau)**.** *If $(\widehat{X}, \widehat{\omega})$ is a compact symplectic manifold, then*

$$H^k_{d+d^\Lambda}(\widehat{X}) \simeq H^{2n-k}_{dd^\Lambda}(\widehat{X}).$$

Definition 23.27. We say that $\widehat{X}$ satisfies the dd^Λ lemma if

1. every form that is both d-exact and d^Λ-closed is dd^Λ-exact,
2. every form that is both d^Λ-exact and d-closed is dd^Λ-exact,
3. every form that is both d-closed and d^Λ-closed is cohomologous to a d-exact form only if it is dd^Λ-exact.

Theorem 23.28. *The dd^Λ lemma holds if and only if the natural map to de Rham cohomology is an isomorphism.*

Remark 23.29. If $\widehat{X}$ is a Kähler manifold, then it satisfies the dd^Λ lemma.

Questions.

- Is there a more natural way to obtain the cohomologies introduced above?
- What is the right notion of symplectic cohomology here?

New proposal of symplectic cohomology.

Introduce a formal variable u and consider the complex

$$d + ud^\Lambda.$$

This gives four possible cohomologies:

- dd^Λ cohomology.
- Periodic cyclic cohomology.
- The symplectic analogue of Bott–Chern cohomology.
- The symplectic analogue of Aeppli cohomology.

Let $N_1 = \ker(d_1)$ and $N_2 = \ker(d_2)$. If $N_1 \subset N_2$, then the inclusion induces two parallel exact sequences.

Lemma 23.30. *There is a canonical isomorphism between the cokernels.*

Theorem 23.31. *Let $(\widehat{X}, \widehat{\omega})$ be a symplectic manifold. Then the symplectic Bott–Chern cohomology has odd and even parts. These are the two graded pieces of the new proposal.*

Let $\widehat{X} \to B$ and $X \to B$ be non-Kähler SYZ mirror partners. Denote by $H_{B;BC,s}$ the cohomology restricted to the basic forms.

Theorem 23.32. *The Fourier–Mukai transform induces an isomorphism*

$$H_{B;BC,c}(\widehat{X}^\bullet) \simeq H_{B;BC,s}(X^\bullet).$$

Lemma 23.33. *Fourier–Mukai gives an isomorphism between the odd and even parts.*

24. Olivier Martin

Speaker.
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Title.
Rationally inequivalent points on complete intersections.

Abstract.
Let $X \subset \mathbb{P}^{n+k}$ be a very general complete intersection of multidegree $(d_1, \dots, d_k)$. Inspired by work of Voisin, Chen– Lewis–Shen conjectured in 2021 that the set of points rationally equivalent to, and different from, a fixed point $x \in X$ has dimension at most

$$2n - \sum(d_i - 1).$$

The authors proved this conjecture for hypersurfaces ($k = 1$), and Riedl– Yang proved it when

$$\sum(d_i - 1) \leq 2n$$

or

$$\sum(d_i - 1) \geq 2n + 2.$$

I will explain how to generalize the technique of Riedl–Yang to tackle the full conjecture and to obtain finer information about rational equivalence of points on complete intersections.

Let X be a smooth projective variety and $Z_0(X)$ a free abelian group on the points of X. $\alpha, \beta \in Z_0(X)$ are *rationally equivalent* if there exist

- S_n projective curves C_i
- morphisms $f_i : C_i \to X$
- a rational function $\phi_i \in \mathbb{C}(t)$

such that

$$\alpha = \beta = \sum If(\text{div}\phi_i)$$

which leads us to the Chow group of X:

$$\text{CH}_0(X) = Z_0(X)/\sim .$$

Example 24.1.
- $x, y \in \mathbb{P}^1$, $x \sim_{\text{rat}} y$, $x, y \in X$ lie on a rational curve in X then $x \sim_{\text{rat}} y$.
- S a K3 surface over $\mathbb{P}^2$ and a smooth sextic $C \subset \mathbb{P}^2$.
- Rational equivalence has more to do with raional curves in $\text{Sym}^{\vee}(X)$ than on X itself.

Now conside

$$X \longrightarrow CH^0(X)$$
$$x \longmapsto [x]$$

The fiber that contains x is called the orbit fiber rational equialences of x and is denoted $R_{X,x}$. This is a countable union of closed subsets.

Theorem 24.2 (Mumord-Roitman 69-71)**.** *If there exists a positive integer $p \leq \dim X$ s.t. $H^\bullet(\Omega^p_X) \neq 0$ then for $x \in X$ very general $codim R_{X,x} \geq p$. In particular, if K_X is effective then X is not covered by positive dimension orbits.*

upshot: if you have forms, orbits are.. small?

Theorem 24.3 (Voisin, 94-96)**.** *If $X \subset \mathbb{P}^{n+k}$ is a very general complete intersection of multidigree $(d_1, \dots, d_k)$ such that $\sum(d_i - 1) \geq 2n-2$ then no two points on X are rationally equivalent to one another. (The Chow map $X \to CH^0(X)$ is injective.)*

It was conjectured in 2001 by Chen-Lewiz-Shein (two from Alberta, one from China) that if $X \subset \mathbb{P}^{n+k}$ v.g. complete intersection of m. d. $(d_1, \dots, d_k)$ then $\forall x \in X$, $\dim R_{X,x} \leq 2n - \sum(d_i - 1)$.

This would imply Theorem 24.3.

Remark 24.4.
- CLS prove it for hypersurfaces.
- Riedl-Yang 22 prove when $\sum(d_i - 1) \neq 2n$.

Expected theorem (M)

Same setup: $X \subset \mathbb{P}^{n+k}$ very general complete intersection of m.d. $(d_1, \dots, d_k)$ then $\dim \tilde{R}_X \leq 4n - 2\sum(d-1)$.

Theorem $\implies$ Conjecture, $R_{X,x} \times R_{X,x} \subset R_X$.

Example 24.5. If $X \subset \mathbb{P}^3$ is a very general quintic surface then CLS says $R_{X,x}$ is a countable union for all x. Theorem says that $\{x : R_{X,x} \neq \{x\}\}$ is also countable.

Rieell-Yang method.

The idea is to use this method to prove the expected theorem. This method should work for a wide variety of other problems [list, including hyperbolicity].

Let B_n be the parameter space for smooth complete intersections in $\mathbb{P}^{n+k}$ of n. d. $(d_1, \dots, d_k)$ and its universal family

$$\begin{tikzcd} \mathfrak{X}_n \arrow[d] \\ B_n. \end{tikzcd}$$

Definition 24.6. A *parametrized* $(n+k)$*-linear subspace* of $\mathbb{P}^{m+k}$ is an injective linear map $\Lambda : \mathbb{P}^{n+k} \to \mathbb{P}^{m+k}$. The set of such is denoted $G_{n,m}$. Λ is called *adapted* to a pointed complete intersection (X, x) if (1) x inn Im(Λ) and (2) $\Lambda^2(X) \subset \mathbb{P}^{n+k}$ is a smooth complete intersection.

$$\Lambda^*(X, x) := (\Lambda^{-1}(X), \Lambda^{-1}(X) \in \mathfrak{X}_n.$$

The idea is to take a complete intersection on the RHS and pull it back.

Definition 24.7. Given $\mathcal{Z} \subset \mathfrak{X}_n$ a construcible subset, we let

$$\mathcal{Z}^+ = \{(X, x) \in \mathfrak{X}_{n+1} : \exists \Lambda \in G_{n,n+1} \text{ adapted to } (X, x) \text{ s.t. } \Lambda^*(X, x) \in \mathcal{Z}\}.$$

Somehow this is a natural operation to study subsets of complete intersections; or even “towers” of subvarieties.

The main property that they proved is

Theorem 24.8 (RY)**.** *If $\mathcal{Z} \subset \mathfrak{X}_n$ is a constructible subset of positive codimension then*

$$\mathit{codim}_{\mathfrak{X}_{n+1}} \mathcal{Z}^+ < \mathit{codim}_{\mathfrak{X}_n} \mathcal{Z}.$$

Let's sketch how to use to prove RY '22 theorem (see above).

Theorem 24.9 (RY 22)**.** *Let $X \subset \mathbb{P}$ be a very general complete intersection w.d. $(d_1, \dots, d_k)$, $\sum_i (d_i - 1) < 2n$ for all $x \in X$. Then*

$$\dim R_{X,x} \leq 2n - \sum (d_i - 1).$$

Proof. (Sketch.) First can assume

$$\sum (d_i - 1) > n.$$

Let $c := \sum (d_i - 1) - n - 1$. $K_{\mathfrak{X}_{n,t}} = \mathcal{O}(c)$, which measures how far we are from a CY.

Suppose towards a contradiction that the following dominates the base B_n

$$\left\{ (X, x) : \dim R_{X,x} > 2n - \sum_i (d_i - 1) \right\}$$

If this is the case, there exists $\mathcal{Z}$ contained in the set above, irreducible with relative dimension $> 2n - \sum_i (d_i - 1)$ over B_n. This implies that

$$\begin{aligned} \mathcal{Z} &\leq n - \left(2n - \sum_j (d_j - 1) + 1 \right) \\ &= \sum_j (d_j - 1) - n - 1 \\ &= c. \end{aligned}$$

Then

$$\mathrm{codim} \mathcal{Z}^{+c} \leq \mathrm{codim} \mathcal{Z} - c = 0,$$

so $\mathcal{Z}^{+c}$ is dense, so for $t \in B_{n+c}$ generic, $\mathcal{Z}_t \subset \mathfrak{X}_{n+c,t}$ is dense.

$$\mathcal{Z}_t \subset \left\{ x \in X : \dim R_{X,p} > 2n - \sum_i (d_i - 1) > 0 \right\}.$$

i.e. $\mathfrak{X}_{n+c,t}$ is covered by positive dimension orbits, contradicting Mumford-Rotman. □

A bipointed RY method.

$\mathcal{Z} \subset (\mathfrak{X}_x \times_{B_n} \mathfrak{X}_n)$, where the latter is a *bipointed complete intersection.*

$$\mathcal{Z} \subset \{ (X, x_1, x_2) : x_1 \sim x_2, x_1 \neq x_2 \}.$$

The idea is that when it "stabilizes" the codimension decreases, namely the naive hope is that $\dim \mathcal{Z}^+ < \mathrm{codim} \mathcal{Z}$. However, this is false. But in a controllable way: we can understand when it fails.

Example 24.10. Suppose $\mathcal{Z}$ is the diagonal,

$$\mathcal{Z} = \Delta = \{(X, x, x)\} \subset (\mathfrak{X}_n \times_{B_n} \mathfrak{X}_n).$$

When we stabilize the diagonal, we get the diagonal again:

$$\Delta_n^+ = \Delta_{n+1}$$

but the codimension on LHS is n, and on the RHS is $n + 1$.

Example 24.11. Suppose

$$\mathcal{Z} = \{(X, x_1, x_2) : \operatorname{span}(x_1, x_2) \text{ is tangent to } X\}.$$

In this case, the codimension does not decrease! That would lead to a contradiction using the tangent lines.

Definition 24.12. $(X, x_1, x_2) \sim (X', x_1', x_2')$ if $\dim \operatorname{span}(x_1, x_2) = \dim \operatorname{span}(x_1', x_2')$.

$$\begin{aligned} \exists \phi : \operatorname{span}(x_1, x_2) &\longrightarrow \operatorname{span}(x_1', x_2') \\ x_i &\longmapsto x_i \end{aligned}$$

$$\phi^* : I_{X'}|_{\ell'} \xrightarrow{\sim} I_X|_{\ell}$$

Suppose, for the purpose of the exposition, that the $\operatorname{codim}\mathcal{Z}^+ < \operatorname{codim}\mathcal{Z}$ does hold for all $\mathcal{Z}$.

We want to show

$$\dim \tilde{R}_X \leq 4n - 2\sum(d_i - 1).$$

Suppose not. Then there exists

$$\mathcal{Z} \subset \{(X, x_1, x_2) : x_1 \neq x_2\}$$

- irreducible
- dominates the base,
- $\dim \mathcal{Z} > 4n - 2\sum(d_i - 1)$.

thus, $\operatorname{codim}\mathcal{Z} < 2n - (4n - 2\sum(d_i - 1)) = 2\sum(d_i - 1) - 2n$.

$$\begin{aligned} \operatorname{codim}\mathcal{Z}^{+c} &\leq 2\sum(d_i - 1) - 1 - \sum(d_i - 1) + n + 1 \\ &= \sum(d_i - 1) - n \\ &= c - 1. \end{aligned}$$

If $pr_1 : \mathcal{Z}^{+c} \to \mathfrak{X}_{n+c}$ is dominant we win because the fiber is positive dimensional.

But saturation and projections are compatible.

$$pr_1(\mathcal{Z}^+) = (pr_1(\mathcal{Z}))^+$$

So it's enough to show $pr(\mathcal{Z})\operatorname{codim} \leq c$. If it has codim $> c$, the fixing $t \in B_n$ generic, $pr_1(\mathcal{Z})$ has dim $\leq n - c$ so the fibers have codim in $\mathcal{Z}$ bigger than …